Econ 139: Labor Economics

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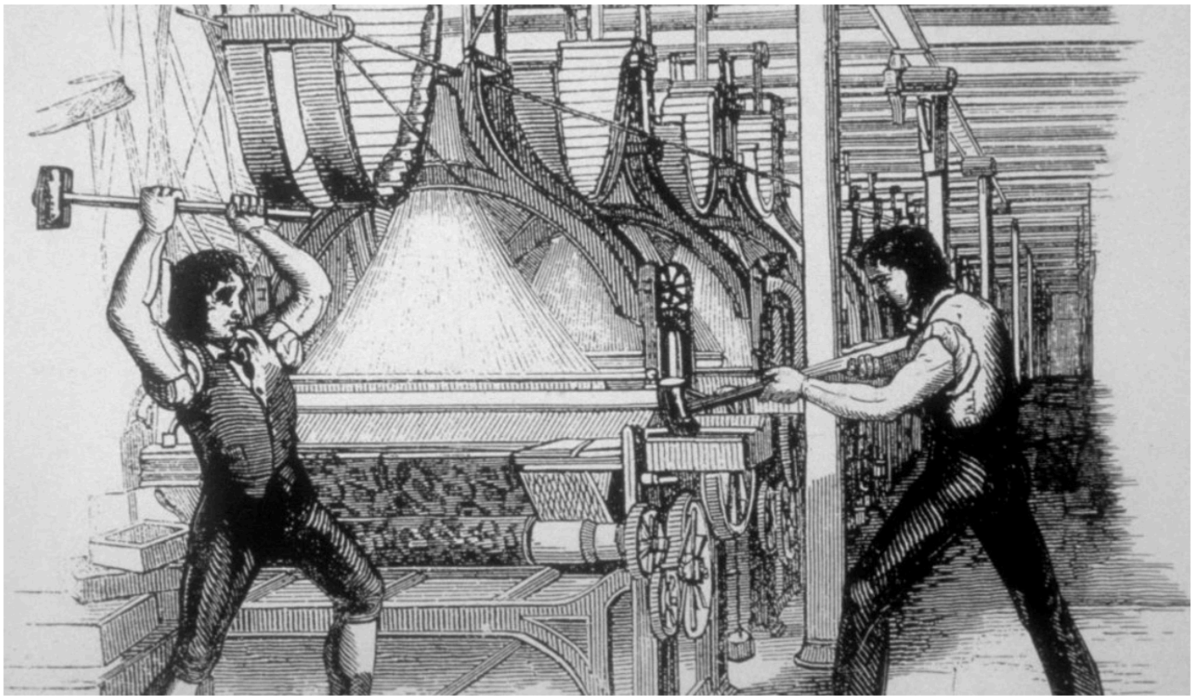
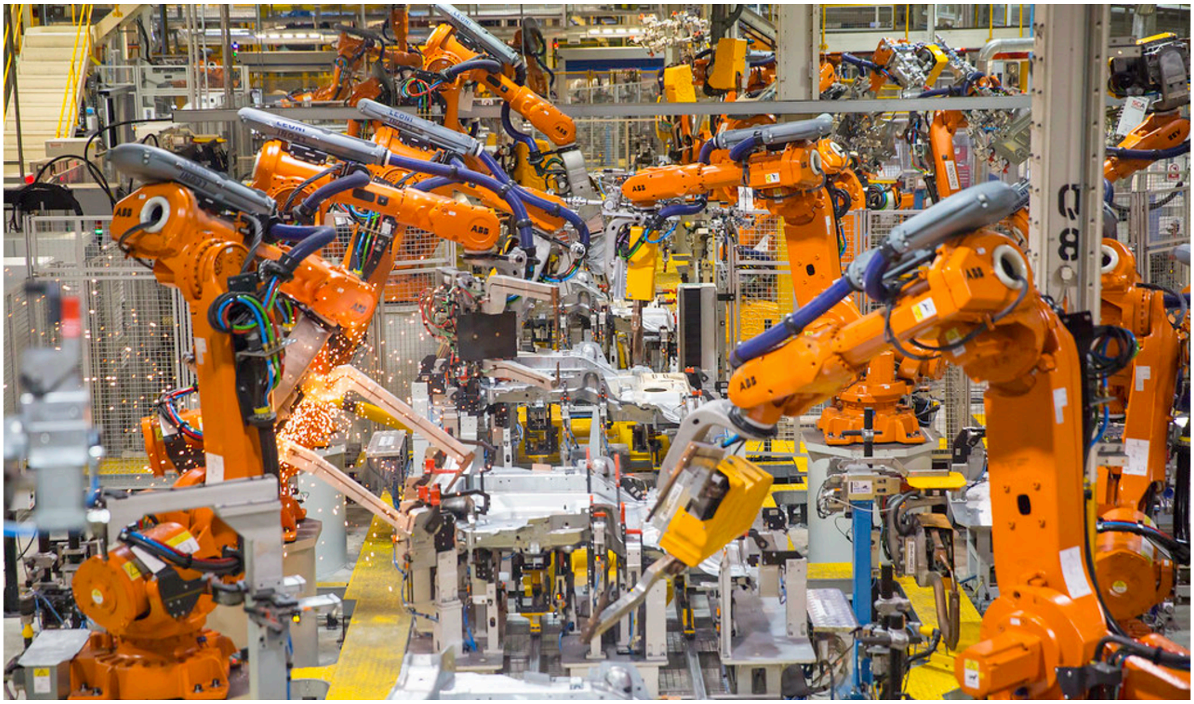
Table of contents

Introduction





In this book, we'll explore some of the biggest questions in labor economics:

1. Why has wage inequality grown so much?

2. What explains the decline in labor's share of income?

3. Why has the labor force participation rate changed so dramatically in the last 50 years?

To answer these questions, we'll use a combination of economic theory and empirical evidence. This will take us many places, to understanding how minimum wages affect employment and inequality, to exploring how technology has impacted the nature of work, to examining how competition between firms, both inside and outside the US, shapes labor market outcomes.

Part I

Main Text

1 Big Questions

1.1 The Rise of Wage Inequality

The early 20th century was marked by the rise of many titans of industry. Figures like John D. Rockefeller, Andrew Carnegie, Henry Ford and J.P. Morgan amassed enormous fortunes. The roaring twentires, a period of massive economic growth, was accompanied by tremendous inequality. For example, in 1929, if you consider the top 1% of all earners, these indviduals took home just over 20% of all income in the U.S. In contrast, the bottom 50% took home just under 15%.

World War II however brought about a dramatic change in the U.S. economy. The post-war boom led to a period of tremendous growth that benefited not only the wealthy, but also dramatically grew the middle class. Throughout the 1950s, for example, the share of income going to the top 1% fell dramatically. By 1970, it was just above 10% of total income, whereas the bottom 50% were earning just above 20% of all income.

However, things started change in the 1980s. The dramatic decrease in inequality brought about by the post-war boom was eroded away. In 2020, inequality reached levels not seen since the 1920s.

Figure 1.1 shows the complete picture over time. This figure was originally produced by economists Thomas Piketty and Emmanuel Saez in their influential paper *Income Inequality in the United States, 1913-1998* (Piketty and Saez 2003). The more recent data comes from the World Inequality Database, which continues to track income inequality in the U.S. and around the world. Our first big question will focus on one aspect of this figure: the rise in inequality since the 1980s. Before we dig into this question, however, there are two important points to make. First, there is no single definition of inequality. In the figure above, we measured inequality as differences in the share of income going to the top 1% vs. bottom 50%. But why not top 10% vs. bottom 50%? Instead of share of income, we can also measure inequality across groups of people. For example, a common measure of inequality measures differences across individuals of different educational backgrounds. We will return to these different measures shortly, but before we do, let's discuss the second important point about inequality. Inequality doesn't actually tell us anything about how well-off people are in general. Even if inequality is rising, it could be true that everyone is getting richer over time.

To study how well people are doing over time, we need a way to compare income over time. The average median household income in the U.S. in 1950 was about $3,000. In 2023, the

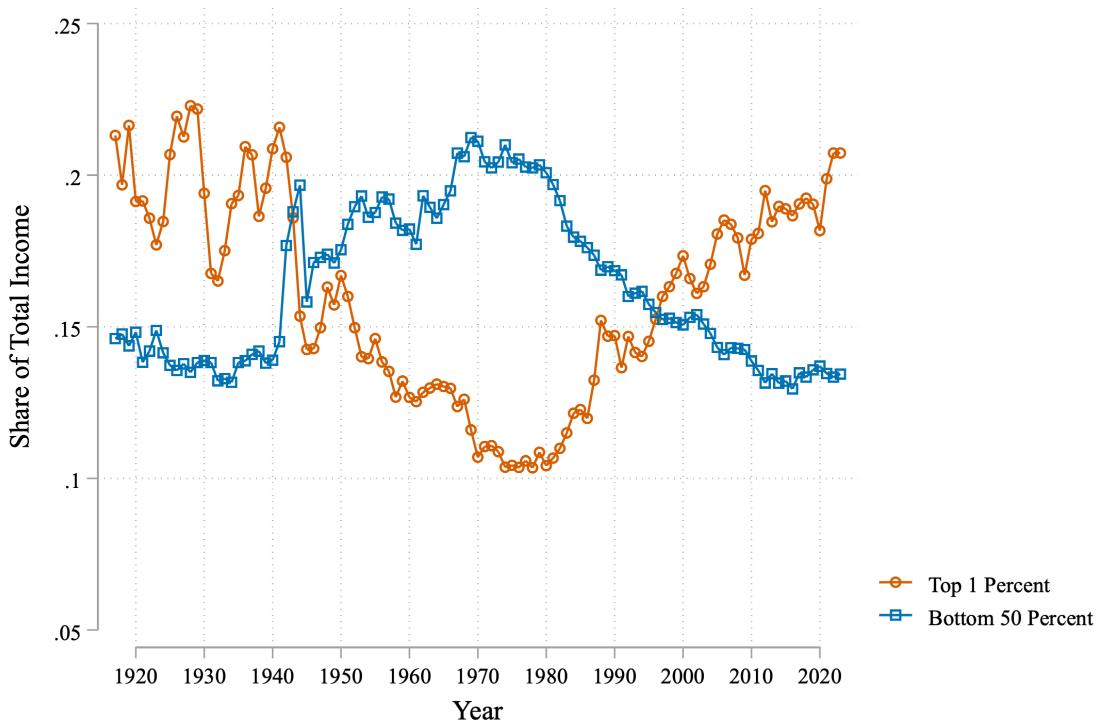


Figure 1.1: Income Inequality in the United States

median household income was $80,000. Does that mean the median household is richer in 2023? Not necessarily. Prices are dramatically different between these two time periods.

The key to comparing income across time is to compute **real income**. This is a measure of income that adjusts for inflation. By computing real income we are measuring how much a household can actually purchase. Therefore, if real income is higher today than in 1950, we interpret this as households being able to purchase more today than in the past. To get a sense of how this works in practice, let's construct a simplified example. Let's assume that households only pay for two things: housing and groceries.

In 1950, the median monthly rent for housing was $42 per month. In 2023, the median monthly rent for housing was about $1,300. Therefore, housing prices have increased by about 30 times since 1950. Now, let's consider groceries. To make things even simpler, let's just look at the case of bread. A single loaf of bread in 1950 cost about $0.30. In 2023, a loaf of bread costs about $2.00. Therefore, the price of bread has increased by about 10 times since 1950. For simplicity, let's assume the price of bread is a good indicator for how grocery prices have changed over time.

So how much have prices risen between 1950 and 2023? Housing has increased 30% but groceries 10%. How do we decide how much prices have risen overall? To do so, we will compute a **price index**. To compute a price index, we weight each price increase by what percent of a household's budget is spent on that good or service. For the sake of this simplified example, let's assume 50% of a household's budget goes to housing, and 50% goes to groceries. Therefore, the aggregate price index is given by:

$$\text{Price Index 2023} = \underbrace{0.5 \times 30}_{\text{Housing Increase}} + \underbrace{0.5 \times 10}_{\text{Grocery Increase}} = 20$$

This means that prices on average have increased by a factor of 20 since 1950. Now, remember our goal here. We want to understand whether $3,000 in 1950 can buy more or less than $80,000 in 2023. Now that we know prices are 20 times higher in 2023, we can take our 2023 income and deflate it by the price index to get real income in 1950 dollars:

$$\text{Real Income 2023} = \frac{\$80,000}{20} = \$4,000$$

So we finally have our answer. The median household income in 2023, when adjusted for inflation, is about $4,000 in 1950 dollars. This is higher than the median household income in 1950 of $3,000. Therefore, we can conclude that the typical household is better off today than in 1950.

Now we have made a number of simplifications. In particular, we have assumed there are only two goods: housing and groceries. We have assumed the cost of bread is a good approximation for the cost of all groceries. This is clearly not a good assumption in practice. In practice, the Bureau of Labor statistics tracks the prices of thousands of goods and services to compute the

Consumer Price Index (CPI). The concept is similar to what we have done here. You create a basket of goods that consumers buy. Then you construct a price change for each specific good, aggregating these price changes in order to get an overall change in prices. When you hear about inflation going up in the news, this is what they are referring to.

Using publicly available data from the U.S. Census Bureau and the Bureau of Labor Statistics (BLS), we can get a more accurate comparison of price levels between 1950 and 2023:

$$\frac{Prices_{2023}}{Prices_{1950}} = \frac{CPI_{2023}}{CPI_{1950}} \approx 12.6$$

So if we take the median household income in 2023 of $80,000 and deflate it by this price index, we get:

$$\text{Real Income 2023} = \frac{\$80,000}{12.6} \approx \$6,350$$

So the real median household income in 2023 is about $6,350 in 1950 dollars. This is more than double the median household income in 1950 of $3,000. So in real terms, households have much more income today than they did in the past. This might be a bit surprising to you. You often hear news about the rising cost of living since the 50s and 60s. But even adjusting for this rising cost of living, households earn more today than they did in the past. As we will see later, however, part of this might be coming from the fact that there are more two-earner households today than in the past.

There is one additional complication we should point out before moving on. To compute changes in prices, we assumed there is some basket of goods that consumers are purchasing. But a lot of what consumers buy today did not exist in the 1950s. Streaming subscriptions, smartphones, laptops, and so on. Even for products that do exists in both time periods, like televisions, the quality is dramatically different. This is a fundamentally difficult conceptual problem with no easy solution. That is one nice feature of how inequality is defined in Figure 1.1, which does not require adjusting for inflation.

Now that we understand how to compare income over time (and the complications that arise), let's get back to our big question: how has inequality changed over time? We are now going to shift from comparing income shares (as in Figure 1.1) to comparing real incomes across different groups. In particular, we are going to compare how real income has evolved for workers with different education levels.

Figure 1.2 plots changes in real income for men across different groups over time (D. H. Autor 2019). Unlike the previous example, we are now plotting income for individuals (rather than households, which may have multiple earners). This figure was produced by economist David Autor in his 2019 paper *Work of the Past, Work of the Future*. The figure shows some remarkable patters over time. First, between 1963 and 1972, incomes for all groups were rising dramatically. Incomes rose nearly 40% for individuals with a graduate degree (e.g. medical doctors, lawyers, professors). For those with either a high school degree or less, incomes rose about 20%. Because individuals with higher education saw their incomes rise faster, inequality between these groups was rising during this period. But again, no one is worse off, at least in monetary terms. Every group can buy more in 1972 than they could in 1963. This is what real income measures.

Throughout the 1970s wages were mostly stagnant. This was a period of high inflation and low growth, commonly known as stagflation, so its not too surprising we don't see rapid increases in real incomes. But then in the 1980s, something dramatic happens.

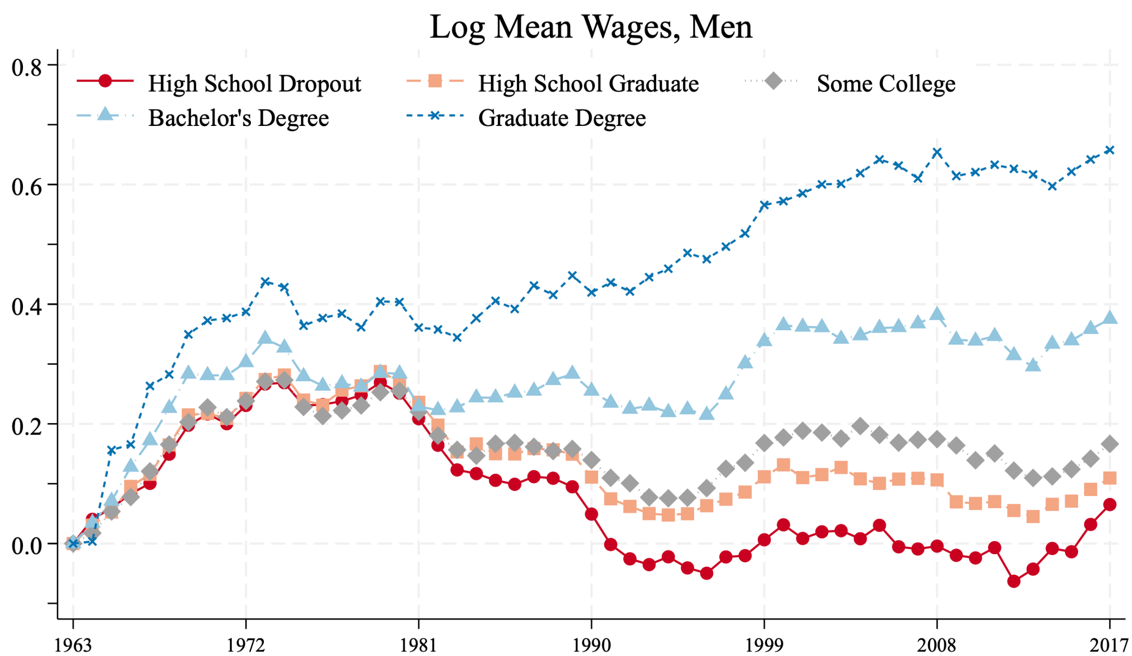


Figure 1.2: Real Income by Education Level in the United States

For individuals with graduate degrees, real incomes continue to rise steadily over time. For individuals with a college degree, there is some lag, but by the mid 1990s, we are again seeing increases in real incomes. Other groups are not doing so well. For example, for high-school dropouts, all of that income growth in the 60s is eroded away by around 1990 with little to no growth since then. This is why inequality skyrocketed in the 1980s. While highly-educated individuals (who already earned more on average) were seeing large increases in real income, less-educated individuals were seeing declines in real income. This is our second big question: what drove this rapid increase in inequality that began in the 1980s?

The McWage

One thing that you may have taken away from this chapter is that comparing income over time is perhaps not as straightforward as you might have thought. The same is true when comparing income across countries. For example, if you want to compare income in the U.S. to income in China, you have a number of hurdles. Just as in comparisons across time, the goods purchased might be very different.

Orley Ashenfelter and Štěpán Jurajda came up with an ingenious way to measure real income differences across countries using the ubiquitous presence of McDonald's (Ashenfelter and Jurajda 2024). Take a McDonald's worker in the U.S. earning $10.00 an hour. Let's assume the average price of a Big Mac in the U.S. is about $5.00 (the actual price varies by location). This means that in one hour, the worker can buy about 2 Big Macs. This is a measure of the worker's purchasing power. Ashenfelter and Jurajda refer to this as the "McWage".

Now, let's consider a worker in China. We can also compute the McWage for this worker. In China, because wages are generally lower, a McDonald's worker earns about half a Big Mac per hour. Why is this useful? Well, we now have a very clean comparison across places. We are considering wages for a highly standardized job. McDonald's uses the same technology and production process in all of its locations. We are also comparing a very similar good: a Big Mac. This allows us to make a very clean comparison of purchasing power of these workers that is not confounded by differences in the types of goods or the quality of goods in the different locations.

1.2 The Fall in the Labor Share

The labor share of income is the fraction of income that is paid to workers, with the rest being paid to capital owners. Labor shares are often reported at the country level, but to understand this concept, let's consider the case of single firm: Apple. The exact numbers we need to compute the labor share for Apple aren't publicly disclosed, but we can get a good approximation from Apple's annual reports.

In 2024, Apple reported about 390 billion dollars in revenue. While we might be tempted to think of this number as Apple's "income", that's not quite right. Our goal in this exercise is to compute what share of income is going to workers at Apple vs. firm owners and investors. But Apple has a lot of costs. It needs to pay other firms for the materials used in its products. This is money that is not going to either workers or owners of capital, so we shouldn't count it as part of Apple's income. Instead, we want to look at Apple's value added, which is the revenue minus the amount they spend on the inputs. In 2024, Apple's value added was roughly 180 billion dollars. According to annual reports, Apple spent roughly 46 billion dollars on worker compensation. Therefore, labor's share of income is given by:

$$\text{Labor Share} = \frac{\text{Total Labor Compensation}}{\text{Value Added}} = \frac{46}{180} \approx 0.25$$

So in 2024, Apple's labor share of income was about 25%. This means that 25% of the value created by Apple was paid to its workers. The rest, 75%, went to owners of capital, which includes the shareholders and other investors.

The labor share of income varies dramatically across companies and industries. For example, take Walmart, the largest employer in the U.S. Walmart had a value added of roughly 100 billion dollars in 2024, and it paid about 75 billion dollars in labor compensation. This means that Walmart's labor share of income is about 75%, much higher than Apple's. This makes some sense. Walmart is a retail company, spending large amounts on its massive workforce. Apple, on the other hand, has fewer employees. Much of the value in Apple is in their brand and intellectual property. The firm owners are the ones who gain the most from this value.

This example illustrates that the labor share of income may vary widely across firms. We, then, might expect that it also varies widely across time. Firms are constantly going in and out of business, industries are constantly evolving, new technologies emerging. But if we look in the data, we find a somewhat miraculous fact: for much of the 20th century, the labor share of income was remarkably stable.

The first person to draw attention to this fact was Nicholas Kaldor, who wrote a paper called *A Model of Economic Growth* (Kaldor 1957). In this paper, Kaldor documented a number of empirical facts that became extremely influential in the field of macroecnomics. Among these facts was the observation that the labor share of income in the U.S. over time had been stable. To compute the labor share of income for a country (rather than for an individual firm), you can add up all the income paid to workers and then divide by the total GDP of the country. The total GDP represents value of everything produced in the country. So at a national level, the labor share is the amount of value produced in the economy that goes to workers. The figure below plots the labor share of income in the U.S. from 1947 to 2019.

Kaldor had much less data to work with, but between 1947 and 1961, you can see in Figure 1.3 that the there is no clear trend in the labor share of income. Yes, it sometimes bounces up, sometimes down, but on average it hovers around 65%. From 1961 to 2000, the labor share of income did fluctuate, but again, it was remarkably stable. While overall there is a slight downward trend, it is not particularly pronounced. In 1961, for example, the labor share was about 65% and in 2000 it was about 63%. This is despite the intervening years being a period of great change for the U.S. economy. The 1960s continued the tremendous growth of the post-war period. In 1970s, the economy was hit by oil shocks that caused a large inflationary period with high unemployment. In 1987, the stock market crashed, with the single largest one-day drop in the history of the stock market. In the mid-1990s, the U.S. economy experienced a period of rapid growth brought on by the rise of the internet. Despite all these massive changes, the labor share of income, on average, remained fairly steady.

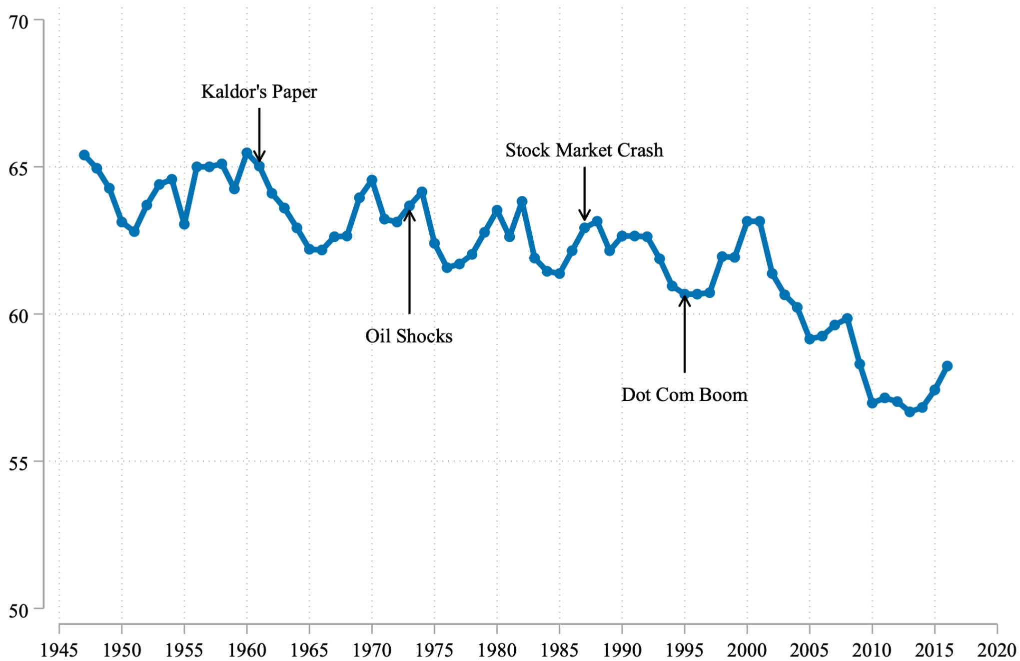


Figure 1.3: Labor Share in the United States

Then, in 2000, the labor share began to drop. Instead of the usual cycles we see before this time, the drop has been steady. This is the second big question we will tackle in this course: why has the labor share of income been falling since 2000?

> **The Labor Share in Ancient Rome**
>
> One of the first sources of any data on labor markets and prices comes from a unique episode of Roman history. In the third century, the Roman empire faced an economic crisis. The government repeatedbly reduced the amount of silver in coins to increase military spending (the modern equivalent is the government printing more money). This caused rampant inflation throughout the Roman Empire.
>
> As a response, in 301, Diocletian, the Roman Emperor, issued a series of edicts that set maximum prices for goods and wages. Stone tablets with these prices were erected in public squares. These edicts have provided economists and historians a rare glimpse into the ancient economy. With these edicts, we can calculate a (very) approximate labor share of income for Ancient Rome.
>
> To do so, let's take a simplified example. Let's consider a single product: wheat. The edicts contained both the price of wheat, as well as the wage for an unskilled agricultural worker.
>
> To compute the labor share of income, let's consider an agricultural worker on a single day. According to Diocletian's edicts, the maximum wage for an unskilled agricultural laborer was 25 denarii per day (the denarii is the currency in Ancient Rome).
>
> But what is the value of what the worker produced? From other ancient sources, we can get a rough sense of the quantity of wheat that could be harvested in a single day. On an average day, a single worker could harvest about 240 kg of wheat. This would sell for around 1,400 denarii according to Diocletian's edicts
>
> So what does this imply for the labor share? Well, a worker can earn 25 denarii per day, and the value of what they produce is 1,400 denarii. The labor share of income is then equal to just 0.02, or 2%, a miniscule fraction. And this is ignoring the fact that many workers were not paid at all.

1.3 Dramatic Shifts in Labor Force Participation

In the news you often see reports about fluctuating unemployment rates. The unemployment rate is the fraction of people (aged 16 or over) who are actively looking for work within the last month, but are not currently employed. It is undoubtedly an important metric of the health of the labor market which we will return to in future sections. However, something that is often overlooked is whether people are even participating in the labor market at all.

In this section, we will explore the labor force participation rate. This is the fraction of people (aged 16 or over) who are either employed or actively looking for work. It is much less responsive to economic fluctuations. If an economy has a recession, many people will be laid off. If all of these individuals continue to search for work, then the labor force participation rate will not change. Therefore, it is much less informative about the short-run health of the labor market. However, this doesn't mean it is unimportant. In the long-run, we care a lot about what fraction of people are engaging with the labor market. And it turns out there are some important, and somewhat puzzling, trends in labor force participation.

Figure 1.4 plots the labor force participation rate in the U.S. from 1948 to 2023, separately for men and women. For women, labor force participation started at around 30% and rose to around 60% by the mid 1990s, an incredibly rapid increase. You are likely at least somewhat familiar with this trend, even if the exact numbers are new.

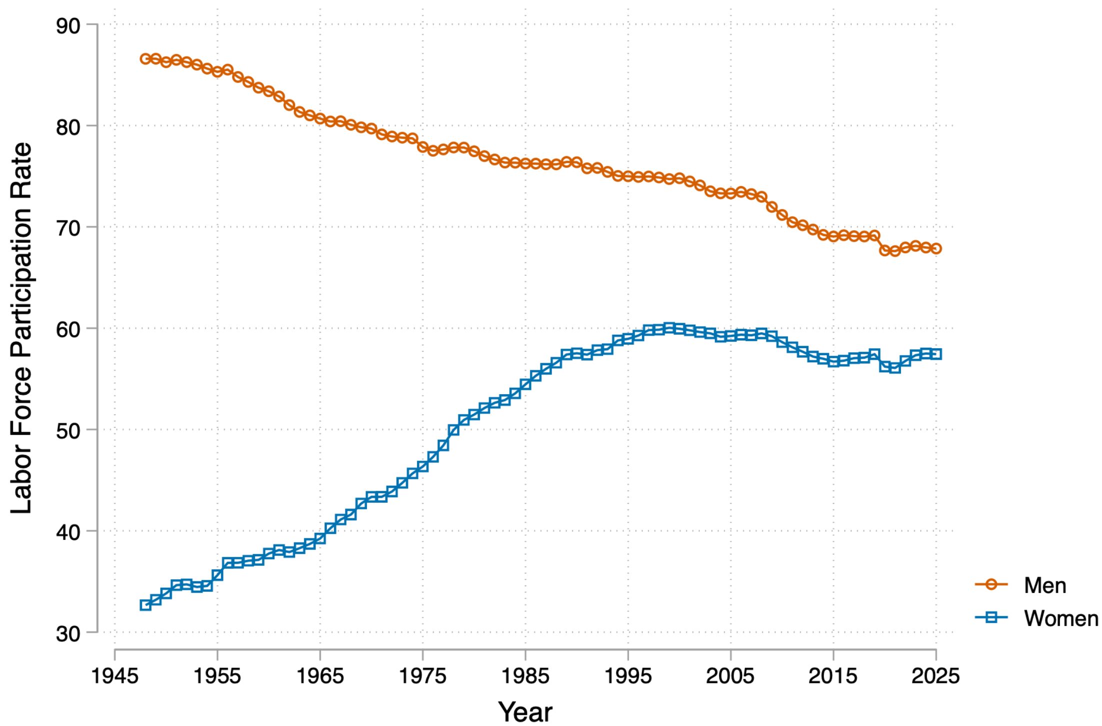


Figure 1.4: Labor Force Participation Rate in the United States

For men, we see a very different trend. The labor force participation rate for men started at just below 90% in 1948 and has gradually declined over time. Today, it is just below 70%. Of course, these two trends could be related. The increase in female labor force participation could cause some men to drop out of the labor force in order to take care of children in the household. We will find that this can't explain the entire trend. Even from the data so far, we can see that since the mid 1990s, female labor force participation has remained pretty stable, even declining somewhat. But male labor force participation continued to decline, even picking up speed in the 2010s, with a noticeable drop during the COVID-19 pandemic.

This is our third big question: what is driving the dramatic changes in labor force participation in the US over time?

1.4 Potential Explanations

As we will see, there are seldom extremely simple answers to these big questions. Throughout the twentieth century and into the twenty-first century, there have been many important changes in the labor market. Many of the changes may contribute to one, two or all of these big trends we've studied so far. To help organize, we are going to organize different explanations into three broad categories:

1. **Technology**: This refers to the introduction of new technologies that change how work is done. For example, the rise of computers and automation has dramatically changed the nature of work in many industries. We will study how these changes have affected the labor share of income, wage inequality, and labor force participation.
2. **Competition**: We will study two facets of competition. First, one of the most-discussed changes in the post-war economy is the rise of globalization. This fundamentally changes what is produced by a country, which can have important implications for the labor market. Second, we will study how competition between firms for workers impacts wages. Throughout much of the history of labor economics, economists have assumed that the labor market is perfectly competitive. We will discuss in detail what this implies, and why an imperfectly competitive labor market may be a better description of the labor market.
3. **Institutions**: Policies and institutions also play a pivotal role in the labor market. Unionization rates, minimum wages, and employment protection legislation all shape who works and how much they earn. It is often hotly debated whether policies like the minimum wage are good or bad for the labor market. We will study a number of important institutions and explore how the impact the labor market and how they have changed over time.

With each topic we study we will return to these big questions.

1.5 Economic Models

To guide our exploration of these big questions, we will introduce a few simple models of the labor market. The labor market, in reality, is quite complex. It is composed of millions of firms and workers, each with their own preferences and constraints. The sum total of all the individual behavior of these firms and workers would be impossible to characterize fully. As an analogy, take Newton's model of gravity. The sun and earth are each composed of billions of individual particles, each with their own gravitational pull. But Newton's model of gravity tells us if we want to understand the motion of the earth around the sun, we don't really need to keep track of all the individual particles. We just need to know the mass of the sun and the earth, and the distance between them. That makes the problem enormously easier. This is, in spirit, the goal of our models of the labor market. Our goal is to understand complex phenomena, like the rise in wage inequality. We want a model that is simple, but allows us to gain insight into the problem, and potentially make predictions.

Generally, models are formalized with math. Some people, even academics, are sometimes inherently skeptical of models. They say they are too simple. They miss important aspects of reality. This is often true. But even if you are against models, and refuse to write one down with math, you often do have a model in your mind. Take the example of the minimum wage. If you think that raising the minimum wage will lead to more unemployment, you have a model in your mind. If you think that raising the minimum wage will lead to higher incomes, with little change in employment, you have a different model in your mind. By making the model explicit, we can better understand the assumptions we are making, and how they lead to different predictions. With a model, you can prove to yourself that your different predictions are at least logically consistent with the assumptions you are making.

In this course we will consider various models of the labor market. The first is a model of a perfectly competitive labor market. This is probably the version you saw in your introductory economics class. For studying technological changes, we will find this is a powerful model. It abstracts from other aspects of the labor market that are perhaps not central to the study of the impact of technology on the labor market. However, it has significant limitations. When turning to institutional factors, like the minimum wage or unionization, it will predict these policies have negative impacts on the labor market. A perfectly competitive labor market is efficient. It's better not to mess with it.

But the reality is that the labor market is not perfectly competitive. The second model we will introduce is a monopsonistic model of the labor market. In this model, we will assume that firms have some market power over their workers. This implies that wages will not necessarily reflect the value of a worker's output. This will also have important implications for how we think about policies like the minimum wage.

×

2 Perfect Competition

2.1 The Wage Level?

All of our big trends depend crucially the wage level. For the labor share of income, if wages go down, the the labor share of incomes will also go down, as the total amount of income paid to workers has decreases. Trends in inequality depend directly on the wages paid to different groups of workers. Labor force participation again depends on the wage level. If the wage level goes up, then more individuals will find it worthwhile to work, and the labor force participation rate will increase.

Therefore, understanding what determines the wage level will be a recurring theme throughout this course and book. In this section, we will first focus on a perfectly competitive model of the labor market, and in particular, consider how firms decide how many workers to hire and how much to pay them. This will be a valuable baseline for studying the potential impacts of technology on the labor market later in the course.

2.2 Definition of Perfect Competition

What does it mean for the labor market to be perfectly competitive? Let's take the perspective of an individual firm. To make it even more concrete, let's consider a specific case study: a cafe that is trying to hire barristas.

Imagine the cafe decides that they need to hire 7 barristas (we will come to how they make this decision in the next section). Well, what wage should they offer? In a perfectly competitive market, this decision is straightforward. The cafe will pay whatever the overall market wage is for barristas. The individual cafe has absolute no control over this.

Perfectly competitive models have some underlying assumptions. We are going to keep it as simple as possible in this course and understand the intuition behind these assumptions, rather than the formal mathematical definitions. The two key assumptions are (1) perfectly competitive models have a large number of workers and large number of firms and (2) there are no frictions for workers that make it costly to move between firms.

This "no friction between moving jobs" is central. What it means is that if a firm offers lower than the market wage, it will immediately lose all of its workers. To take the case of our cafe, if the market wage for barristas is $15 an hour and the cafe wants to hire 7 barristas, then if the cafe offers \$14 an hour, it will not be able to hire any barristas. All of the barristas will immediately leave the cafe and go work for another cafe that is paying the market wage of \$15 an hour.

Will the cafe ever decide to pay more than the market wage? No, because if it did, it would be paying more than it needs to. If the cafe offers \$16 an hour, it will still be able to hire 7 barristas, but it will be paying \$1 an hour more than it needs to. Throughout we will assume that firms are profit maximizers. This means the decisions of the firms will entirely be driven by whatever decisions maximize their total profits. In a perfectly competitive market, this means that firms will always pay the market wage.

The way this situation above is often described is that the firms are "price takers." This means that they take the price of labor as given. They don't actually choose the wage they pay to workers, this value is determined by the overall market. The firm can only choose how many workers to hire, and it will always pay the market wage for those workers.

This will be our starting point for understanding how the labor market works. You might think there are aspects of what we have assumed that don't appear realistic to you. For example, when we said that if the cafe pays a dollar under the market wage, it will lose all its workers. In reality, there are firms that pay lower wages than other firms, and it's not true that they have zero workers. To understand many aspects of the labor market, it will be important to relax these assumptions, and we will get to that in time. But for now, we will be able to make significant progress by starting with a perfectly competitive model.

2.3 Single-Input Production Function

In economics, the production function is a central concept. Firms, in essence, take inputs and produce some output. Apple takes in raw inputs like aluminum and silicon, and produces iPhones. Cafes take in inputs like coffee beans, milk, and barristas and then produce lattes. The production function is a mathematical representation of this process. To begin, let's start with the simplest possible case: a production function with labor as the only input.

We will assume that the output produced by a firm Y is a function of the number of workers L that it employes:

$$Y = f(L)$$

To make this more concrete, let's again consider the case of our cafe. Let's assume the cafe produces a single item: lattes. The number of lattes it can produce is a function of the number of barristas it employs. Let's assume that if the cafe has 1 barrista, it can produce 10 lattes in an hour, it it has 2 barristas, it can produce 19 lattes in an hour, and so on. In the graph below the x-axis is the number of barristas and the y-axis is the number of lattes produced. This is a visual representation of the production function for our cafe.

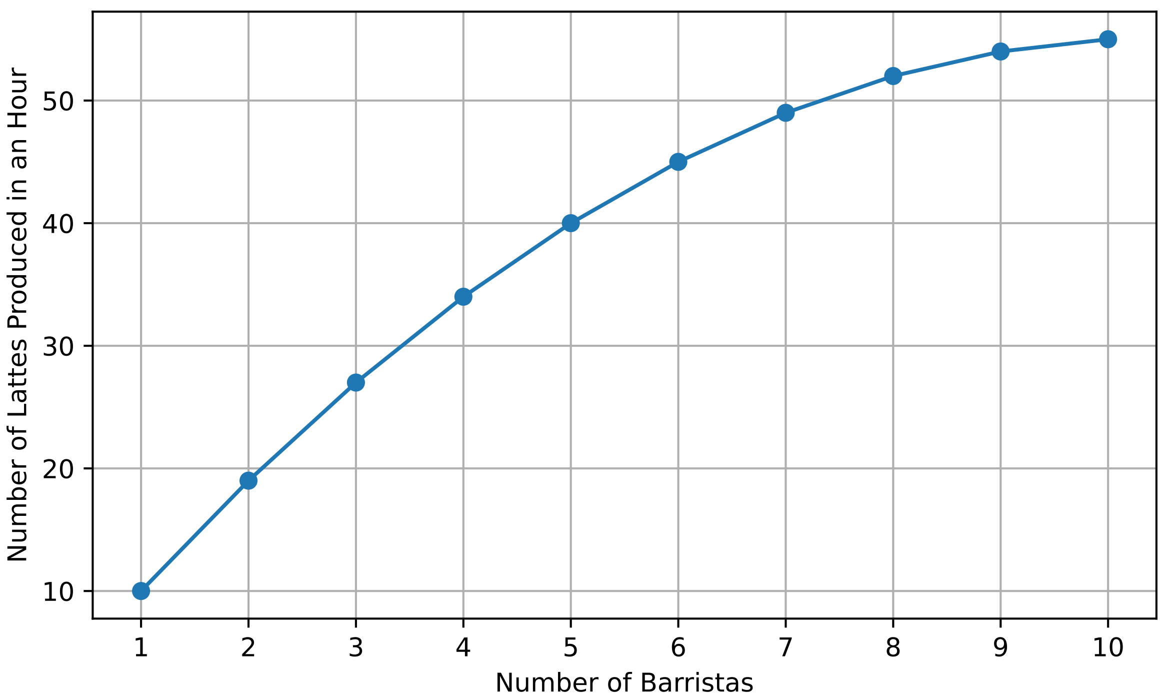


Figure 2.1: Production Function: Barristas vs. Lattes

Now the shape of the production function wasn't an accident. For this production function, we have drawn, going from zero to workers to one worker increases the number of lattes produced by 10. But going from one worker to two workers increases the number of lattes produced by 9, and going from two workers to three workers increases the number of lattes produced by 8. With each additional worker, the number of lattes produced increases, but at a decreasing rate. This is called **diminishing marginal returns**.

In the case of our cafe, this makes a lot of sense. Maybe the coffee shop only has one espresso machine. If it hires a second barrista, then yes, you will likely see an increase in the total number of lattes produced, but there is still only one espresso machine. To take it to the extreme, imagine there are 100 barristas, but only one espresso machine. A lot of those barristas will probably just be standing around. So even though you have more workers, you don't have any more output.

So when does the firm stop hiring barristas? What does it depend on? Well, for each worker, the firm must pay a wage. So let's assume that the market wage for barristas is $15 an hour. Now, we need to compare the cost of hiring an additional worker to the additional output produced by that worker. The additional output produced by that worker is called the **marginal product of labor**. In our case, the marginal product of labor is the additional number of lattes produced by hiring one more barrista.

So going from 0 to 1 barrista, the marginal product of labor is 10 lattes. What is the value of these additional 10 lattes? Well, it depends on what these are selling for. Let's assume that the cafe sells each latte for $5. Then the value of the additional 10 lattes is $50. Therefore, the **marginal revenue product of labor** is equal to $50. The marginal product of labor is how many additional lattes, while the marginal revenue product of labor is how much those additional lattes are worth in terms of revenue.

So the first decision is clear, the $50 in revenue from the first barrista is greater than the $15 cost of hiring that barrista, so the cafe should hire that barrista.

Now, what about the second barrista? The marginal product of labor is 9 lattes, and the marginal revenue product of labor is $45. Again, this is greater than the cost of hiring that barrista, so the cafe should hire that barrista as well. Instead of going through each additional barrista, let's summarize the marginal revenue product of labor and the marginal revenue product of labor in a table:

Table 2.1: Marginal Product and Marginal Revenue Product of Labor

Table 2.1

Number of Workers	Marginal Product (Lattes)	Marginal Revenue Product ($)
1	10	50
2	9	45
3	8	40

Number of Workers	Marginal Product (Lattes)	Marginal Revenue Product ($)
4	7	35
5	6	30
6	5	25
7	4	20
8	3	15
9	2	10
10	1	5

Given this table, when does it make sense for the cafe to stop hiring barristas? Well, it should keep hiring barristas until the marginal revenue product of labor is less than the cost of hiring that barrista. In our case, the cost of hiring each barrista is $15. The marginal revenue product of labor is equal to or greater than $15 for the first 8 barristas, but for the 9th barrista, the marginal revenue product of labor is $10, which is less than the cost of hiring that barrista. Therefore, the cafe should hire 8 barristas.

It is often helpful to visalize this decision graphically. Below, we plot the marginal revenue product of labor and the cost of hiring each barrista. The point where the two lines intersect is the point where the cafe should stop hiring barristas.

2.4 Labor Demand Curve

So far, we have figured out the demand for the firm assuming the market wage is $15. A firm's overall demand curve is the demand for any given wage $L(w)$. Since we know the number of workers hired for a wage of $15 is 8, then $L(15) = 8$. To construct a firm's demand curve, we can simply repeat the process above for different wages.

For example, if the hourly wage is $20, then the cafe will now hire 7 barristas. This is because the value of hiring the 7th barrista is $20, but the value of hiring the 8th barrista is $15. Therefore the cafe will hire 7 barristas.

Let's repeat this process for a range of wages. Below we plot a table where we show for each wage, how many barristas the cafe will hire.

Table 2.2: Labor Demand: Number of Baristas Hired at Different Wage Levels

Table 2.2

Hourly Wage ($)	Number of Baristas Hired
50	1
45	2

Hourly Wage ($)	Number of Baristas Hired
40	3
35	4
30	5
25	6
20	7
15	8
10	9
5	10

Now, let's read through this carefully. When the hourly wage is $50, the cafe will hire 1 barrista. Before, what did we say the marginal revenue product of labor was for the first barrista? It was $50. This makes perfect sense. If the wage is $50, you will hire until the marginal revenue product of labor is less than the wage. So if the wage is $50, you will hire 1 barrista, because the marginal revenue product of labor for any following worker will be less than $50.

Next, we are going to plot this data in our figure. Note that when we think about the labor demand curve, the input is a given wage, and the output is a number of workers hired. Therefore, if we were to plot the demand curve, the wage would be on the horizontal axis and the number of workers hired would be on the vertical axis. However, because of some longstanding conventions in economics, we will actually plot the number of workers hired on the horizontal axis and the wage on the vertical axis. This is a bit counterintuitive, but it conforms to standard economic conventions. Because the wage is on the vertical axis, you will sometimes hear this curve referred to as the inverse labor demand curve. Whether you call it the labor demand curve or the inverse labor demand curve, it is the same concept, it tells you for each wage, how many workers will be hired.

One thing you might notice is that the labor demand curve looks exactly like the marginal revenue product of labor curve we plotted earlier. This is not a coincidence. The labor demand curve is exactly equal to the marginal revenue product of labor curve in perfectly competitive models of the labor market. This is because the firm will hire workers until the marginal revenue product of labor is equal to the wage. Therefore, for any given wage, the number of workers hired is determined by the marginal revenue product of labor.

One important conceptual point here that can get lost sometimes. The demand curve for labor is not literally the marginal revenue product of labor. The demand curve for labor is the relationship between the wage and the number of workers hired. The marginal revenue product of labor is the value of the additional output produced by hiring an additional worker. In a perfectly competitive market, these two are equal to each other. But if you deviate from the perfectly competitive model, then the demand curve for labor and the marginal revenue product of labor will not be equal to each other.

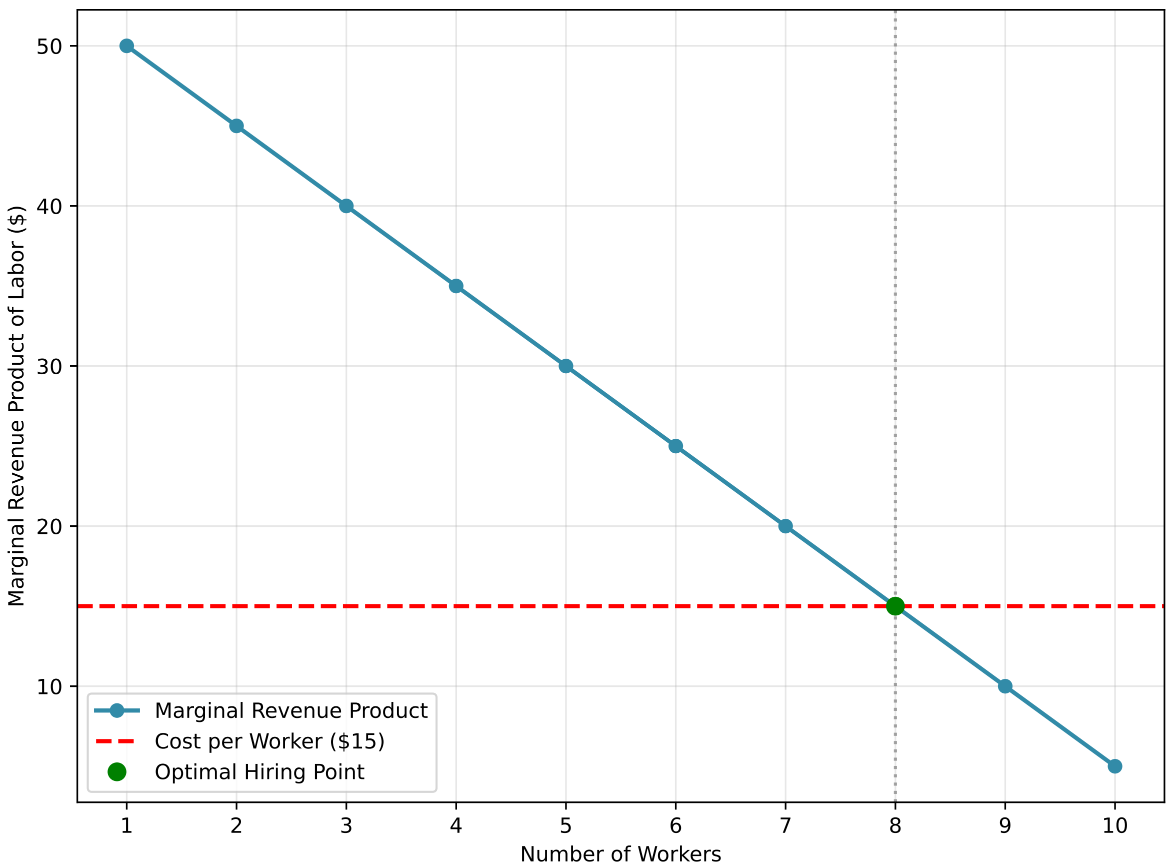


Figure 2.2: Marginal Revenue Product of Labor vs. Cost of Hiring

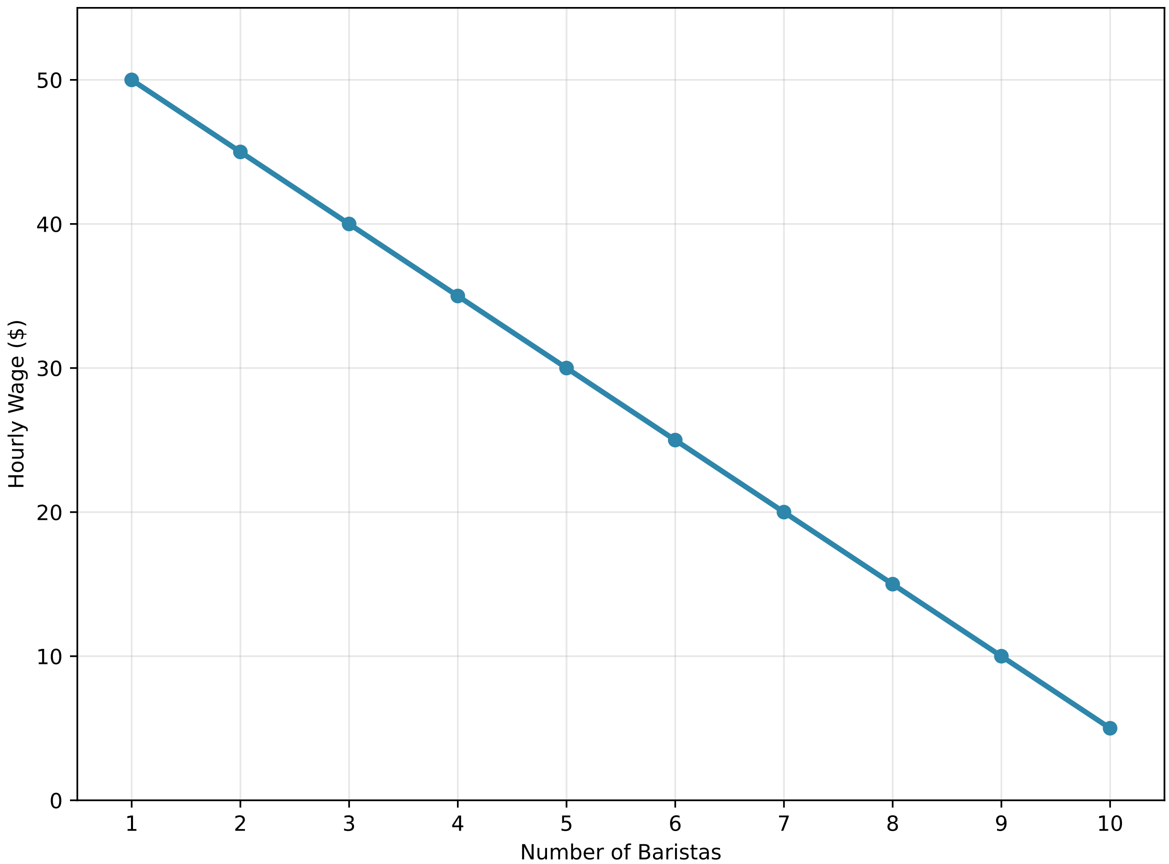


Figure 2.3: Labor Demand Curve for the Cafe

2.5 Market-level Labor Demand Curve

So far, we have figured out the demand for a single cafe. But what about the overall market? What is the demand curve for labor in the market?

To keep things simple, let's first assume there are only two cafes. Our second cafe may now have the exact same production function as the first cafe. The layout of the cafe might be different, it might have different espresso machines. Therefore, the marginal revenue product of labor might be different.

Below, we show the marginal revenue product of labor for both the first and the second cafe. The first cafe is shown in blue, and the second cafe is shown in orange. As can be seen in the figure, the marginal revenue product of labor for the second cafe is lower than the first cafe. This means that the second cafe will hire fewer barristas at each wage level.

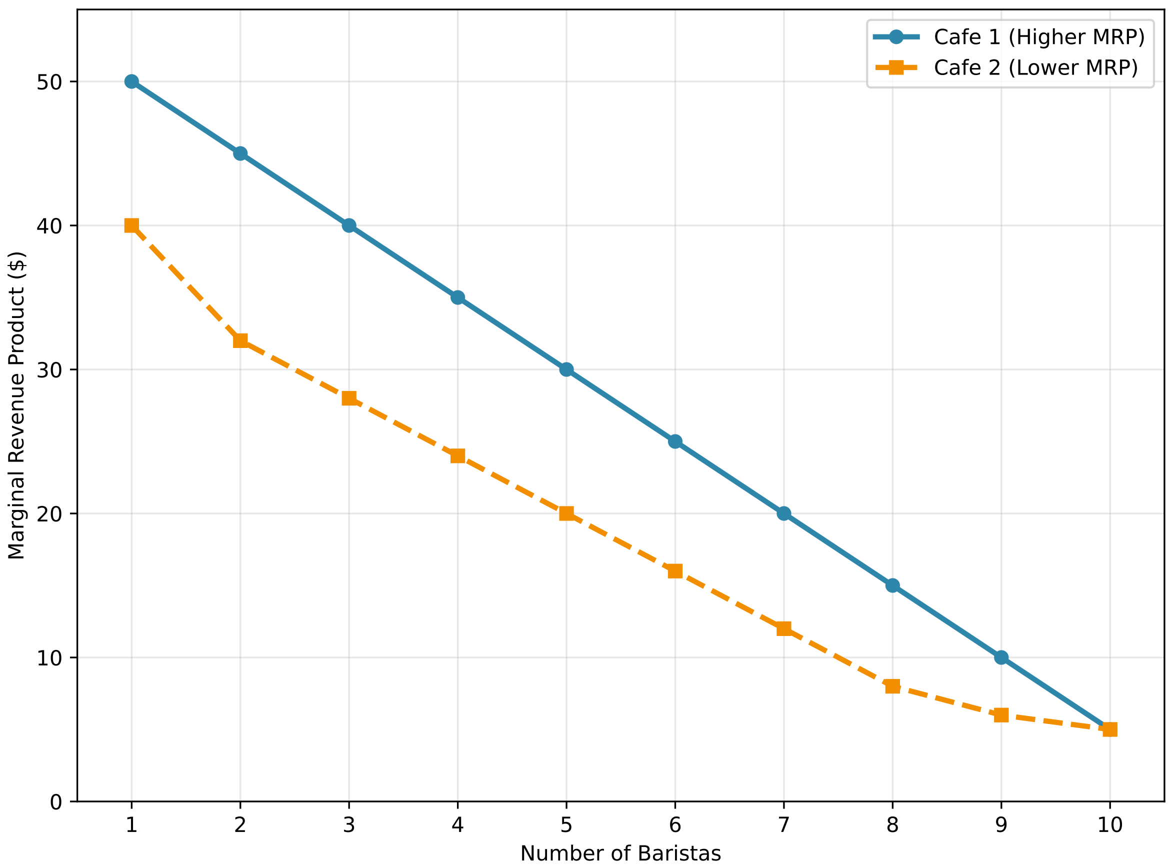


Figure 2.4: Marginal Revenue Product for Two Cafes

So how do we go from these marginal revenue product of labor curves to the overall market demand curve? Well, let's start at a wage of 50. If the wage is 50, then the first cafe will hire 1 barrista, and the second cafe will hire 0 barristas. Therefore, at a wage of 50, the total number of barristas hired in the market is 1.

Let's go down to a wage of 40. At a wage of 40, the first cafe will hire 3 barristas, and the second cafe will hire 1 barrista. Therefore, at a wage of 40, the total number of barristas hired in the market is 4.

Let's continue this process for all the wages. Below, we show a figure that takes these two marginal revenue product of labor curves and combines them into a single market demand curve. The market demand curve is shown in green, and it is the sum of the two individual cafes' demand curves. For example, you can see at the wage of 40, the green line is equal to 4, which is the sum of the 3 barristas hired by the first cafe and the 1 barrista hired by the second cafe.

At a wage of $5, both cafes would hire their maximum of 10 barristas each. Therefore, at a wage of $5, the total number of barristas hired in the market is 20.

The logic here that was applied for two cafes can be applied to any number of cafes. If there are 100 cafes, then we can simply sum the number of workers hired by each cafe at each wage level to get the overall market demand curve.

There are two key features that it is important to take away now that we've constructed the market demand curve. First, the market demand curve is downward sloping. This means that as the wage goes down, the number of workers hired goes up.

Second, the market demand curve is the sum of the individual firms' demand curves. These demand curves are in turn determined by the marginal revenue product of labor. So when we think about how changes in the economy impact the wage level, it will be useful to think about how it impacts the marginal revenue product at a given firm. For example, if technology increases the productivity of every worker, then this will shift the MRPL curve for every firm. This will be important in considering how such shifts are expected to impact the wage level in the economy.

2.6 Labor Supply

Let's take stock of what we know so far. For any given wage, we can figure out how many workers will be hired in the labor market overall. But what will the actual wage be? Will it be $15 an hour? $20 an hour? $50 an hour?

To determine this, we need to think about the other side of the market: the workers. Later in the book and course, we will dig into deeper concerning an individual's choice to work. For now, we will make one simple and realistic assumption: there will be more workers in the labor market the higher the wage.

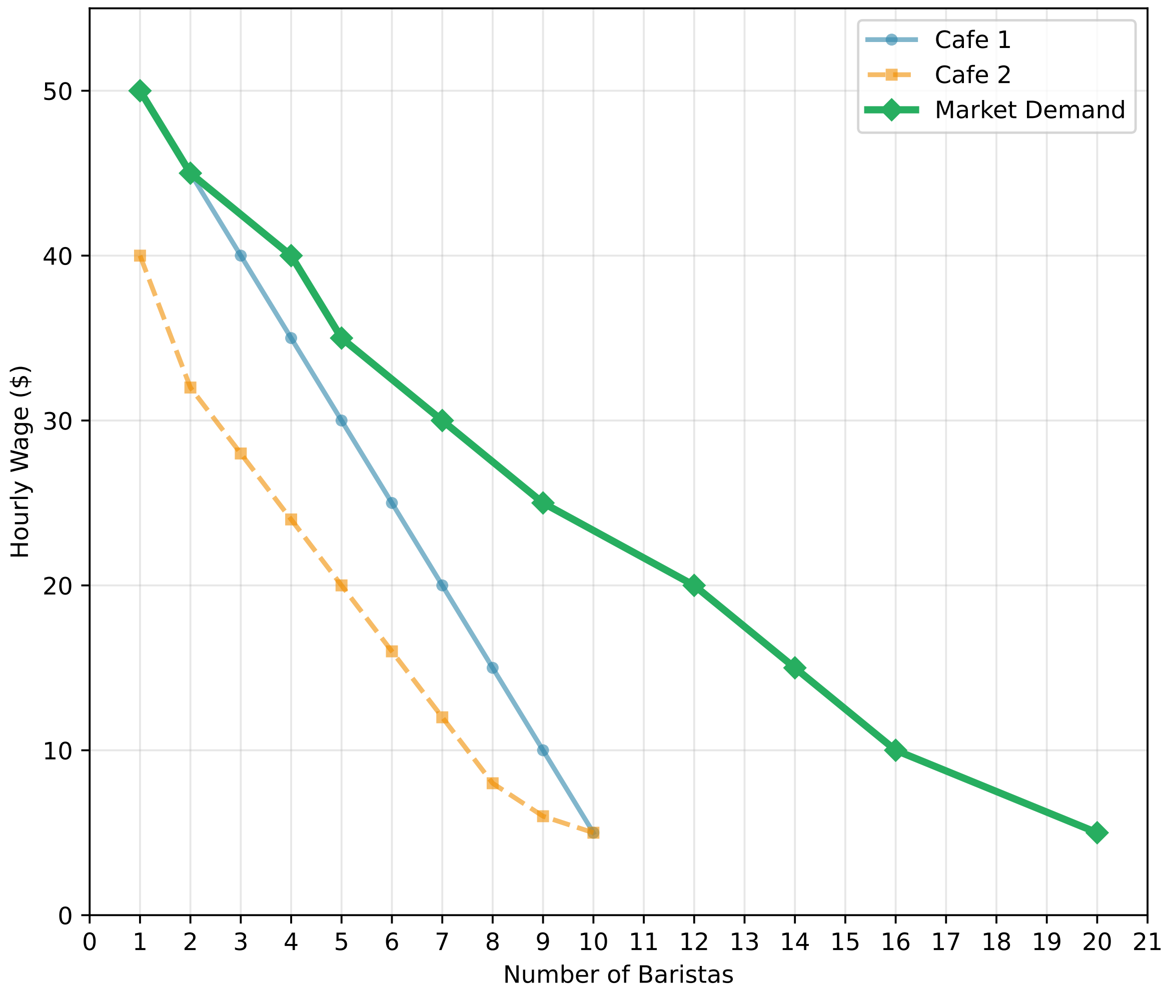


Figure 2.5: Market Labor Demand: Aggregating Two Cafes

For our discussion of labor demand, we built up the demand curve for labor by considering how many workers individual firms would hire at each wage level. We can do the same thing for workers as well. Instead of maximizing profits, workers will be maximizing their utility. This means that they will choose to work if the wage is high enough to make it worth their time.

For now, we are not going to go through the details of how to construct the labor supply curve. Instead, we will simply assume that the labor supply curve is upward sloping. For example, let's assume that the labor supply curve for barristas is the red dashed line given in Figure 2.6. This curve is upward sloping, meaning that as the wage increases, the number of workers willing to work also increases.

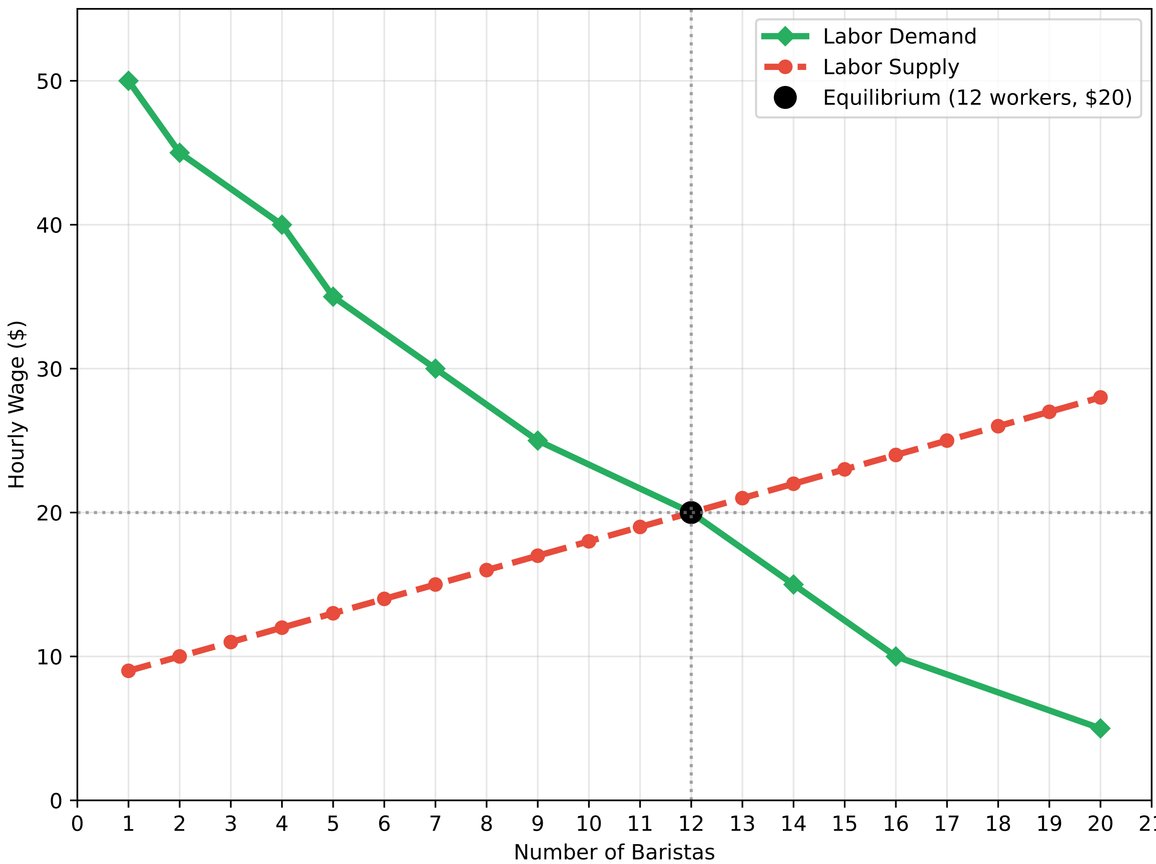


Figure 2.6: Labor Market Equilibrium: Supply and Demand

Now, what determines the market wage? Well, let's see if a $10 wage is a market equilibrium. At a wage of $10, the labor demand curve tells us that firms want to hire about 16 workers. However, at that wage, only about 2 workers are willing to work. Therefore, the labor market is experiencing a shortage.

Is this an equilibrium? No, it's not. In an equilibrium, the quantity of labor supplied equals the quantity of labor demanded. But why can't there be a situation in which a shortage exists in equilibrium? We regularly hear about shortages in the news.

The reason in our setting that a shortage cannot exist is that the firms have an incentive to raise the wage. Let's think back to our original cafe. If the wage is $10, then our original cafe would like to hire 9 workers. Clearly it will not be able to hire 9 workers, because only about 2 workers total are willing to work at that wage across the entire market. So what will the cafe do?

Well, it can increase the wage. Is this profitable to do? Let's say the cafe raises the wage to $12, then it will attract workers from other cafes (remember we assumed perfect competition, so workers can move freely between cafes). At a wage of $12 it will also attract workers who were not working before. The marginal product of labor is still above the wage, so the cafe prefers this situation.

The equilibrium occurs when no firm has an incentive to change the wage. This occurs when the quantity of labor supplied equals the quantity of labor demanded. In our case, this occurs at a wage of $20 and 12 workers. In this case the first cafe wants to hire 7 workers and the second cafe wants to hire 5 workers, for a total of 12 workers.

Does either cafe have an incentive to change the wage? Let's take cafe 1's decision. If it raises the wage it will be able to attract more workers from cafe 2. But what is the marginal revenue product of the 8th worker? It is $15, which is less than the wage of $20. So if cafe 1 raises the wage, it will hire more workers, but it will be losing money on those additional workers. Therefore, it does not want to raise the wage. The exact same logic applies to cafe 2.

2.7 Summary

In this chapter, we explored the labor market through the lens of a perfectly competitive model. We began by understanding how individual firms determine the number of workers to hire and the wage to pay in order to maximize profits. Key to this decision is the concept of **marginal revenue product of labor**, which is the additional revenue generated from hiring one more worker.

We then aggregated the demand from individual firms to understand the **market-level labor demand curve**. This curve is derived from the **marginal revenue product of labor** curves of all firms in the market and shows the total number of workers hired at each wage level.

On the supply side, we made a simplifying assumption that the **labor supply curve** is upward sloping, meaning that higher wages attract more workers to the labor market.

Finally, we combined the labor supply and demand curves to find the **equilibrium wage** and **quantity of labor** in the market, where the quantity of labor supplied equals the quantity of labor demanded.

This framework provides a foundational understanding of how wages and employment levels are determined in a perfectly competitive labor market.

3 Minimum Wage

3.1 A Brief History

In the US, a federal minimum wage was first passed in 1938 as part of the Fair Labor Standards Act. The Fair Labor Standards Act was a key part of the New Deal, a large economic stimulus bill aimed at pulling the US out of the Great Depression.

But ever since its inception, minimum wages have been hotly debated. Figure 3.1 shows a few famous pairs of politicians that had opposing views on the minimum wage. Franklin D. Roosevelt, the 32nd president of the United States, was a strong supporter of the minimum wage. The mastermind of the New Deal, Franklin D. Roosevelt felt that minimum wages were necessary to protect the most vulnerable workers in society, and would provide a much-needed boost to the economy by increasing the spending power of workers. At the time, there were many critics of the minimum wage, even those who supported the broader New Deal legislation. For example, Ellison D. Smith was a prominent Democratic senator from South Carolina who opposed the minimum wage, arguing that it would lead to economic collapse in the South. His argument was that his constituency in the South paid lower wages and would have a hard time adjusting to the new minimum wage. Businesses would fail. The workers the minimum wage was supposed to help would actually be hurt the most.

Later, in the 1970 and 80s, two prominent Republicans had somewhat different views. Richard Nixon was skeptical of large minimum wage changes (he might protest being labelled as pro-minimum wage). However, in 1974, he signed a bill that would raise the minimum wage from \$1.60 to \$2.30 over the course of three years, a roughly 43% increase. After signing the bill, Nixon stated, "raising the minimum wage is now a matter of justice that can no longer be fairly delayed."

His opinions however do stand in stark contrast to another prominent Republican, Ronald Reagan. Reagan was a strong opponent of the minimum wage. In an address to Congress, he stated he would vigorously oppose any attempt to increase the minimum wage, arguing "Raising the level of the minimum wage would cause many adult workers to lose their jobs."

Today we have similar debates. Bernie Sanders, a Democratic senator from Vermont, is a strong proponent of increasing the minimum wage to \$15 per hour. He argues that the current federal minimum wage of \$7.25 is insufficient to provide a decent standard of living for workers and that raising it would help reduce poverty and inequality. On the other hand, Senator Ted

Cruz is staunchly opposed. He argues, "Every time you raise the minimum wage, the people who are hurt the most is the most vulnerable."

There are two perhaps surprsing things about these opposing views. First, the arguments have been basically the same for nearly a century. Individuals who support the minimum wage argue that employers need to provide a living wage to their workers. Having a minimum wage ensures those most vulnerable are not taken advantage of by employers. Those who oppose the minimum wage argue the exact opposite. The minimum wage is not protection for vulnerable workers, but reduces employment and destroys businesses. The individuals who are the worst off are those workers who lose their jobs as a result of the minimum wage.

Therefore, the key question is this: does increases in the minimum wage lead to job losses? We will explore this question in steps. First, to better understand the minimum wage, we will study how it has changed over time. Figure 3.2 plots a timeline of major events that we will consider. First, the federal minimum wage was established in 1938. Since that time, there have been many increases, but also periods of stagnation. The 1980s, in particular, were a period in which the real value of the minimum wage fell significantly. This suggests that a minimum wage may be a contributor to one of our three big trends: the rise in wage inequality. If the minimum wage has fallen in real terms, then this may be one reason why the wages of low-income workers have stagnated.

However, to have the complete picture, we need to understand the impacts on employment. It could be that even if the minimum wage has fallen in real terms, low-wage workers are still better off because there are more jobs available. To understand the relationship between minimum wages and employment, we will explore the seminal study of David Card and Alan Krueger (1994). While there are prior studies, Card and Krueger (1994) introduced a new empirical approach that not only led to important insights into the employment effects of the minimum wage, but has made a lasting impression on empirics in economics. It is one of the first and most famous examples of a "natural experiment" in economics. "Natural experiments" are now ubiquitous in economics, and are useful for identifying causal effects, something that is notoriously difficult in practice.

A Maximum Wage in the Middle Ages

In the 14th century, Black Death swept through much of Europe, killing large portions of the population. This led to a large labor shortage in England. The labor supply curve in medieval England shifted massively to the left. This resulted in widespread wage increases.
Landowners were unhappy with the rising wages and used their influence in politics to pass the Statute of Laborers in 1351. This law set the wage for laborers equal to the wage rate that had been paid before the Black Death. In other words, instead of a minimum wage, medieval England had a maximum wage. Essentially, the landowners were unhappy with competition in the labor market and so passed legislation to limit competition among owners of land. If there is a law that sets a maximum wage, then

PRO	AGAINST


Franklin D. Roosevelt (D)	Ellisson D. Smith (D)

PRO	AGAINST


Richard Nixon (R)	Ronald Reagan (R)

PRO	AGAINST

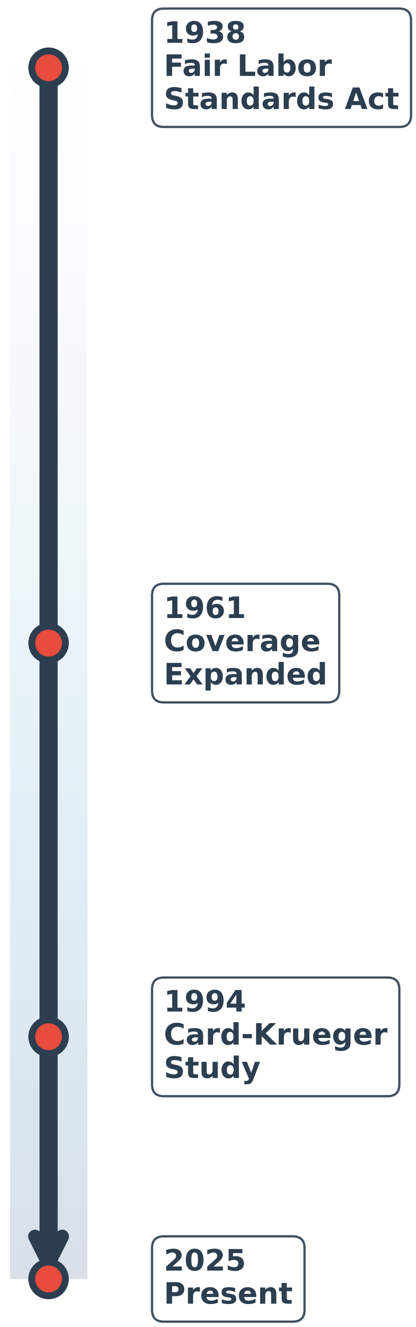



Figure 3.2: Major Minimum Wage Events

you know your competitor will not outbid you for labor.
You can probably guess that a maximum wage was not very popular among workers. It was one of the main contributing factors that led to the Peasants' Revolt of 1381, a major uprising in the history of England.

3.2 Theory

While politicians have more-or-less always been divided, economists, for a long time, were actually in broad agreement. In 1980, economist Charles Brown, in a Journal of Economic Perspectives article, asked the question: "Are minimum wage laws overrated?" At the time, the consensus was generally, "yes."

To see why this was such a widely-held view, let's again consider the case of a perfectly competitive labor market. In the figure below, we plot a labor-demand curve vs. the labor supply curve. Let's assume that the equilibrium wage is $5.00 in this market. Now, suppose the federal minimum wage is set to $7.25.

Well at $7.25, firms are not willing to hire as many workers. The quantity of labor demanded falls from Q1 to Q2. However, at this higher wage, more workers are willing to work. The quantity of labor supplied increases from Q1 to Q3. The result is a surplus of labor, or unemployment. The number of unemployed workers is the difference between the quantity supplied and the quantity demanded, or Q3 - Q2.

The key insight here is that employment does drop from Q1 to Q2. This is a key reason why many economists and politicians believed that minimum wages are not particularly effective policies. It is true you are helping some vulnerable workers (those who keep their jobs are better off), but what about those who lose their jobs? The minimum wage did not help these indivduals.

While the theoretical prediction is clear here, the theory doesn't tell us how much employment drops by. This depends on the shape of the supply and demand curves. Therefore, this is an empirical question.

3.3 Confounding Factors

How do we measure the impact of the minimum wage on employment? One simple idea is to use variation across states. In the US, some states pass higher minimum wages than others. For example, in 2025, the state with the highest minimum wage is Washington, at $16.66 per hour. Alabama, in contrast, has no state-level minimum wage. Therefore, the prevailining minimum wage in Alabama is $7.25, the federal minimum wage.

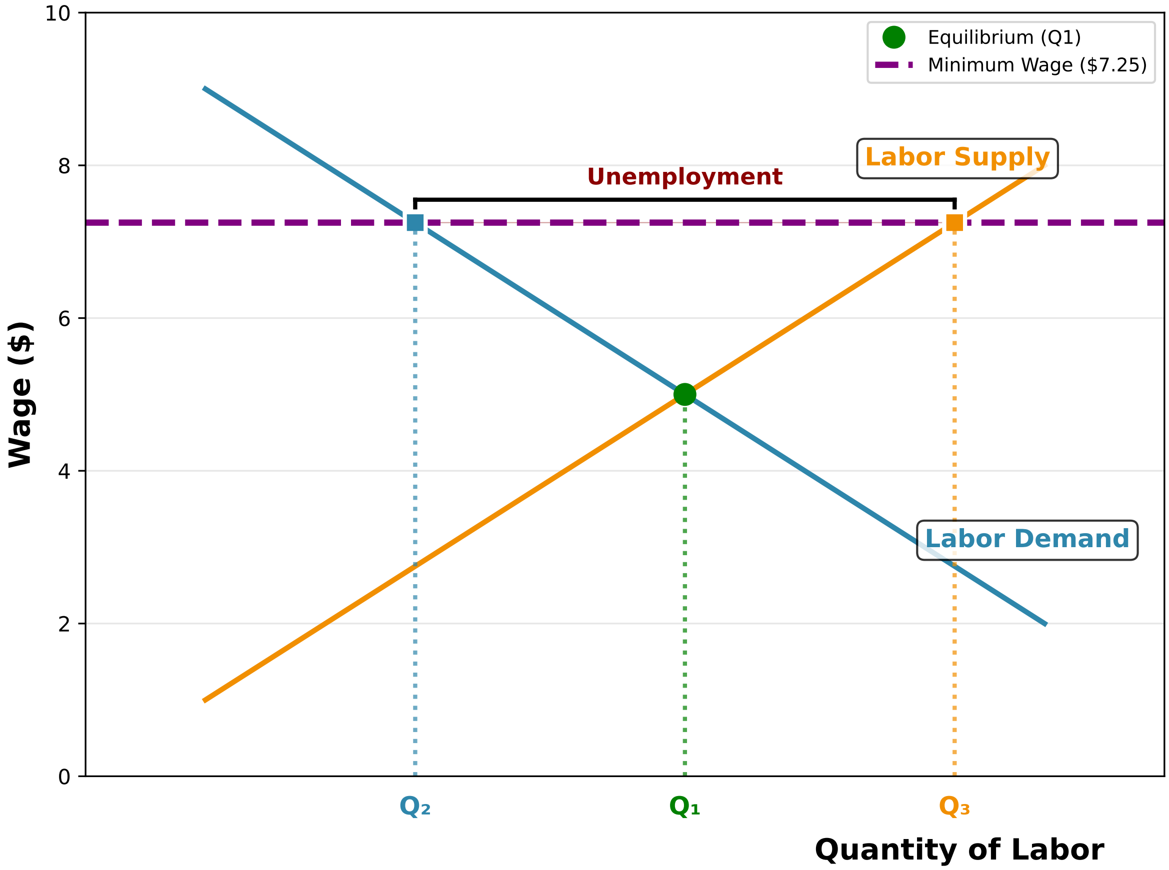


Figure 3.3: Minimum Wage in a Perfectly Competitive Labor Market

If we look the current unemployment numbers, the unemployment rate in Washington is equal to 4.5%, while in Alabama it is equal to 3.3%. What can we conclude from this exercise? Can we say that a minimum wage of $16.66 relative to $7.25 causes a 1.2 percentage point increase in unemployment?

Not really. There are massive differences between the two states. For one, the cost of living in Washington may be very different than in Alabama, so maybe we should attempt to control for local prices. But are there other differences? Well, the types of jobs in the two locations may also be very different. Alabama and Washington have different industries present. The individuals who work in these locations may also be very different. They may have different education, different skills, and different preferences.

This is a perennial issue in economics and science more generally. Our goal here is to understand the **causal effect** of a minimum wage increase on employment. However, we cannot simply compare two states with different minimum wages because there may be **confounding factors** present. A confounding factor is a variable that is correlated with our variable of interest (the minimum wage) and our outcome of interest (employment).

The idea of confounding factors is an extremely important concept, not only studying the minimum wage, not only in all of empirical economics generally, but also more broadly to science. Many questions in science can be understood in a causal framework. We want to know whether a change in X has a causal impact on Y. However, we cannot simply compare two groups with different X values because there may be confounding factors.

So what do we do? How can we measure the causal effect of a minimum wage increase on employment? In medicine, researchers have found a solution to this problem: the randomized controlled trial (RCT). In an RCT, individuals are randomly assigned to a treatment group or a control group. In the context of our example, this would be akin to randomly assigning some states to a high-minimum wage and others to a low-minimum wage. Because the assignment is now random, the two groups will be similar on average.

In the context of minimum wage, conducting an RCT is impossible. We cannot randomly assign individuals to different minimum wage levels. However, instead, researchers have taken advantage of **natural experiments**. The key idea in a natural experiment is that sometimes you can look out into the world and find a situation that looks similar to an RCT. You can identify a group of individuals, firms, or states that receive a policy, but another group of individuals, firms, or states that do not. Let's study how we might find a natural experiment in the context of minimum wage.

3.4 Empirics: Difference-in-Differences

In 1992, New Jersey raised its minimum wage from $4.25 per hour to $5.05 per hour. At the time, this gave New Jersey the highest minimum wage in the country. Our goal here is to understand how the increase from $4.25 to $5.05 impacted employment in New Jersey.

Our goal is to understand how the increase from \$4.25 to \$5.05 impacted employment in New Jersey. To focus our discussion, we will consider employment in fast-food restaurants. These are a natural choice. Fast-food restaurants are a larger industry and are more likely to be affected by the minimum wage change than other industries. Our goal is to measure the change in employment in New Jersey fast-food restaurants before and after the minimum wage increase.

A simple way to potentially do this would be to compute the employment rate in New Jersey before and after the minimum wage increase. How is this exactly useful? Well, remember all those confounding factors we discussed? For example, the cost of living, the education level, the industries present. All of these things may vary between New Jersey and other states, but within New Jersey, these factors are likely to be similar before and after the minimum wage increase. The magic of comparing across time is that we don't actually need to figure out the entire list of confounding factors. We just need to assume that they are constant over time. The change in employment in New Jersey before and after the minimum wage increase is likely to be due to the minimum wage increase, not these confounding factors.

Let's write out the change in employment in New Jersey as follows:

$$\Delta E_{NJ} = E_{NJ,1992} - E_{NJ,1991}$$

Now, before we take E_{NJ} as the causal impact of the minimum wage, let's think carefully about what this would imply. If we were to make this leap, then we would be saying that any change in employment in New Jersey between 1991 and 1992 is due to the minimum wage increase.

This is a strong assumption. Employment changes all the time through the years even though there are often no policy changes. By taking this difference we have controlled for confounding factors specific to New Jersey, but now that we are comparing across time, we have a new set of confounding factors. For example, what if the economy went into a recession in 1992? Then we might accidentally attribute a fall in employment to the minimum wage increase, when in fact it was due to the recession.

To control for this, we essentially need to understand what would have been the change in employment in New Jersey if the minimum wage had not changed. This is referred to as a **counterfactual**.

One way to construct a counterfactual is to find a state that is similar to New Jersey, but did not change its minimum wage. In nearby Pennsylvania, the minimum wage was \$4.25 in 1991 and did not change in 1992. New Jersey and Pennsylvania, especially on the border of the two, are quite similar. It is often difficult to tell that you've crossed the state line when travelling between the two states. Therefore, maybe we can use changes in Pennsylvania to understand what the time trend would have looked like in New Jersey if the minimum wage had not changed.

Let's write out the change in employment in Pennsylvania as follows:

$$\Delta E_{PA} = E_{PA,1992} - E_{PA,1991}$$

Now let's go back to New Jersey. Let's form a **counterfactual**, which is the change in employment in New Jersey if the minimum wage had not changed. To do this, let's take some concrete numbers. Let's say that on average in 1991, employment in New Jersey fast-food restaurants was 20 workers. Our counterfactual asks if the minimum wage had not changed, how many workers would there have been in 1992? We can't look in the actual data for this. The fact of the matter is that the minimum wage did change.

However, to get a clue about what the counterfactual would have been, we can look at Pennsylvania. Let's say that in Pennsylvania, employment in fast-food restaurants was 25 workers in 1991 and 22 workers in 1992. So, on average, employment in Pennsylvania fell by 3 workers.

Therefore, let's use this as our counterfactual. We will assume that if the minimum wage had not changed in New Jersey, then employment would have fallen by 3 workers, just like it did in Pennsylvania. Now, we can look in the data to see what actually happened in New Jersey. Let's say that in 1992, employment in New Jersey fast-food restaurants was 15 workers. In this case, the fall in employment in New Jersey was 5 workers. Therefore, we would conclude that the minimum wage decreased employment in New Jersey by 2 workers. Based on Pennsylvania, we would have expected employment to fall by 3 workers, but in fact it fell by 5 workers. Therefore, the minimum wage must have caused a fall in employment of 2 workers.

Now let's be a bit more general. Let's call ΔE, the effect of the minimum wage on employment. To estimate this, we take two differences. First, we take the difference in employment between New Jersey before and after the minimum wage change (ΔE_{NJ}). To understand what would have happened in New Jersey if the minimum wage had not changed, we take the difference in employment in Pennsylvania before and after the minimum wage change (ΔE_{PA}). To compute the effect of the minimum wage on employment in New Jersey, we can then write this as follows:

$$\Delta E = \Delta E_{NJ} - \Delta E_{PA}$$

This is referred to as a **difference-in-differences** (or DiD) estimator. The name stems from the idea that we are taking two differences: the difference in employment in New Jersey before and after the minimum wage change, and the difference in employment in Pennsylvania before and after the minimum wage change.

A common way to visualize this is to plot the employment in New Jersey and Pennsylvania before and after the minimum wage change. In the figure below, we plot the employment in New Jersey and Pennsylvania before and after the minimum wage change for the illustrative example described above. The blue line represents New Jersey, while the orange line represents Pennsylvania. The vertical dashed line represents the time of the minimum wage change.

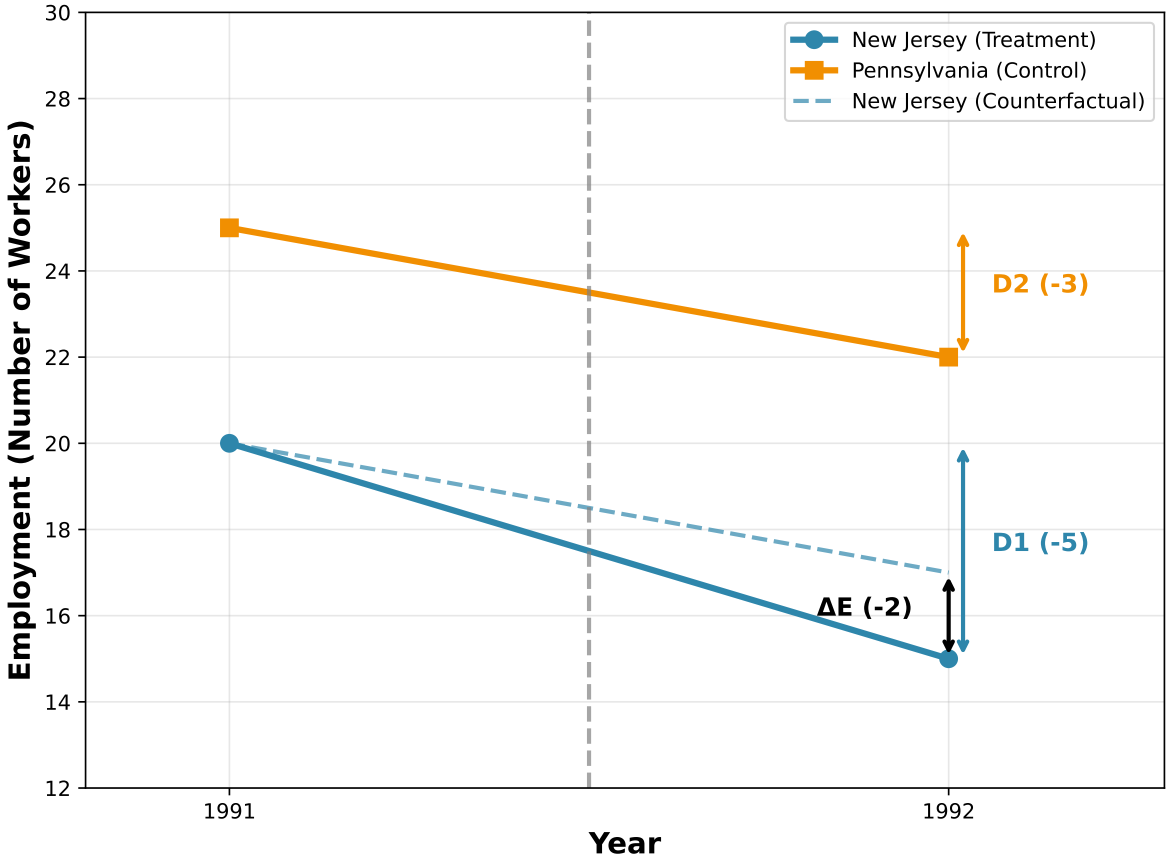


Figure 3.4: Difference-in-Differences Illustration: Employment in New Jersey and Pennsylvania

To solidify the intuition here, let's walk through another example. This time, let's assume employment is growing rapidly in the area. Let's assume that in Pennsylvania, employment increased from 25 workers in 1991 to 30 workers in 1992 (an increase of 5 workers). In New Jersey, let's assume employment increased from 20 workers in 1991 to 21 workers in 1992 (an increase of 1 worker). Now let's plot the trends over time and compute the difference-in-differences estimator.

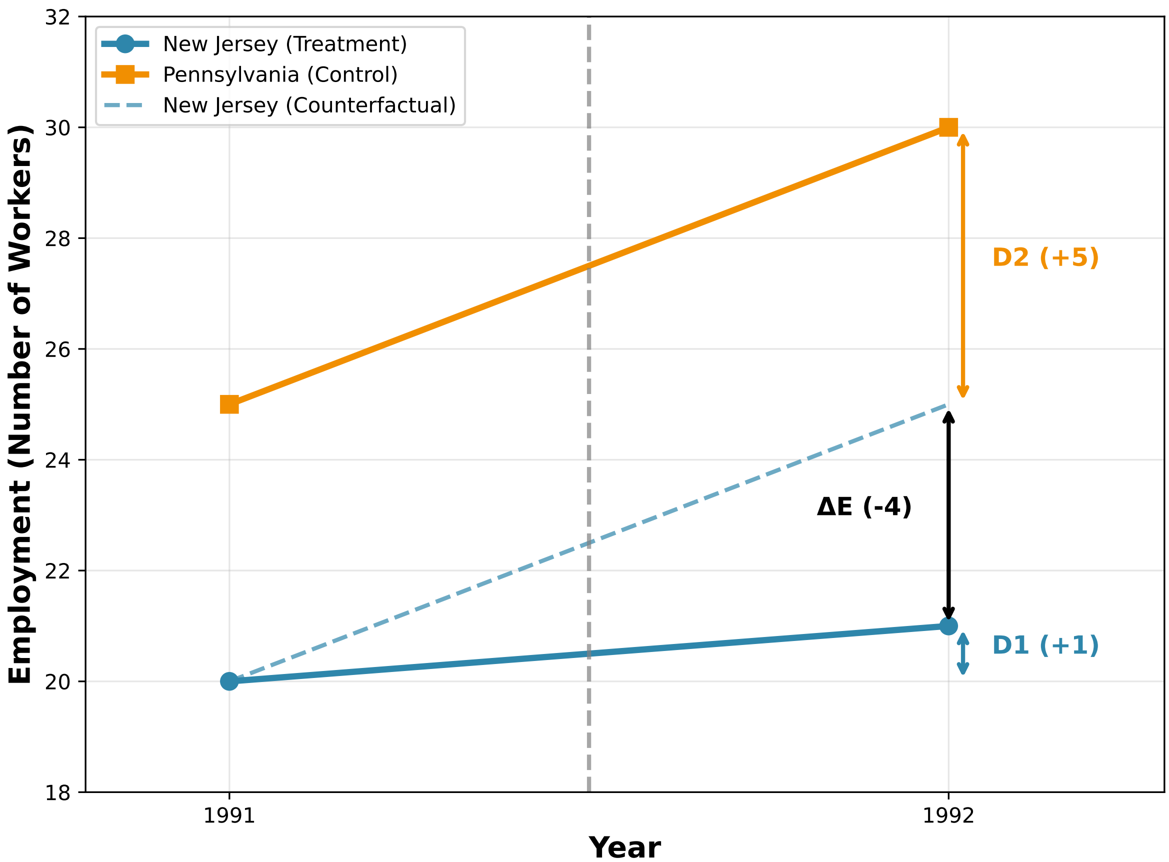


Figure 3.5: Difference-in-Differences with Growing Employment Example

If we use Pennsylvania as our counterfactual, we would have expected New Jersey employment to grow by 5 workers (from 20 to 25). However, it only grew by 1 worker (from 20 to 21). Therefore, the difference-in-differences estimator is ΔE = D1 - D2 = 1 - 5 = -4 workers. This suggests that the minimum wage reduced employment in New Jersey by 4 workers relative to what would have happened without the policy change, even though employment still increased in absolute terms.

This example illustrates why the difference-in-differences approach is so powerful. If we use only variation across states in the minimum wage, we would get the wrong answer. Pennsylvania has higher employment both before and after the minimum wage change, so at least a portion of this difference is due to differences across the states. If we use only variation across time, then we might falsely conclude here that minimum wages actually increase employment. But by using both dimensions of variation, it allows us to separate the effect of the policy from differences across states and differences across time.

3.5 Card and Krueger (1994)

Card and Kruger (1994) used this difference-in-differences approach to study the impact of the minimum wage increase in New Jersey on employment in fast-food restaurants. This is probably one of the most famous studies in all of labor economics. We will get to why soon.

Now, let's recreate the figures above, but using the actual data from Card and Krueger (1994), which performed a survey of fast-food restaurants in New Jersey and Pennsylvania both before and after the minimum wage increase. In the figure below, we plot the average level of employment in fast food restaurants in New Jersey and Pennsylvania before and after the minimum wage change.

The actual data tell a very different story than the theoretical prediction. To start, let's consider employment in New Jersey and Pennsylvania before the minimum wage change. In New Jersey, employment was 20.44 workers per fast-food restaurant, while in Pennsylvania it was 23.33 workers per fast-food restaurant. Therefore, employment in Pennsylvania was higher than in New Jersey by about 3 workers per restaurant. This is one reason we might not want to use a simple comparison across states. Before the policy, employment was already higher in Pennsylvania than in New Jersey. Instead, we are going to look how their changes over time differ.

In Pennsylvania, employment dropped from 23.33 to 21.17 workers (D2=-2.16). If we assume that New Jersey would have experienced the same trend, then we would have expected employment in New Jersey to fall by 2.16 workers, from 20.44 to 18.28 workers. This is our counterfactual dashed line in the figure.

However, you can see the actual change in employment in New Jersey was quite different. Employment in New Jersey actually increased from 20.44 to 21.03 workers (D1 =0.59). Therefore, the difference-in-differences estimator is ΔE = D1 - D2 = 0.59 - (-2.16) = 2.75 workers. This suggests that the minimum wage actually increased employment in New Jersey by about 2.75 workers per restaurant, contradicting the theoretical prediction of job losses.

We should take a moment here to pause and appreciate how surprising this result is. The theoretical prediction from a perfectly competitive labor market is that the minimum wage should reduce employment. Intuitively, if you raise the price of something, then you should buy less of it, right? However, the empirical evidence from Card and Krueger (1994) suggests the opposite: the minimum wage increase in New Jersey actually increased employment in fast-food restaurants.

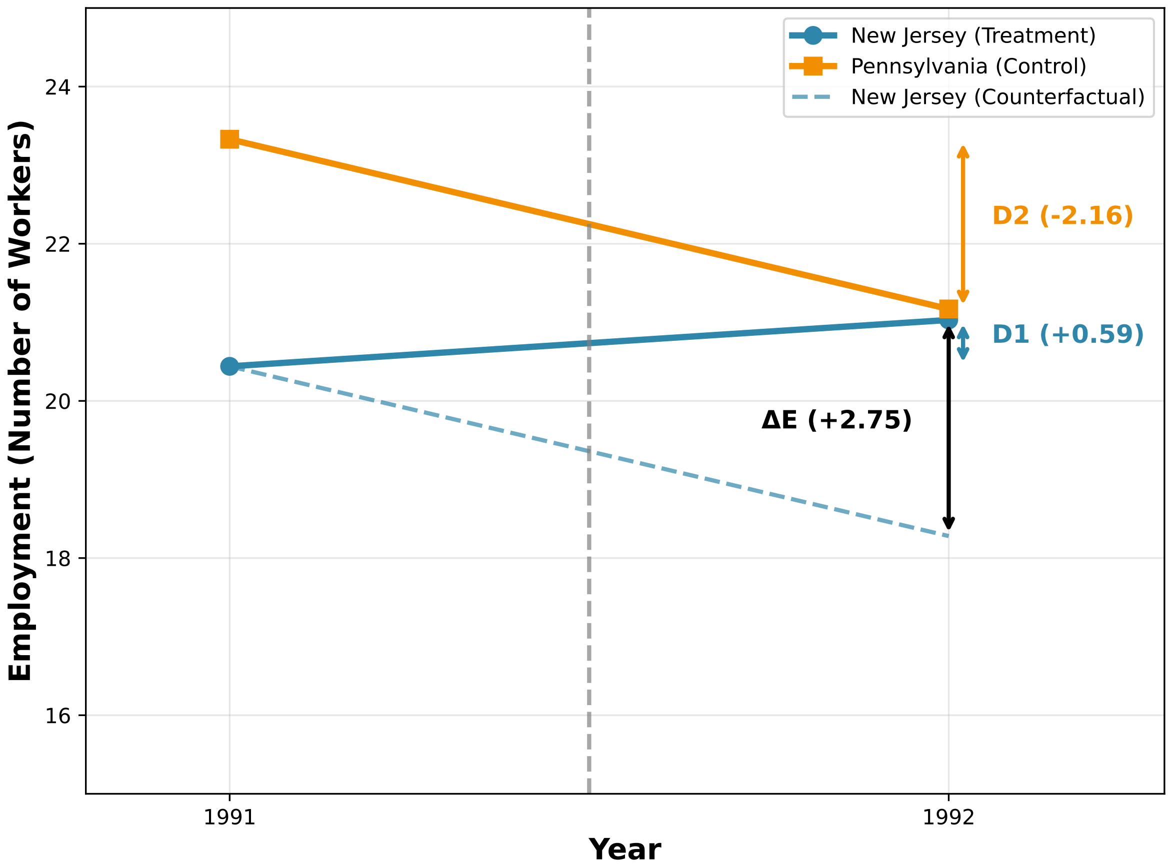


Figure 3.6: Card and Krueger (1994) Data: Employment in New Jersey and Pennsylvania Fast-Food Restaurants

The Card and Krueger (1994) result was extremely influential for two reasons. First, it challenged the conventional wisdom among economists that minimum wages reduce employment. Second, it is considered one of the most influential examples of a difference-in-differences design in economics. While this method had been around for some time (see Ashenfelter (1978) for an early example estimating the impact of a job training program), the Card and Krueger (1994) study is often credited with popularizing it in economics. Today, many applied papers in economics use a difference-in-differences design to estimate causal effects.

3.6 Critiques

In economics, and science more generally, when a very surprising result is found, it is often analyzed very carefully. For the minimum wage result in Card and Krueger (1994), this is especially true. Since the publishing of this article, there have many, many articles studying the minimum wage.

Before getting to the more specific critiques, we can first talk about critiques of difference-in-differences designs more generally. The key assumption in a difference-in-differences design is that the treatment and control groups would have experienced the same trends over time in the absence of the treatment. This is referred to as the **parallel trends assumption**. In our example, one critique of the study could be while New Jersey and Pennsylvania are similar, this doesn't necessarily prove that they would have experienced the same trends over time in the absence of the minimum wage. We already showed that employment was higher in Pennsylvania than in New Jersey before the minimum wage change. Maybe the trends in employment were also quite different.

In our example, we have only two time periods: 1991 and 1992. But what if Card and Krueger had collected data from 1988 onwards? How could we use this data to assess the credibility of the parallel trends assumption? Let's take some hypothetical examples and see how they would change our understanding of the Card and Krueger (1994) results. Let's consider the figure below which plots hypothetical employment data from 1988 to 1992 for New Jersey and Pennsylvania.

How would a picture like this change our interpretation of the Card and Krueger (1994) result? In this example, we see that employment in New Jersey was increasing steadily from 1988 to 1991. However, in nearby Pennsylvania, employment was decreasing steadily from 1988 to 1991. This violates the logic of our difference-in-differences design. If I just plotted the data between 1988 and 1991 and asked you, do you expect employment in New Jersey to increase in 1992, you would probably say yes. If I asked you, do you expect employment in Pennsylvania to increase in 1992, you would probably say no. Therefore, we can't use Pennsylvania as a control group for New Jersey. The trends in employment were already quite different before the minimum wage change.

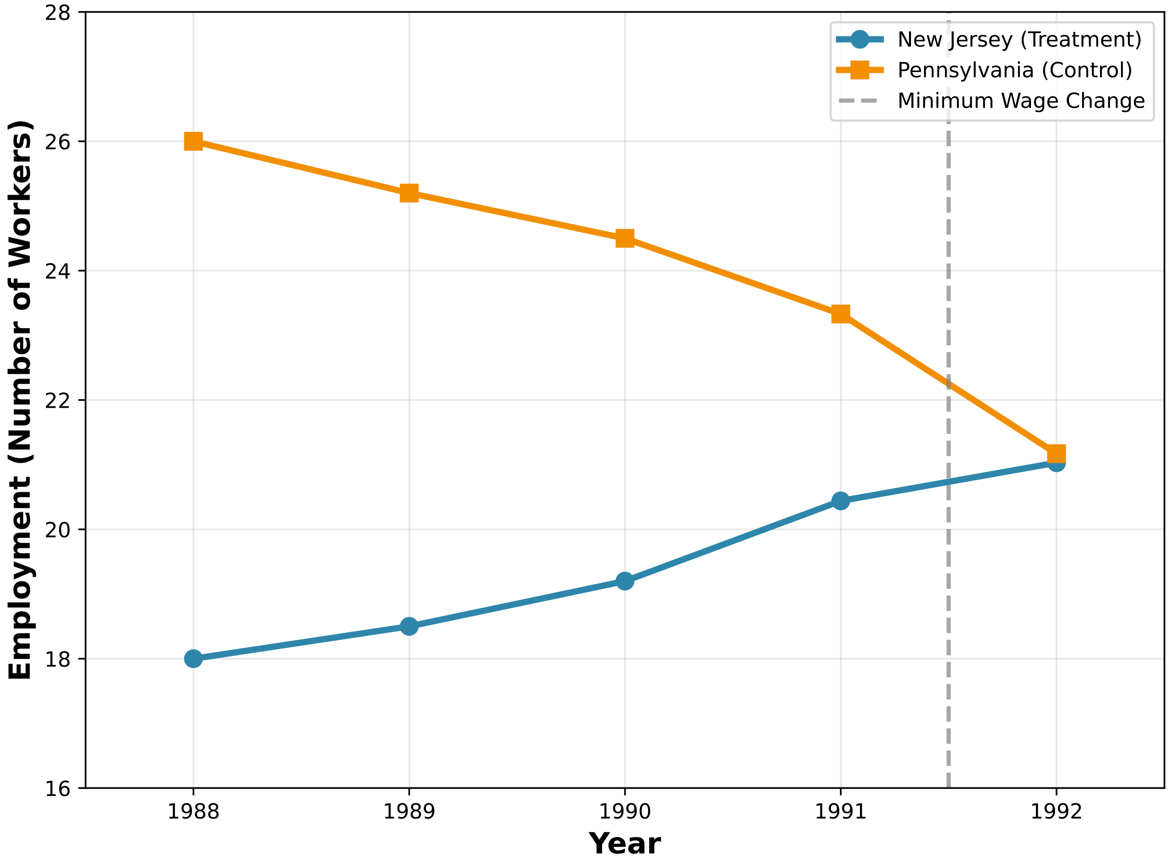


Figure 3.7: Hypothetical Pre-Trends Assumption Violation

Let's take another hypothetical example. This time, let's assume that employment in both New Jersey and Pennsylvania were stable from 1988 to 1991.

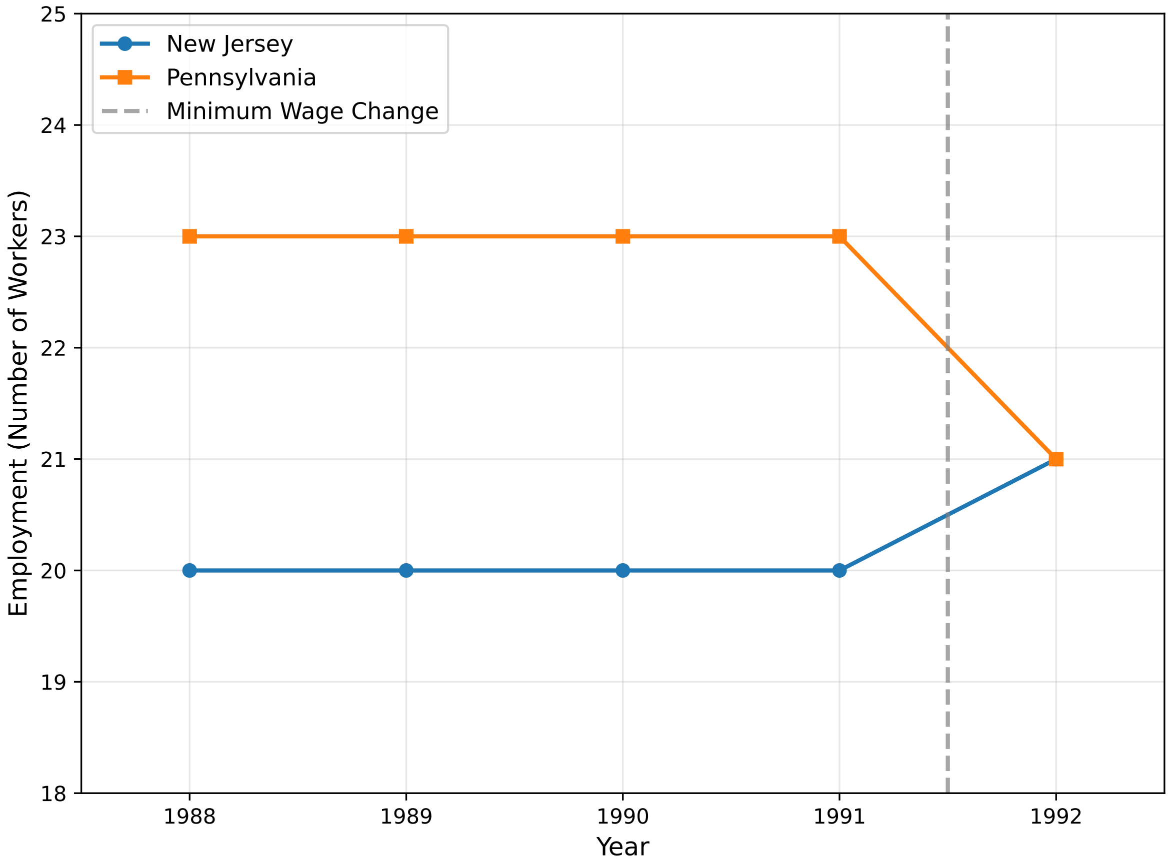


Figure 3.8: Hypothetical example with stable pre-trends (1988-1991)

In this case, the trends in New Jersey and Pennsylvania are very similar. It is fine that Pennsylvania has higher employment than New Jersey, as long as the trends are similar. In this case, it looks like Pennsylvania provides a reasonable way to construct a counterfactual for New Jersey. Therefore, this hypothetical example supports the credibility of the parallel trends assumption. For this set of hypothetical data, we would be more confident that our difference-in-differences estimator is valid and therefore identifies the causal effect of the minimum wage on employment.

In modern studies that use difference-in-differences designs, studying the trends before the event is essential. It is unlikely that results will be taken seriously if the researchers are unable to establish that the treatment and control groups had similar trends before the event. For minimum wage studies, the concern over this issue is especially severe. Minimum wage changes are not random. They are passed in response to potentially changing economic conditions. For example, if a state is experiencing a boom in the economy, then politicians may be more likely to pass a minimum wage increase. If the economy is booming, then employment may be increasing. Therefore, if we see employment increasing after a minimum wage increase, it may be due to the booming economy, not the minimum wage increase.

3.7 Modern Studies

Since the publishing of Card and Krueger (1994), there have been a range of more modern studies of the minimum wage. In the public discourse, the opinions on the minimum wage and its impact on workers is still extremely divided. Well, it turns out in this case, the academic literature is also divided.

3.7.1 Meta-Analyses

Recently, there have been many meta-analyses of the minimum wage literature. A meta-analysis is a study that looks across many different studies to summarize the overall evidence. For example, Neumark and Shirley (2022) is a recent survey of the minimum wage literature. They review 67 studies on the minimum wage to summarize overall impacts. The studies they review differ along some important dimensions. Some use variation in the federal minimum wage over time, while others use variation across states. Some studies look at the impact of small changes in the minimum wage, while others look at large changes. Some studies look at teen workers, while other looks at low-wage industries (like fast-food restaurants), or individuals with low education levels.

Overall, their reading of the literature does not support the results of Card and Krueger (1994). They find that just under 80% of the studies they review find negative employment effects of minimum wage increases. The remaining 20% of studies find either no effect or positive effects.

Even among meta-analyses, there is disagreement on what one should take away from the literature on the minimum wage. Wolfson and Belman (2019) is another recent meta-analysis that uses a different methodology to aggregate the results of many different studies. They find that the minimum wage has negative impacts on employment overall, but the magnitude is quite small. They report that they find "no support for the proposition that the minimum wage has had an important effect on U.S. employment."

3.7.2 Cengiz et al. (2019)

Cengiz et al. (2019) is another important contribution to the minimum wage literature. One argument in the paper (which occurs in other papers by the authors as well) is that considering many different studies at a time can be misleading. Some of the studies cited in Neumark and

Shirley (2022) look at small changes in the minimum wage, while others look at large changes. Some look at teen workers, while others look at low-wage industries. Some use different controls.

Cengiz et al. (2019) provide a new framework for understanding the impact of the minimum wage overall. Their methodology does not require picking a specific group of workers they think are targetted by the policy, such as restaurant workers or teen workers. Instead, they look at the entire distribution of wages. They find that minimum wage changes have negligible impacts on employment overall. The results are quite similar qualitatively to the original results in Card and Krueger (1994).

3.7.3 Summary of Current Evidence

To summarize, there is still some disagreement over the overall impact of the minimum wage on employment. However, the predictions of large negative employment effects from minimum wage increases seem to be largely unsupported by the evidence. Most studies find small negative effects or no effects at all. Some studies even find small positive effects.

There are two important points to keep in mind when interpreting the evidence. First, many of the minimum wage changes studied are relatively small. One recent political movement has called for a $15 minimum wage (the Fight for $15). This would be an increase from $7.25 to $15.00, much larger than most of the changes studied in the U.S. context. It is unclear whether the results from small changes in the minimum wage can be extrapolated to large changes. Of course there will be a point in which the minimum wage is so high that it will cause large employment losses. However, we really don't know where this point is at the moment.

Second, we have studied one margin of adjustment: employment. However, firms have many margins through which they can adjust to the minimum wage. They can reduce hours, increase prices, reduce benefits, or change the types of workers they hire. We will discuss these other margins next.

3.8 Non-Employment Margins

The (potential) employment effects of the minimum wage dominate headlines. However, firms have many margins through which they can adjust to the minimum wage. While theoretical models often conceptualize firms as choosing employment, they actually choose a wide range of things. They make a choice on what type of job requirements to include when hiring workers. They don't just pay wages, but they also offer amenities, like health insurance. They choose the prices that they charge to consumers. The minimum wage could also impact these other margins of adjustment. Clemens (2021) offers a recent survey of the literature on how firms respond to minimum wage increases.

We can think through how these margins might impact workers with our simple supply and demand framework. First, it is helpful to review marginal revenue product of labor. The marginal revenue product of labor is the additional revenue that a firm earns from hiring one more worker. This can be decomposed into two parts: the marginal product of labor (MPL) and the price (P) that the firm charges. In Chapter 2, we showed that a firm's demand for labor is equal to the marginal revenue product of labor.

Figure 3.9 shows our standard analysis of the labor market that we previously discussed in Figure 3.3. This figure is taken from Clemens (2021) and studies a change in the minimum wage from w_1 to $w_{min,2}$. In this graph, the minimum wage causes a decline from L_1 to L_2. Next, we are going to discuss how other margins of adjustment could impact this analysis, and in particular, the magnitude of the employment decline.

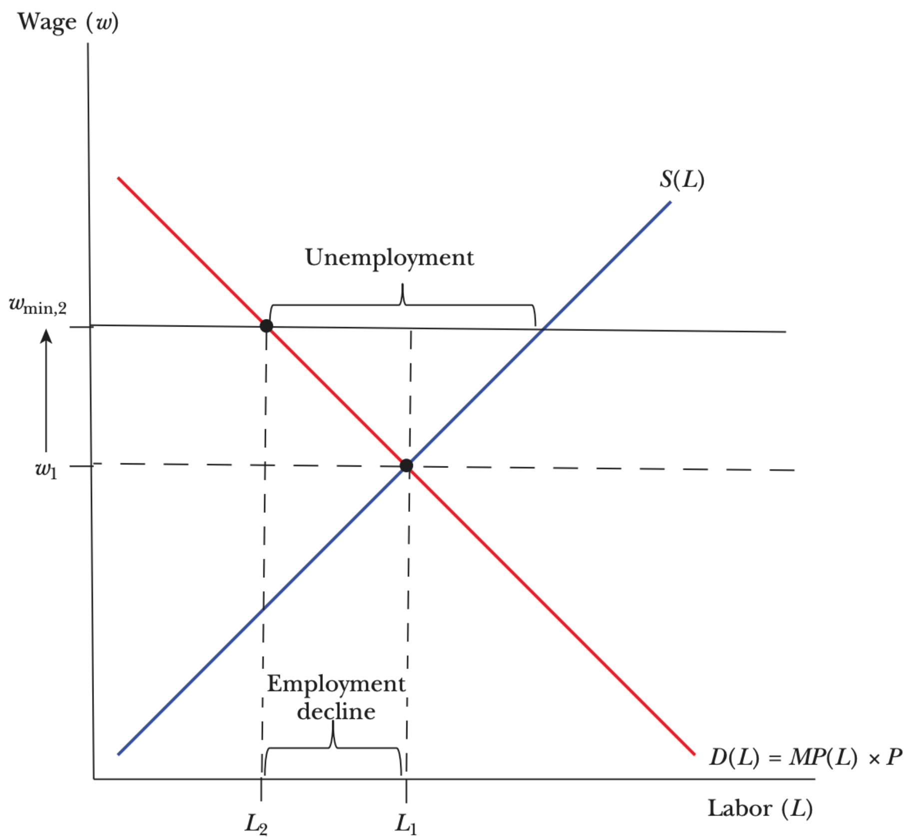


Figure 3.9: Textbook Minimum Wage Analysis

3.8.1 Prices

In the standard analysis, we assume that the demand curve for firms remains the same when the minimum wage changes. However, what if price also changes? This is often discussed by detractors of the minimum wage. How will an increase in price impact the impact of the minimum wage on employment?

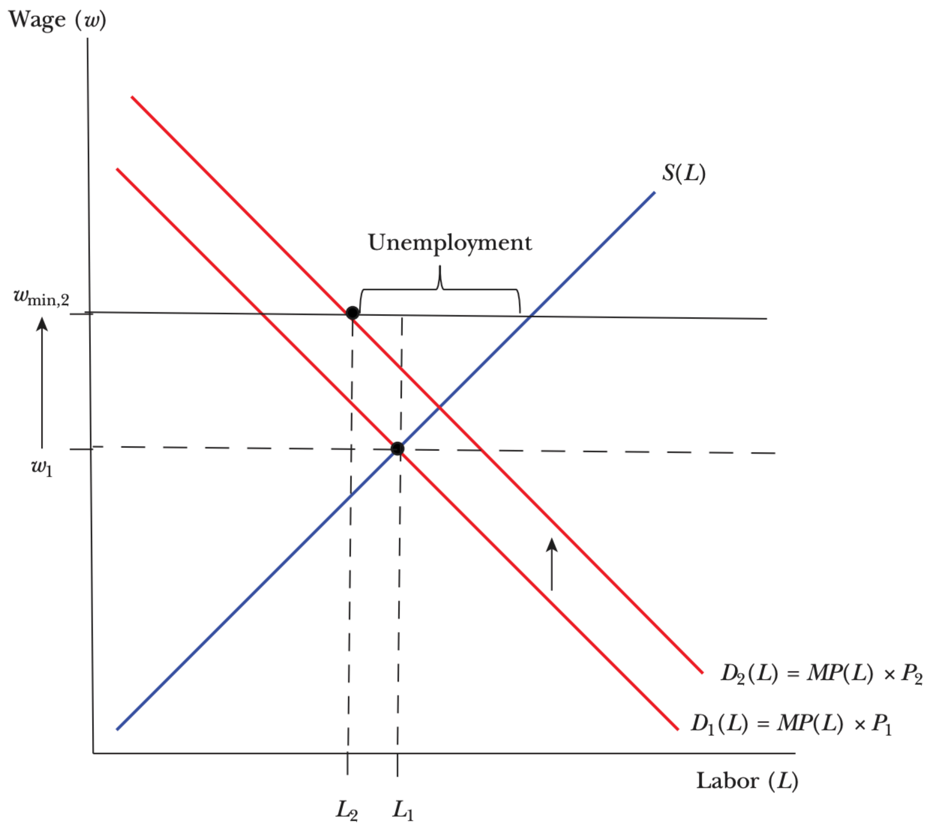


Figure 3.10: Partial Offset Through Price Pass-Through

If prices increase, this will shift the demand curve outwards. Why? Because if price is higher, then for each additional amount of output, the firm can earn more. This increases the marginal revenue product of labor. In the cafe example from Chapter 2, if the price of a latte increases, then the money earned from the workers also increases. Therefore, the marginal revenue product of labor increases.

As can be seen in the graph, there is still a decline in employment from L_1 to L_2. However, the decline is smaller than in the original analysis. Therefore, if firms can pass on some of the cost of the minimum wage to consumers through higher prices, then this will reduce the negative employment effects of the minimum wage. This could be one reason why the empirical evidence does not find large negative employment effects of the minimum wage.

But this might be different than what proponents of the minimum wage had in mind. If the minimum wage is paid by consumers, then this is basically a transfer from consumers to workers. Most proponents envision that it is a transfer from firm owners to workers.

So what is the empirical evidence on price pass-through? Harasztosi and Lindner (2019) is one of the most compelling studies on this question. They study the impact of a large minimum wage increase in Hungary. In 2001, the minimum wage in Hungary increased by a substantial amount. Before the increase, the minimum wage in Hungary was about 35% of the median wage. In the US in 2024, it was estimated that the median weekly earnings was around $1,134, so 35% of the median weekly earnings would be about $399. This would be equivalent to a wage of about $10 per hour for a full-time worker.

After the increase, the minimum wage in Hungary was about 55% of the median wage. Therefore, converting to US dollars this would be about $623 per week, or about $15.60 per hour for a full-time worker. So this is a uniquely large increase in the minimum wage.

Harasztosi and Lindner (2019) use adminstrative data to document a large increase in the cost of labor after a minimum wage change. They then ask, who pays for this increase in labor costs? They find that about 75% of the increase in labor costs is passed on to consumers through higher prices. The remaining 25% is paid for by the firm owners. Therefore, this suggests that in this context, price pass-through is an important margin of adjustment for firms in response to minimum wage increases.

3.8.2 Nonwage Benefits

Another margin of adjustment for firms is nonwage benefits. Firms don't just pay wages, they also provide other benefits to workers. This could include health insurance, paid leave, retirement plans, or even amenities like free food or gym memberships. If the minimum wage increases, firms may be able to reduce other forms of compensation to offest the increase in wages.

To be more general, let's write out compensation as follows:

$$C = w + A$$

where C is total compensation, w is the wage, and A is the value of amenities, which could be things like health care and overtime pay, but also more abstract things like workplace culture. If the minimum wage increases, then w increases. If firms can decrease A, and workers are indifferent between w and A, then total compensation C may not change. The presence of amenities is a much deeper point. It applies to many aspects of labor economics. Many of the models we use abstract from amenities, but if you think about the real world, amenities are important for deciding where to work.

To see how this works, we can go back to our cafe example. Imagine the cafe provides free food to its workers. They are paid a daily wage of $50, but also receive $10 worth of free food each day. Therefore, their total compensation is $60. If the minimum wage increases to $60, then the cafe may be able to reduce the value of free food provided to workers from $10 to $0. Therefore, total compensation remains at $60, but the cafe is able to comply with the minimum wage law.

How empirically relevant is this margin of adjustment? Clemens, Kahn, and Meer (2018) study the impact of minimum wage laws on employer-provided health insurance. They find that low-wage occupations are about 2 percentage less likely to receive employer-provided health insurance after a minimum wage increase. This decrease in health insurance offsets about 9% of the increase in wages from the minimum wage.

3.9 Inequality

Let's return to the original question we asked at the beginning of this chapter: does the minimum wage reduce inequality? We are going to focus on the period between 1979 and 1989, which saw some of the largest increases in inequality in the U.S. This was also a period in which the real value of the minimum wage declined. Basically, at the federal level, the minimum wage was not changed much during this period. Due to inflation, the real value of the minimum wage declined.

As we discussed in a previous chapter, there are many ways to measure inequality. In this section, we are going to focus on the 90/10 wage ratio. This is the ratio of the wage at the 90th percentile of the wage distribution to the wage at the 10th percentile of the wage distribution. A higher 90/10 ratio indicates more inequality. If you are above the 90th percentile of wages, that means you make more than 90 percent of workers. If you are below the 10th percentile of wages, that means you make less than 90 percent of workers. Therefore, the 90-10 ratio tells you about the spread in wages between relatively rich workers and relatively poor workers.

In 1979, the 90/10 wage ratio was about 3.6. This means that the wage at the 90th percentile was about 3.6 times higher than the wage at the 10th percentile. Figure 3.11 shows how the 90/10 wage ratio changed over time. Throughout the 1980s, the 90/10 wage ratio increased substantially, reaching about 4.5 in 1989.

Now, the 90/10 ratio can go up for two reasons. First, wages at the top could be going up. For example, the 1980s were a time in which computers were introduced in the workplace. If high-wage workers were more likely to use computers, then this could have increased their productivity and therefore their wages. We will explore this channel in a future chapter. Second, wages at the bottom could be going down. For example, if the real value of the minimum wage declined, then this could have reduced wages for low-wage workers. In this case, the 90/10 ratio would increase even if the wages of high-wage workers remained constant.

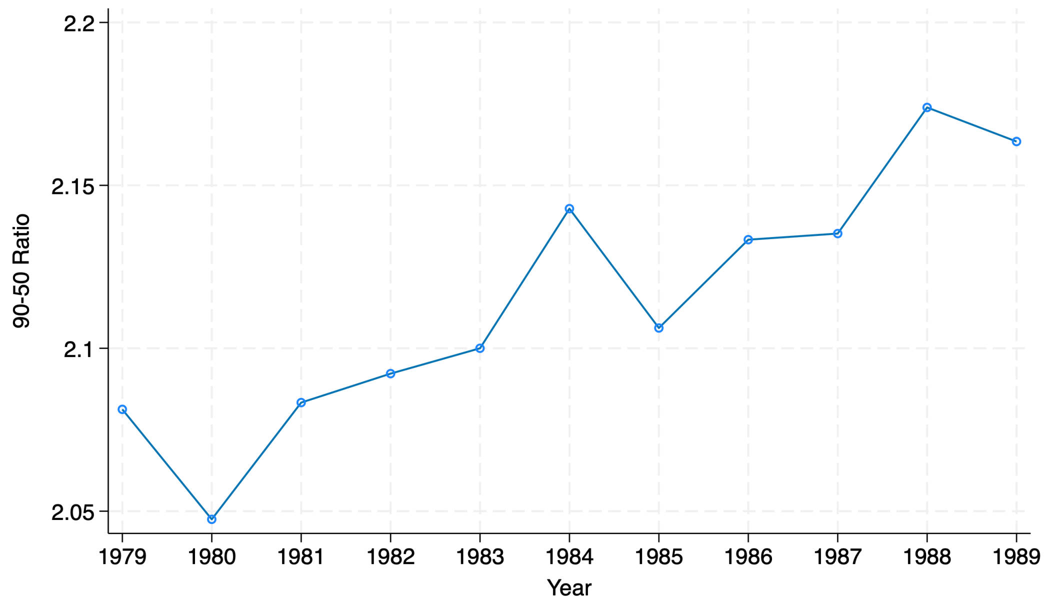


Figure 3.11: 90-10 Ratio Over Time

Our goal is to understand potentially how much of the increase in the 90/10 ratio was due to changes at the bottom of the wage distribution.

To understand how to do this, we are going to plot the distribution of wages in 1989. In this plot, We are using data from the March Current Population Survey (CPS). The CPS is a large, nationally representative survey of households in the U.S. It is conducted by the U.S. Census Bureau and is the primary source of data on employment and unemployment in the U.S. The data presented here uses additional cleaning made availably in D. H. Autor, Manning, and Smith (2016).

In Figure 3.12, we are plotting the distribution of hourly wages in 1989. The x-axis shows the hourly wage, while the y-axis shows the frequency of workers at each wage level. The higher the bar, the more workers in that range.

The red dashed line shows the actual minimum wage in 1989, which was $3.35 per hour. As you can see, there are a few workers who are paid below the minimum wage. This is likely due to some combination of non-compliance with the law and measurement error in the data.

Our goal with this figure is to ask: what would this look like if the minimum wage had kept up with inflation since 1979? In 1979, the federal minimum wage was $2.90 per hour. If we adjust this for inflation, the minimum wage would have been about $4.77 per hour in 1989. The orange dashed line shows this counterfactual minimum wage. Therefore, the distance between the red and orange dashed lines shows how much the real value of the minimum wage declined between 1979 and 1989.

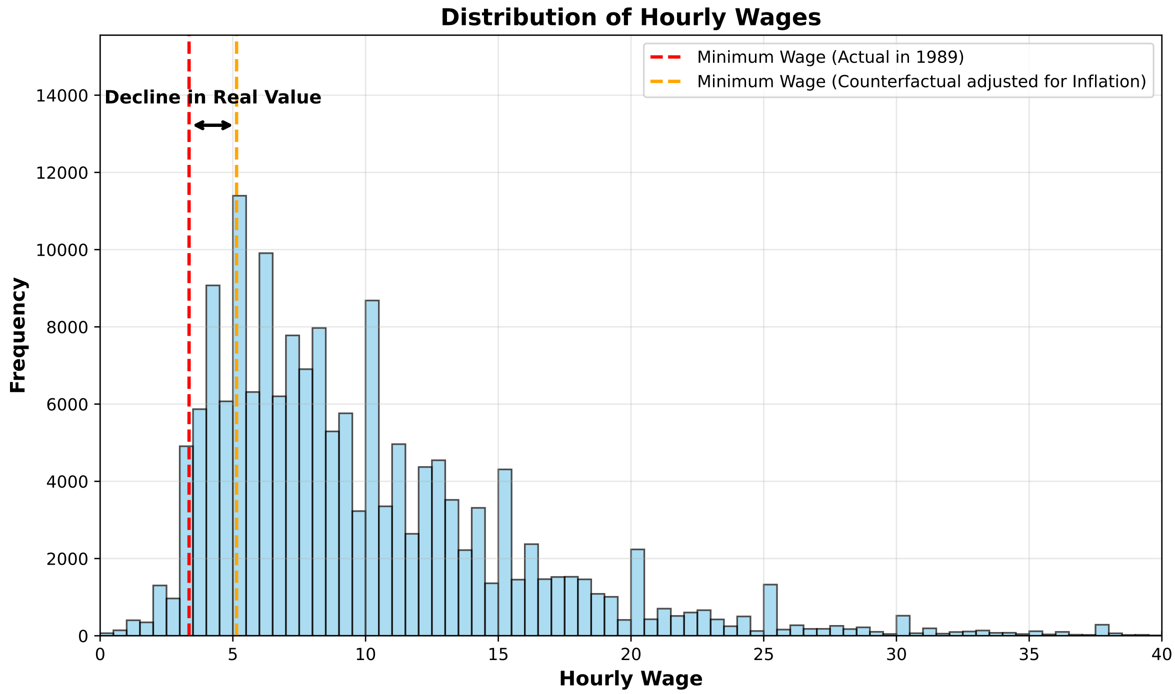


Figure 3.12: Distribution of Hourly Wages in 1989

Next, we are going to form a counteractual wage distribution in 1989. To do so, we are going to make some strong assumptions. First, we will assume that everyone below the counterfactual minimum wage of $4.77 per hour is paid exactly $4.77 per hour. This is a strong assumption, but it allows us to form a clear counterfactual wage distribution. What are potential issues with this assumption? First, as we have discussed in length, the minimum wage may have negative employment effects, so if the minimum wage was in fact $4.77 per hour, then there may be fewer workers overall. We are going to take the view of the literature that these effects are small.

Second, it is possible that there are spillovers on other workers. There are strong norms in the workplace that may generate spillovers. Earnings often grow with tenure at the firm. If a new minimum wage makes it so that new hires are making the same amount as existing workers, there may be pressures to increase the wages of existing workers. This could push up wages for workers above the minimum wage. We are going to ignore this possibility in our counterfactual, although there is evidence showing this is an important margin of adjustment for firms.

Third, firms may adjust in other ways, such as reducing hours or benefits, as we have discussed in the prior section. Lastly, we assume compliance is now perfect, meaning no workers are paid below the minimum wage.

Below we plot the counterfactual wage distribution in 1989 if the minimum wage had kept up with inflation since 1979.

This figure shows how the wage distribution would have looked if the minimum wage had kept up with inflation since 1979. In the counterfactual distribution, all workers who were earning less than $4.77 per hour are now earning exactly $4.77 per hour. This creates a large spike at the minimum wage level, showing how many workers would have been directly affected by the higher minimum wage.

Now let's go back to our original question: how much of the increase in the 90/10 wage ratio between 1979 and 1989 was due to changes at the bottom of the wage distribution? In 1989, the actual 90/10 wage ratio was about 4.5. In our counterfactual scenario, where the minimum wage had kept up with inflation, the 90/10 wage ratio would have been about 3.8. This is quite close to the 90/10 wage ratio in 1979, which was 3.6. Therefore, according to this simple exercise, the falling minimum wage can explain a significant portion of the increase in inequality during this period ((4.5-3.8)/(4.5-3.6) $\approx$ 78%).

This is an extremely suprising result. The rise in inequality during the 1980s led to a swarm of research tyring to understand the sources. Most papers studied supply and demand factors. Supply-side factors include things like changes in immigration. Demand-side factors include things like technology change. Supply and demand-side factors certainly impact wage levels and therefore, likely impact inequality in wages. However, this simple exercise suggests that the falling minimum wage can explain a large portion of the increase in inequality during this period. Policy-induced changes are often referred to as institutional factors. Other institutional factors that may impact inequality include changes in unionization rates and changes in tax policies.

Now, as we have discussed before, this is a simplified example. We assumed there would be no employment effects, no spillovers, and perfect compliance. So what do more careful studies find? DiNardo, Fortin, and Lemieux (1996) is a seminal study in the literature. They provide methods to perform counterfactual exercises like this one, but with more advanced statistical techniques than we will cover in this class. They find that about 40-65 percent of the rise in the 90/10 wage ratio between 1979 and 1988 can be explained by changes in the minimum wage. More recent analysis by D. H. Autor, Manning, and Smith (2016) finds similar results. They find that about 30-40 percent of the rise in the 90/10 wage ratio between 1979 and 2012 can be explained by changes in the minimum wage. Therefore, while it is unlikely that minimum wage can explain all of the rise in inequality, it seems to be a relatively important factor.

3.10 Summary

Minimum wage research has a long history in economics. The original theoretical predictions seemed clear: minimum wages should reduce employment.

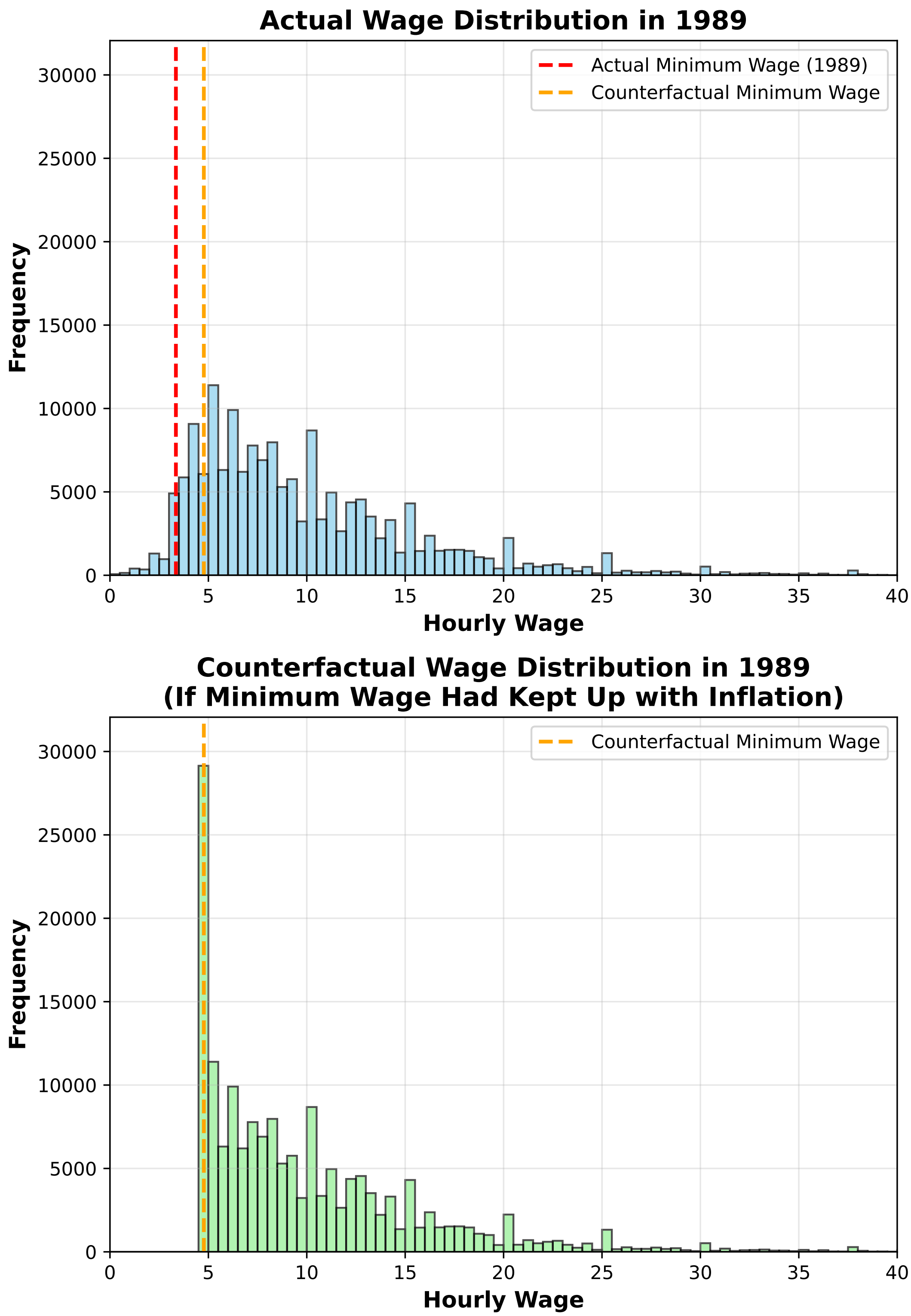


Figure 3.13: Counterfactual Wage Distribution in 1989 with Minimum Wage Adjusted for Inflation

However, the empirical evidence has been surprising. The seminal study by Card and Krueger (1994) found that a minimum wage increase in New Jersey actually increased employment in fast-food restaurants. This challenged the conventional wisdom among economists and led to a resurgence in minimum wage research. Overall, the evidence suggests having a nuanced view. The large employment effects of detractors do not seem to manifest in practice. However, there are other effects that may not be intended. In certain settings we have evidence of large price increases paid for by consumers.

In this chapter, we viewed the minimum wage through the lens of a perfectly competitive labor market. However, the finding in Card and Krueger (1994) and subsequent findings has led some to question whether this is really the right model for understanding the labor market. In the next chapter, we will deviate from perfect competition and turn to an alternative, arguably more realistic way to think about the labor market: monopsony.

4 Monopsony: Theory

4.1 A Brief History

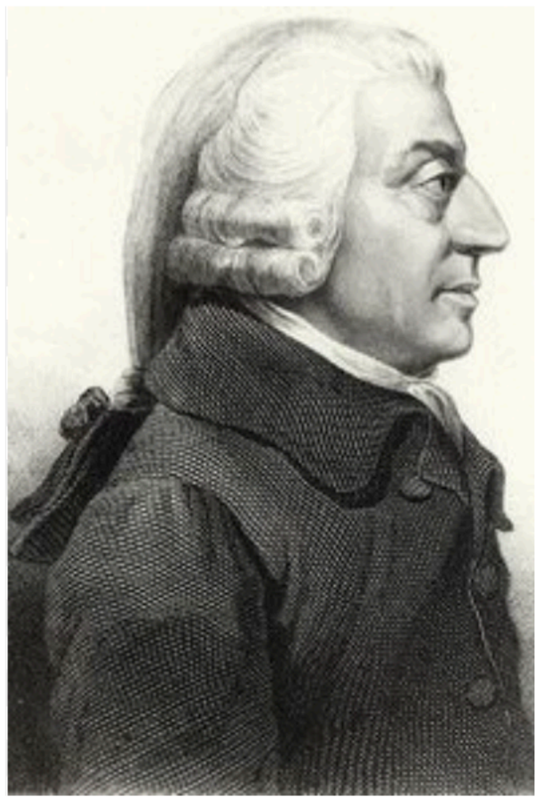


Figure 4.1: Adam Smith

Adam Smith is often called the “father of economics”. In 1776 he published his magnum opus *The Wealth of Nations*. The goal of the book was to understand how nations build wealth. In it, he introduced a number of concepts that became extremely influential in modern economics. One of the most famous concepts is the “invisible hand”. If markets are free and unregulated, Smith argues that individual self-interest will lead to efficient outcomes. Even though there is not an overall planner of the economy, this invisible hand will guide the economy to an efficient outcome.

But Adam Smith recognized that there are settings in which these effcient outcomes won't be realized. In particular, he writes the following about the labor market:

> What are the common wages of labor, depends everywhere upon the contract usually made between [employers and employees], whose interests are by no means the same… The masters, being fewer in number, can combine much more easily … Masters are always and everywhere in a sort of tacit, but constant and uniform, combination, not to raise the wages of labor above their actual rate.
>
> —*Adam Smith, The Wealth of Nations*

This aspect of the labor market, that employers may exercise market power over their workers, did not really receive much attention. The next big development came in 1933, when Joan Robinson published *The Economics of Imperfect Competition*. Robinson coined the term "monopsony" to describe a market with a single buyer. She used the term to describe the labor market, where employers are the buyers of labor. In Adam Smith's quotation, he is arguing that employers have monopsony power, although the term did not yet exist at the time.

Again, after Joan Robinson's work, studying monopsony power in labor markets did not really catch on. While it generated some initial interest, for much of the 20th century, the dominant paradigm in labor economics was a framework of perfect competition. Robinson herself said in 1969, "All this had no effect. Perfect competition, supply and demand…and marginal products still reign supreme in orthodox teaching." (See Card (2022) for more discussion on Robinson's work and its impact.)

After all, there are a lot employers. That's really a core assumption of the competitive model. However, are you interested in working at every single firm? Is every firm interested in hiring you? Of course not. Assuming the number of firms is the size of the labor market for an individual is clearly not a good assumption.

To see another potential problem with the competitive model, consider the following scenario: a firm is going through a downturn in sales and decides that it can't afford to offer a 3% raise to its employees as usual, and so offers no wage increase that year. Other firms in the area, however, are doing well and are offering 3% raises. Do you think the firm will lose all of its employees to other firms?

Probably not. Workers have some attachment to their current employer. They can't costlessly quit their job and find a new one that pays better. But in perfectly competitive models, that is the assumption. There is a market wage. If you pay below that wage (i.e. don't meet that 3% raise), you lose all your workers. For our textbook model in the next section, we will change this assumption.

4.2 Equilibrium in Monopsony

In perfect competition, we assumed firms are wage takers. This implies that if a firm decides to pay a wage below the market wage, it will immediately lose all of its workers. In this section, we will make a very simple change to this assumption. We will assume that the wage a firm offers depends on the amount of labor it wants to hire. In particular, we will assume that the higher the wage, the more workers a firm can attract. We will denote $w(L)$ as the inverse labor supply curve. This function tells the firm that how much it needs to pay in order to attract a certain number of workers. For example, if a firm needs to pay a wage of \$15 to attract 5 workers, then $w(5) = 15$.

Now you might hear some arguments against monopsony power because it is assumed to be a monopsonist you need to be the only employer in the area. This is the classic "factory town" example, where there is literally only one employer in the town. But monopsony power is much more general than this. Monopsony power can exist even when there are many employers. The key distinction will be whether the firm faces an upward sloping labor supply curve. If a firm can hire as many workers as it wants at a fixed wage, then it is a wage taker and does not have monopsony power. If a firm has to raise the wage in order to attract more workers, then it has some monopsony power.

Let's go back to the example we considered in the perfect competition section. Suppose we have a cafe hiring baristas. First, we are going to specify the labor supply curve to the cafe. We will assume that the labor supply curve is given in the following figure:

Now let's consider the decision of the cafe owner of hiring the first worker. To hire one worker, the cafe needs to pay a wage of \$5. Is this worth opening at all. To answer this question, we need to know the marginal revenue product of the worker. How much revenue is going to be brought in by the first worker? Let's assume that the worker can make 13 lattes an hour, each of which sells for \$5, implying the worker generates \$65 in revenue per hour. The cost of hiring the worker is \$5, so the cafe owner should definitely hire the first worker.

Now let's consider the decision of hiring a second worker. To hire two workers the firm needs to pay a wage of \$10. Is this worth it? Well, it depends again on the marginal revenue product of the second worker. Let's assume that the second worker produces 12 lattes an hour, so \$60 in revenue per hour.

Now, what do we compare the \$60 in revenue to? One common pitfall is to compare it to the wage you are paying: \$10. This is what we did in the perfectly competitive model. But this isn't right anymore. What you need to compute is the additional cost of hiring the second worker. To hire the second worker, you need to pay that worker \$10, but you also need to pay the first worker \$5 more (from \$5 to \$10).

The increase in cost of hiring an additional worker is referred to the marginal factor cost (MFC). The MFC of the second worker is \$15 (\$10 + \$5). The marginal revenue product of

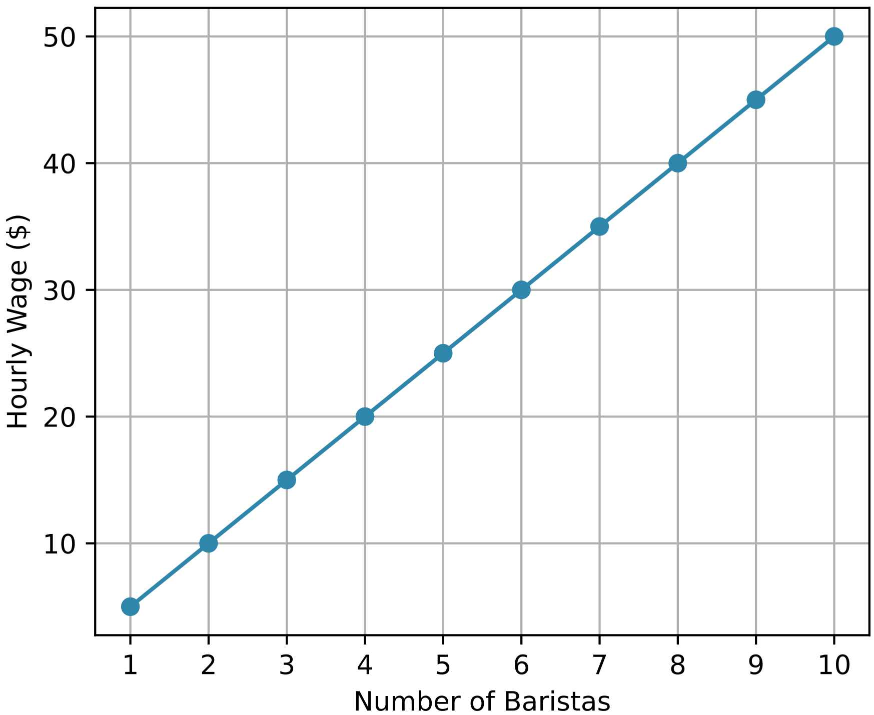


Figure 4.2: Labor Supply Curve to a Cafe

the second worker is \$60, which is greater than the MFC of \$15, so the cafe owner should hire the second worker.

What's the marginal factor cost of the third worker? To hire the third worker, you need to pay that worker \$15, but you also need to pay the first two workers \$5 more than they were previously being paid (from \$10 to \$15). So the MFC of the third worker is \$25 (\$15 + \$5 + \$5). Now that we understand conceptually how to compute the MFC, let's compute it for the entire range of the possible number of workers.

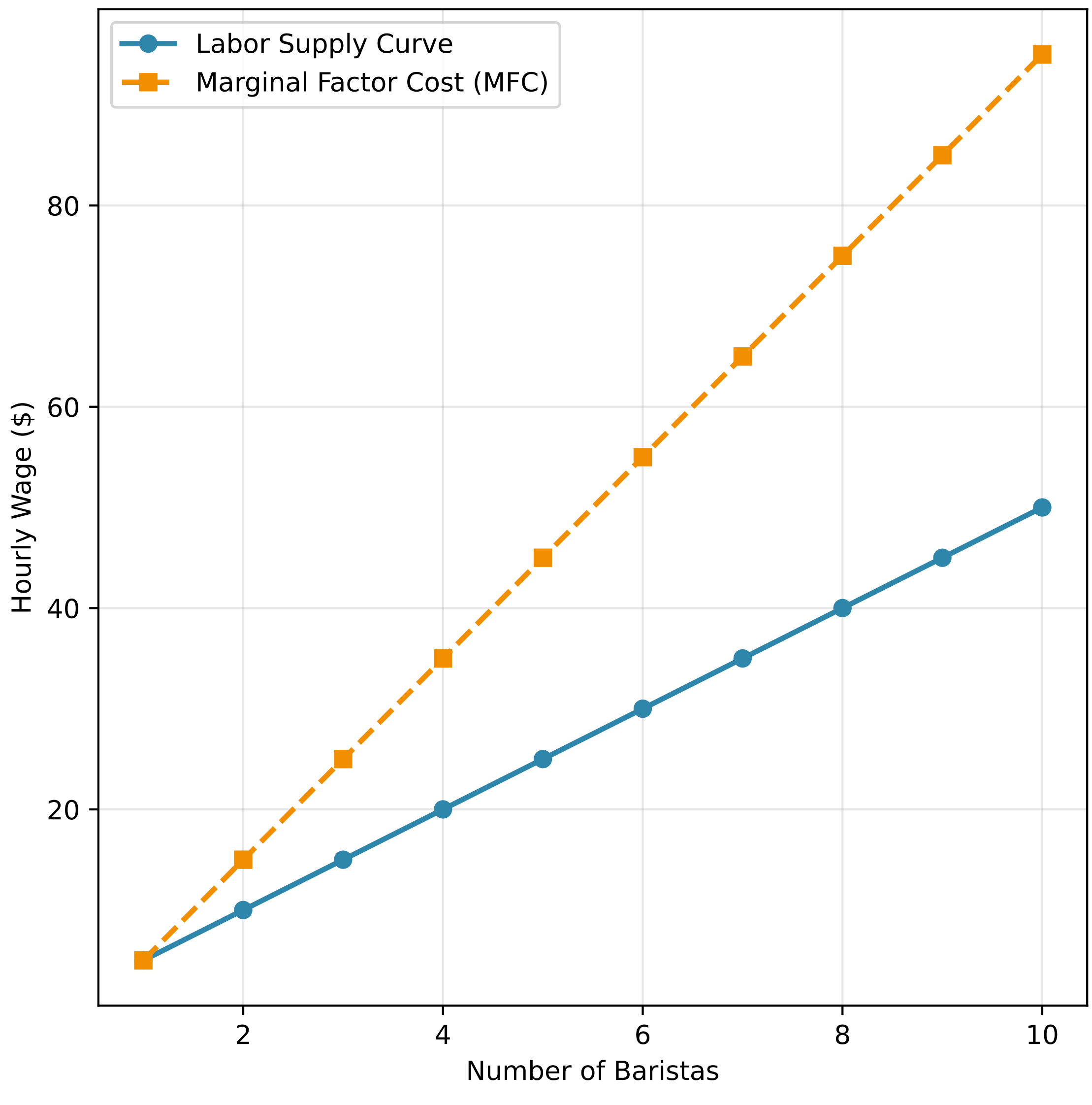


Figure 4.3: Labor Supply Curve and Marginal Factor Cost Curve

Now, to determine how many workers the cafe will hire, we need to find where the marginal revenue product of labor (MRPL) equals the marginal factor cost (MFC). Again, the MRPL is the additional revenue generated by hiring one more worker. We've already assumed the MRPL of the first worker is \$65 and the second worker is \$60 Let's assume that the MRPL of each additional worker is given in the figure below.

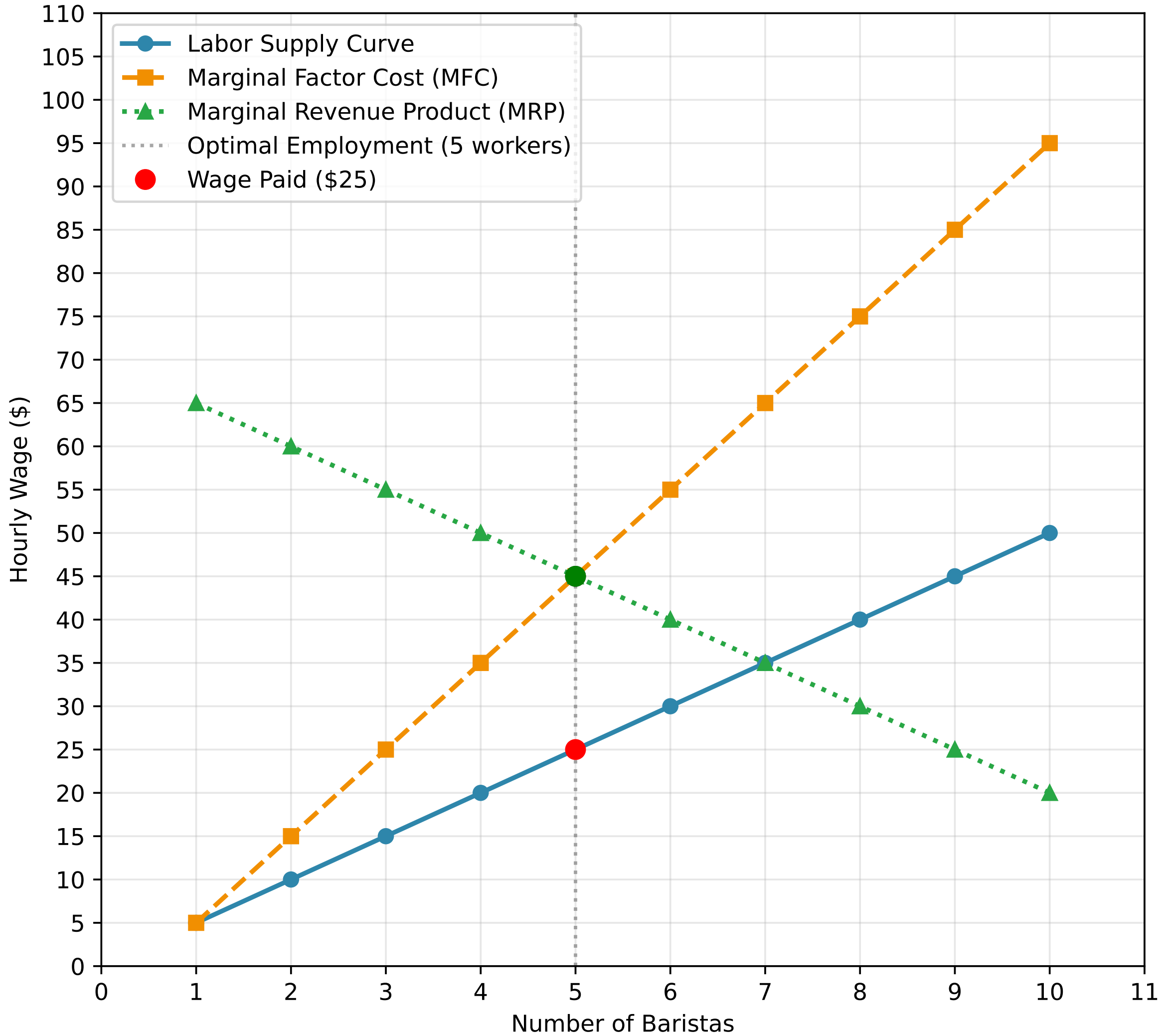


Figure 4.4: Monopsony Equilibrium: MRP, MFC, and Labor Supply

When will a firm stop hiring workers? As soon as the marginal factor cost (MFC) of hiring an additional worker is greater than the marginal revenue product of labor (MRPL) of that worker. In our case, the firm will hire 5 workers. For the 6th worker, the marginal factor cost is \$55. This is because the firm needs to pay \$30 to attract this 6th worker, plus the increase in pay for all the previous workers (from \$25 to \$30). Therefore the total increase in costs of hiring the 6th worker is \$30+\$25=\$55. The MRPL of the 6th worker is only \$40, so from the firm's perspective, it is not worth hiring the 6th worker. It will be losing money by hiring that worker.

So we have figured out equilibrium employment. But what about the equilibrium wage? Well, if the firm is hiring 5 workers, then it needs to pay a wage high enough to attract 5 workers. In our case, this is a wage of $25. So the equilibrium wage is $25 and the equilibrium employment is 5 workers. In our case that value comes directly from the inverse labor supply curve. To attract 5 workers, the firm needs to pay a wage of $25.

4.3 Inefficiency in Monopsony

Now that we've figured out the equilibrium employment and wage, let's try and understand why monopsonistic markets are inefficient.

To do so, let's consider the case of hiring the 6th worker. The 6th worker is willing to work for a wage of $30. This tells us that the 6th worker would prefer to work for $30 than not work at all. Let's consider the value the 6th worker brings to the firm. The marginal revenue product of the 6th worker is $40. This means that the firm would gain $40 in revenue from hiring this worker.

So if we were to consider this in isolation, it seems like a good idea to hire the 6th worker. So why doesn't the firm hire the 6th worker? Again, the firm takes into account that it can't offer the 6th worker a wage of $30 without also raising the wages of the first 5 workers. The increase in wages of the first 5 workers is referred to as an inframarginal cost. An inframarginal worker is one that is already working at the previous wage. So increasing the wage does not attract this worker.

What is the size of the inefficiency? It depends on two things. First, how many workers are not being hired even though the marginal revenue product of labor is higher (or equal to) than the wage they are willing to work for. In our case, this is the 6th and 7th worker. It also depends on the gap between the marginal revenue product of labor and the wage they are willing to work for. In our case, the 6th worker is willing to work for $30 and has a MRPL of $40, so there is a large gap. For the 7th worker, the MRPL is equal to the wage ($35), so there is no inefficiency from not hiring this worker.

Often we allow fractional employment. For example, what if you could hire 5.1 workers. Well there would be a large inefficiency from not hiring this 0.1 worker because the MRPL at 5.1 is much higher than the wage of $25. This is often a background assumption in introductory economics courses. It allows us to calculate the inefficiency of monopsony more easily.

For example, in the figure below we have shaded in the inefficiency of monopsony. These are all the units in which the MRPL is greater than the wage, but are not being hired by the firm.

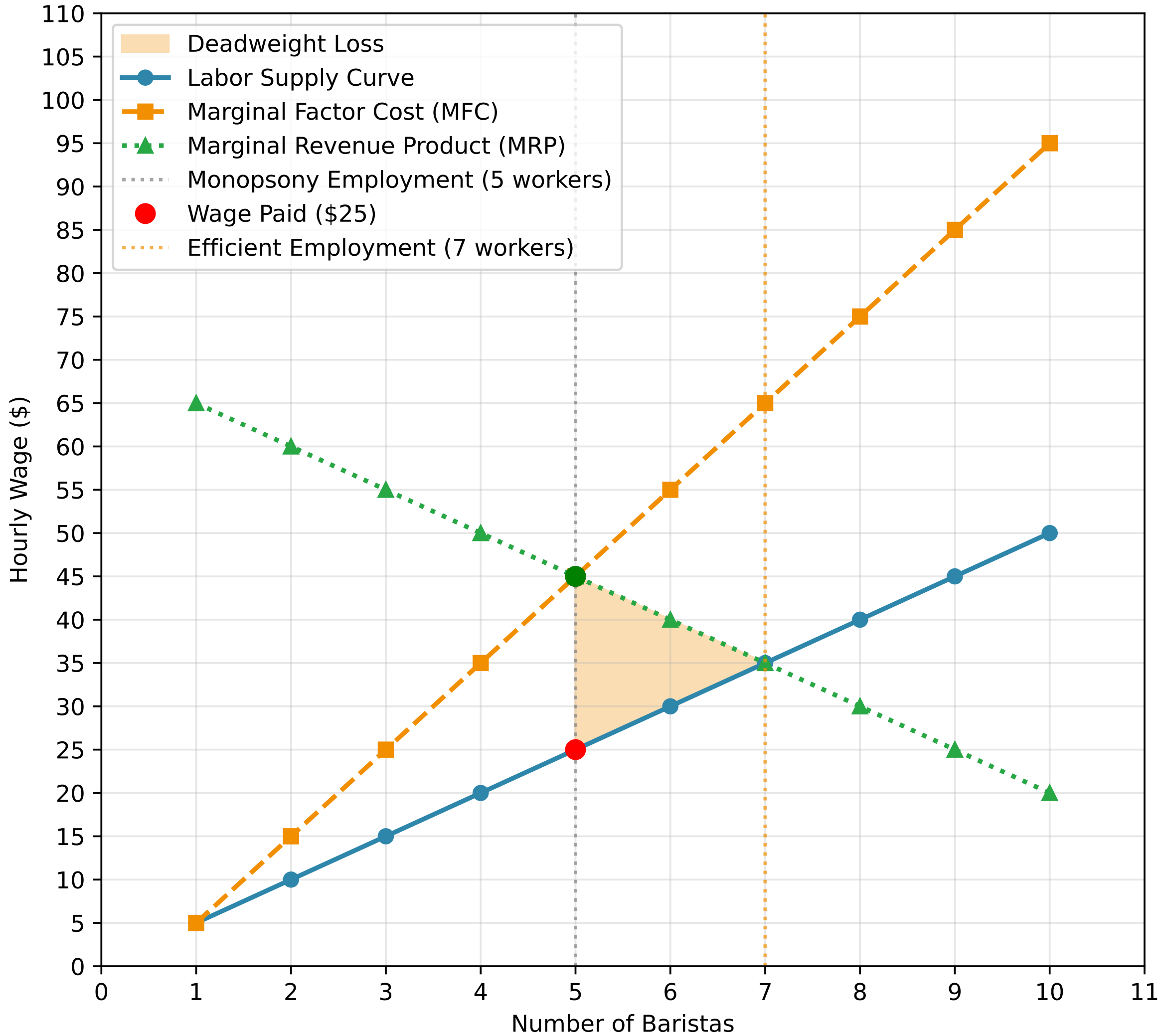


Figure 4.5: Deadweight Loss of Monopsony

Now, let's see how the inefficiency of monopsony changes when we change the labor supply curve. In particular, let's see how the inefficiency changes when we make the labor supply flatter.

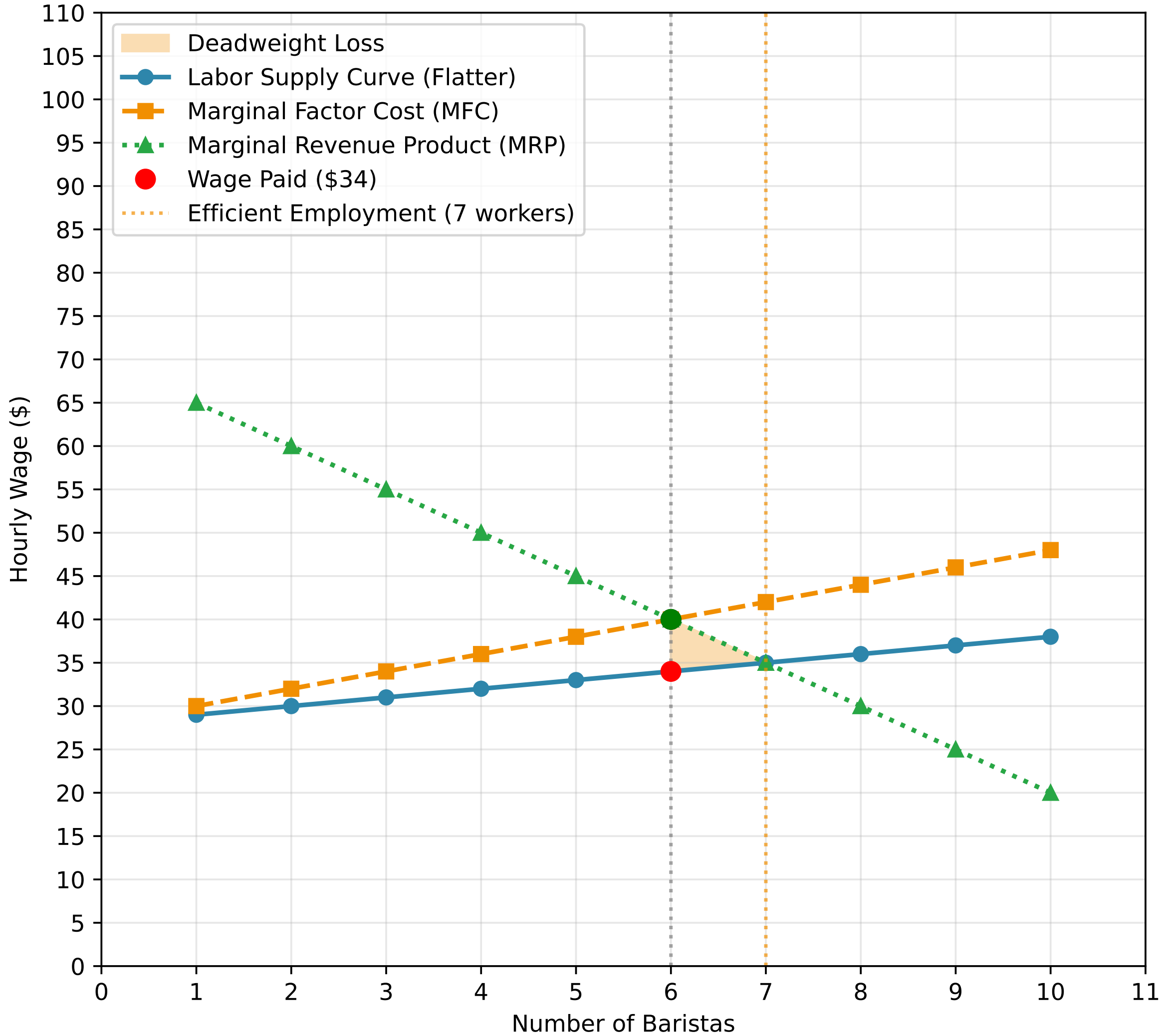


Figure 4.6: Monopsony with Flatter Labor Supply Curve

Notice that with the flatter labor supply curve, the deadweight loss area is much smaller compared to the steeper supply curve in the previous figure. Why does this make sense?

Well, let's again think about the marginal factor cost. To go from 5 to 6 workers, you need to pay a slightly higher wage (about $33 to $35). But you also need to pay the first 5 workers a higher wage. But this inframarginal cost is not as large anymore. Instead of having to pay an additional $5 to each of the first 5 workers, you only need to pay an additional $2. The gap between the labor supply curve and the marginal factor cost curve is much smaller when the labor supply curve is flatter. This means that our employment levels and wages will be closer to the efficient levels.

Intuitively, what is going on here? Let's take the firm's perspective. In the extreme case of perfect competition, the firm can hire as many workers as it wants at a given wage. This means that the firm does not have to worry about inframarginal costs. There is no inframarginal cost because you don't need to hire the wage to attract more workers. For a very flat labor supply curve, the inframarginal cost is small. You need to increase the wage a little bit to attract more workers, but you don't need to increase it by much. The fact that inframarginal costs are small limits the inefficiencies of monopsony.

The key parameter here is the elasticity of labor supply. The elasticity of labor supply is a measure of how responsive the quantity of labor supplied is to changes in the wage. If the elasticity of labor supply is high, then small changes in the wage will lead to large changes in the quantity of labor supplied. What does this mean in terms of the slope of the labor supply curve? A high elasticity of labor supply means that the labor supply curve is flat. In contrast a low elasticity of labor supply means that the labor supply curve is steep.

In math, the elasticity of labor supply to the firm is defined as:

$$\eta = \frac{\Delta L/L}{\Delta w/w} = \frac{\%\Delta L}{\%\Delta w}$$

where L is the quantity of labor supplied, w is the wage, and Δ indicates a change in the variable. This is a central parameter in the literature on monopsony in labor economics. Let's again consider perfect competition. In a perfectly competitive market, if there is a tiny change in the wage, the firm either loses all its workers or gains all the workers in the market. In other words, a small change in the wage leads to an infinite change in the quantity of labor supplied to the firm. Therefore, under perfect competition, $\eta = \infty$.

The smaller η is in magnitude, the more monopsony power the firm has. Why, because even if the firm decreases its wage substantially, it will not lose many workers. This will tells us the scope of monopsony power in the labor market.

4.4 Measuring Monopsony Power

In actual empirical studies of monopsony power, it is often difficult to estimate the entire supply curve to the firm. While we have plotted linear supply curves, they could be more general in practice. Identifying the entire supply curve is extremely difficult.

Instead, researchers often focus on a statistics that are informative about the degree of monopsony power. The statistic that has been most commonly estimated is the elasticity of labor supply to the firm. Again, a low elasticity of labor supply to the firm indicates that the firm has a lot of monopsony power, so it makes sense to focus on this statistic.

However, there are other potential statistics. Some researchers have focused on the gap between the marginal revenue product of labor and the wage. This is often referred to as the wage markdown. Mathematically, the wage markdown is often defined as:

$$\text{Wage Markdown} = \frac{MRPL - w}{MRPL}$$

So in our example, at the equilibrium, the MRPL is \$45 and the wage is \$25. Therefore, the wage markdown is (45-25)/45 = 0.444. This means that workers receive 44.4% of the marginal revenue product of labor.

I personally like this statistic because of its interpretability. The elasticity of labor supply to the firm is certainly useful, and in simple models, the elasticity of labor supply to the firm directly determines the wage markdown.

Wage markdowns, instead, have some intuitive appeal. If the wage markdown is going up over time, for example, then we can interpret this as workers receiving a smaller and smaller share of the marginal value they create. If the elasticity of labor supply to the firm is going down over time, then the scope for monopsony power is increasing. However, we really need the model to be correct. If our model is wrong, then it isn't actually clear if workers are receiving a smaller share of the marginal value they create or not.

Lastly, other measures that have been used in the literature are things that are relatively easy to measure, but require perhaps more assumptions on how they translate to monopsony. Concentration indices, for example, are commonly used. The idea is that the scope for monopsony power is higher when there are fewer firms in the market. Therefore, if concentration is going up over time, then we might expect monopsony power to be increasing as well.

4.5 Sources of Monopsony Power

So far, we have assumed that the labor supply curve to the firm is upward sloping and that this leads to lower employment, lower wages, and inefficiency. But why is the labor supply curve upward sloping? What are the sources of monopsony power?

When you think about sources of monopsony power, you should think about why a worker may not leave his or her firm if the firm lowers the wage. That really is what is generating these upward sloping labor supply curves. So let's consider a few that have been discussed in the literature.

4.5.1 Search Frictions

The assumption of perfect competition is that workers can effortlessly move between firms. They know what other firms in their area are paying and they can immediately start working at a new firm if they are offered a higher wage.

This is clearly not true in practice. Workers need to search for jobs. They need to apply for jobs and go through interviews. These **search frictions** are a source of monopsony power. The presence of search frictions became particularly influential in the 1980s with the work of Peter Diamond, Dale Mortensen, and Christopher Pissarides. Their goal was less about monopsony power and more about understanding aggregate unemployment dynamics. A consequence of their work, however, is that workers are paid less than their marginal revenue product of labor. Hence, there is monopsony power in the labor market.

Some search frictions occur naturally, like the job search process, while others are created by firms. For example, some firms require non-compete agreements. These agreements prevent workers from working for a competitor for a certain period of time after they leave the firm. This is a potential source of monopsony power because it again limits the ability of workers to move between firms.

4.5.2 Differentiated Jobs

Another source of monopsony power is job differentiation. So far, we have assumed that all jobs are the same and that workers prefer the job with the highest wage. But this is not true in practice. Some individuals may prefer a job because it is closer to their house. Some may particularly like the work environment or colleagues at a certain firm. All of these factors can lead to monopsony power. Why? Because these factors tie individuals to their jobs. They won't move in response to a small change in the wage because they like their job for other reasons.

Differentiated products is a classic way to model imperfect competition in product markets. The same idea can be applied to labor markets. If jobs are differentiated, then firms have some monopsony power over their workers. This idea is clearly illustrated in Card et al. (2018).

4.5.3 Concentration

We started this chapter from a quote from Adam Smith. He argued that employers have monopsony power because they are few in number and can collude to keep wages low. This is another potential source of monopsony power.

Outright collusion is illegal in the United States, but we do have some evidence that it occurs in certain settings. For example, in 2010 the Department of Justice brought a case against many of the major firms in Silicon Valley, including Apple, Google, and Intel. The case alleged that these firms had agreed not to poach each other's workers. In the end, the firms settled the case and agreed to pay hundreds of millions of dollars.

Even without explicit collusion, a small number of firms can lead to monopsony power. Oligopsony models are a workhorse set of models from industrial organization. In these models, there are a small number of firms that compete with each other. These markets lead to higher prices and lower output. Antitrust in the United States is explicitly concerned with markets getting too concentrated because of the belief that concentrated markets lead to higher prices and lower output.

4.6 Policy and Monopsony Power

Why does it matter whether we think the labor market is perfectly competitive or monopsonistic? One very good reason is that our policy decisions depend on the answer to this question. Even if politicians don't necessarily say the words "perfect competition" or "monopsony", they are implicitly making assumptions about the labor market when they make policy decisions.

For example, let's return to the minimum wage. As we discussed in a previous chapter, the standard model of perfect competition predicts that a minimum wage will lead to unemployment. Increasing the cost of labor will lead to firms demanding less of it. Makes perfect sense.

But let's see what happens in a monopsonistic labor market. Recall in our original cafe example in this chapter, the firm was hiring 5 workers at a wage of $25. The efficient level of employment was 7 workers at a wage of $35. What happens if we impose a minimum wage of $35?

The key in analyzing this is to consider the marginal factor cost curve. Let's consider the marginal factor cost of the 6th worker. To hire the 7th worker, the firm needs to pay that worker $35 to attract the worker. In the previous setup, that would mean increasing the wage for the previous 6 workers by $5. But now the firm is already paying the $35 in order to meet the minimum wage. Therefore, the marginal factor cost of the 6th worker is $35. There are no inframarginal costs anymore. This is true for every single worker below the minimum wage. The marginal factor cost of every worker below the minimum wage is just equal to the minimum wage.

But once you get above the minimum wage, then you need to again consider inframarginal costs. For example, to hire the 8th worker you need to pay that worker $40, but you also need to pay the first 7 workers $5 more. Therefore, the marginal factor cost of the 8th worker is ($40 + $7*5) = $75. So the marginal factor cost curve with a minimum wage is flat at the minimum wage until you reach the number of workers that are willing to work at that wage. After that, the marginal factor cost curve is equal to the original marginal factor cost curve.

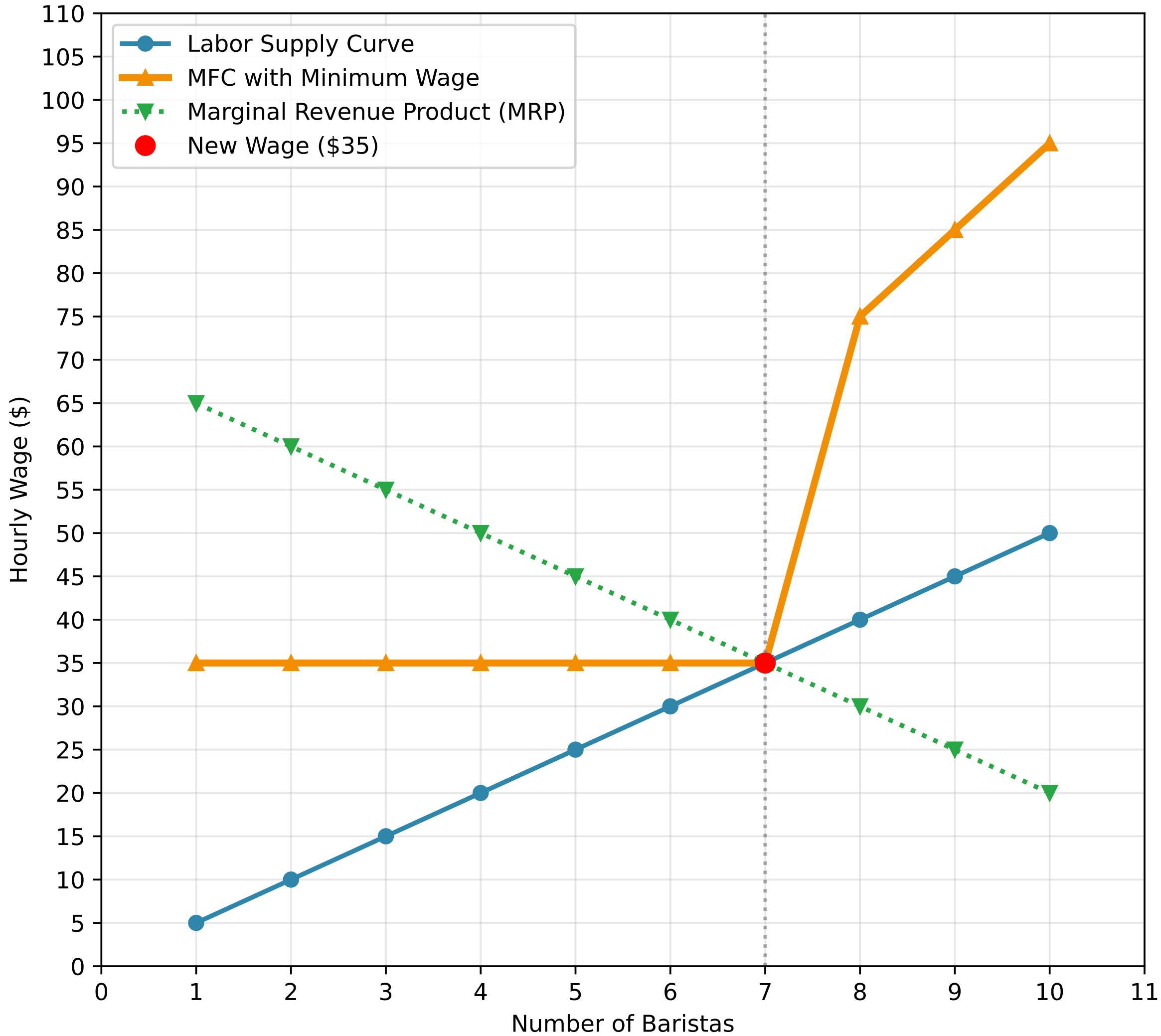


Figure 4.7: Marginal Factor Cost with Minimum Wage

So what will the firm decide to do. It's decision follows the exact same process. Keep hiring workers until the marginal factor cost of hiring an additional worker is greater than the marginal revenue product of that worker. In our case, the firm will hire 7 workers at a wage of \$35, because that is where the marginal factor cost curve intersects the marginal revenue product curve.

This is a very surprising result! We've made a seemingly simple change to our model: firms face upward sloping labor supply curves. This change has led to a completely different prediction about the effects of a minimum wage. In this model, we can actually get that an increase in the minimum wage can **increase** employment. If you recall, that is exactly what the original Card and Krueger (1994) paper found.

However, we have to a bit careful here. Increases in the minimum wage **can** increase employment in monopsonistic models, but they don't have to. Imagine the minimum wage in this market was set to \$50. Let's see what happens in this case.

Now the firm will hire only 4 workers at a wage of \$50. This is fewer workers than both the original monopsony equilibrium (5 workers) and the efficient outcome (7 workers). When the minimum wage is set too high, it can actually reduce employment below even the monopsony level, demonstrating that there is an optimal range for minimum wage policy in monopsonistic markets.

4.7 Summary

Monopsony power in labor markets can stem from many sources. The consequence of monopsony power is that firms will pay lower wages and hire fewer workers than in a perfectly competitive market. Modelling labor markets as monopsonistic is important because it can lead to very different predictions about the effects of labor market policies.

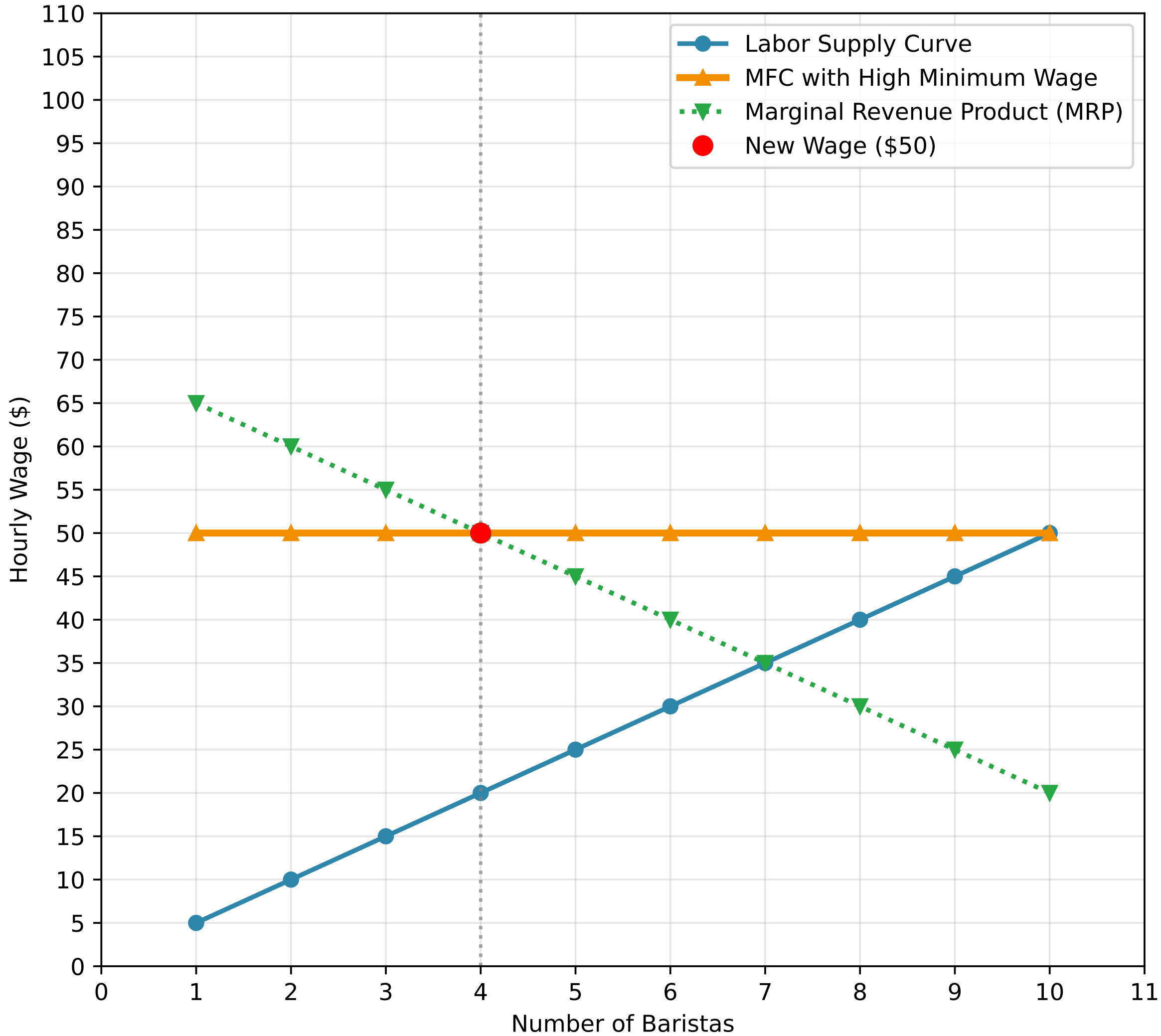


Figure 4.8: Minimum Wage Set Too High

5 Monopsony: Empirics

5.1 The Evidence

We are going to discuss three separate sets of empirical evidence that support the existence of monopsony power in labor markets. The first set of evidence is most closely linked to the theory that was presented in the previous chapter. This evidence seeks to identify the elasticity of labor supply to the firm. The elasticity of labor supply to the firm is how changes in wages affect employment at the firm. A firm that can lower its wages without losing workers has a very inelastic labor supply curve. A firm that loses its workers for even small decreases in wages faces a very elastic labor supply curve. In a perfectly competitive labor market, the elasticity of labor supply to the firm is infinite.

The second set of evidence discusses other aspects of the labor market that appear inconsistent with the perfectly competitive model. This includes differences in wages across firms, the presence of the shortage of workers, among others. The last set of evidence are case studies of documented collusion in labor markets. In this section we will also consider the implications of monopsony power for antitrust policy.

Lastly, we will again link our discussion of monopsony power to one of our big motivating questions: what explains changes in labor market inequality over time in the US?

5.2 The Elasticity of Labor Supply to the Firm

To understand how difficult it is to estimate the elasticity of labor supply to the firm, let's consider what we would actually observe in the data about a firm. To make this concrete, let's consider a hypothetical example. Let's imagine a software engineering firm has 100 employees and pays a salary of \$100,000 per year in 2023. Imagine in 2024 the firm raises the salary to \$110,000 and we observe that the firm now has 120 employees.

We might naively attempt to use this data to estimate the elasticity of labor supply to the firm. In this case, there was a 10% increase in wages and a 20% increase in employment. Therefore, we might conclude that the elasticity of labor supply to the firm is:

$$\eta = \frac{\Delta L/L}{\Delta w/w} = \frac{20\%}{10\%} = 2$$

But why might this be the wrong conclusion? To figure this out, let's consider the plot below, which simply plots the number of employees at the firm against the wage.

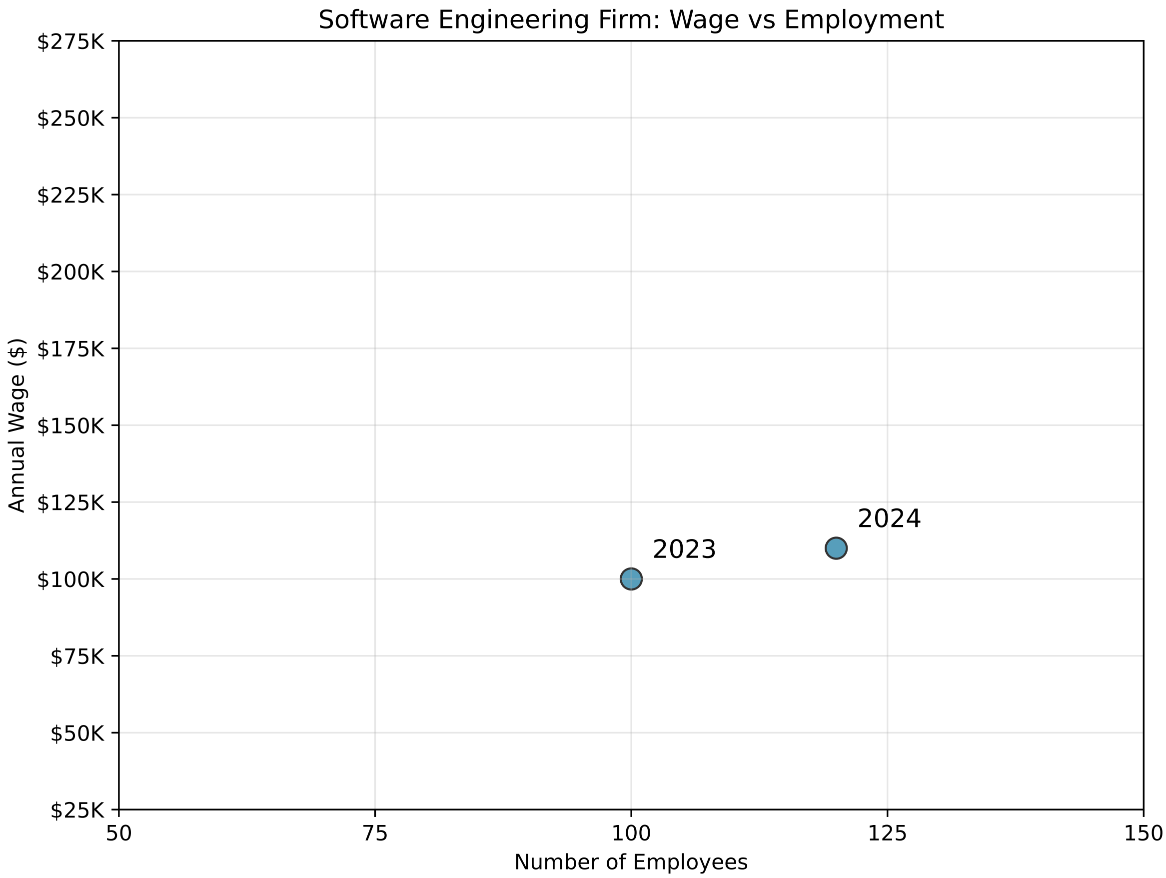


Figure 5.1: Observed Wage and Employment Data

Now, when we use these two estimates to estimate the elasticity of labor supply to the firm, what are we actually doing? We are assuming that these two points are on the labor supply curve to the firm. If we assume it is linear, then we can simply draw a line between the two points and we've identified the entire labor supply curve to the firm.

Let's draw this supply curve to the firm below.

Now, let's think through why the firm might have raised wages. As we discussed in the previous chapter, the firm continues to hire until marginal revenue product of labor (MRPL) equals the marginal factor cost (MFC). Therefore, if these two points are equilibrium points, then these must be points in which the MRPL equals the MFC. Since the marginal revenue product of labor curve is downward sloping, it must be that the curve shifted upwards between 2023 and 2024 to sustain these as equilibrium points (it is impossible to draw a downward sloping curve

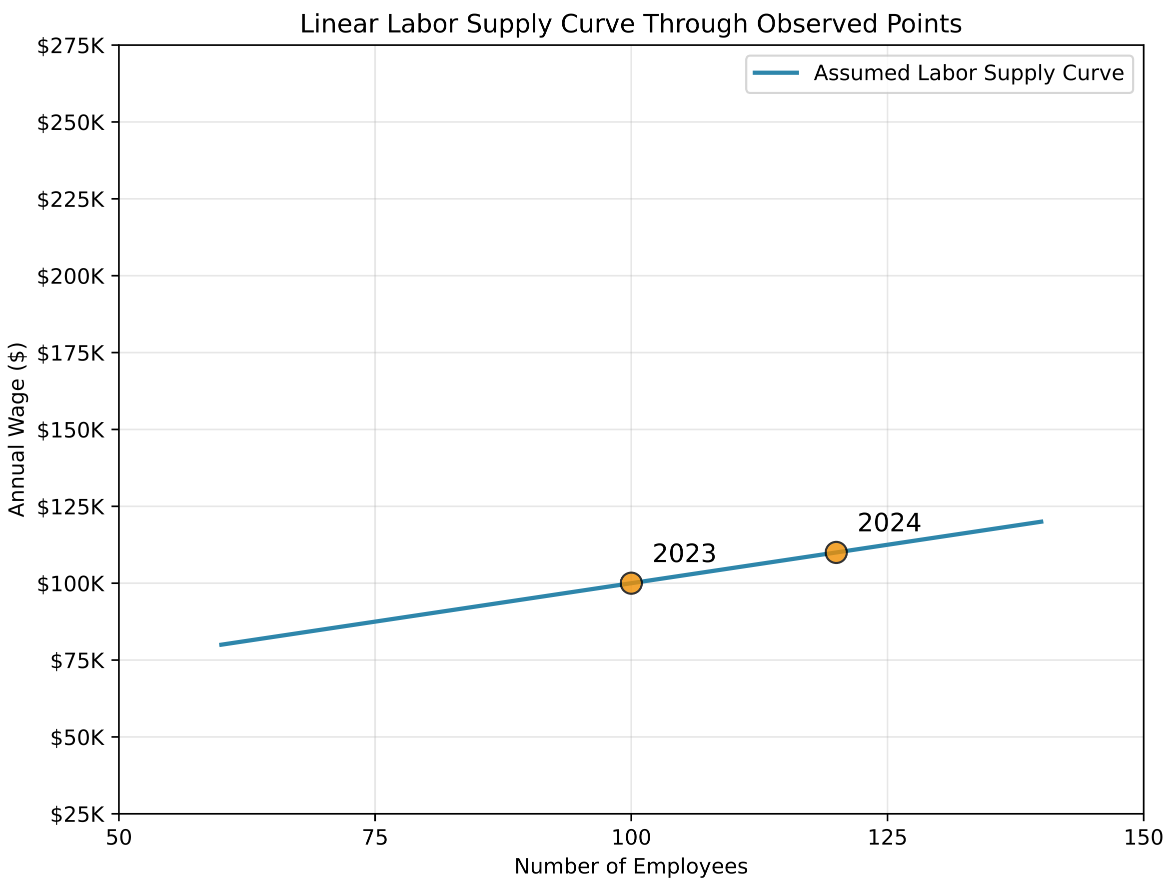


Figure 5.2: Assumed Labor Supply Curve to the Firm

that intersects the MFC at these points). This makes perfect sense. The reason the firm is increasing the wage is due to an increase in the MRPL of its workers. Let's draw these marginal revenue product of labor curves and the associated marginal factor cost (MFC) curves below. Why would the MRPL curve shift? It could be many things. For example, maybe the price of the firm's product increased, so that each additional unit now sells for more. Maybe demand for the product increased overall.

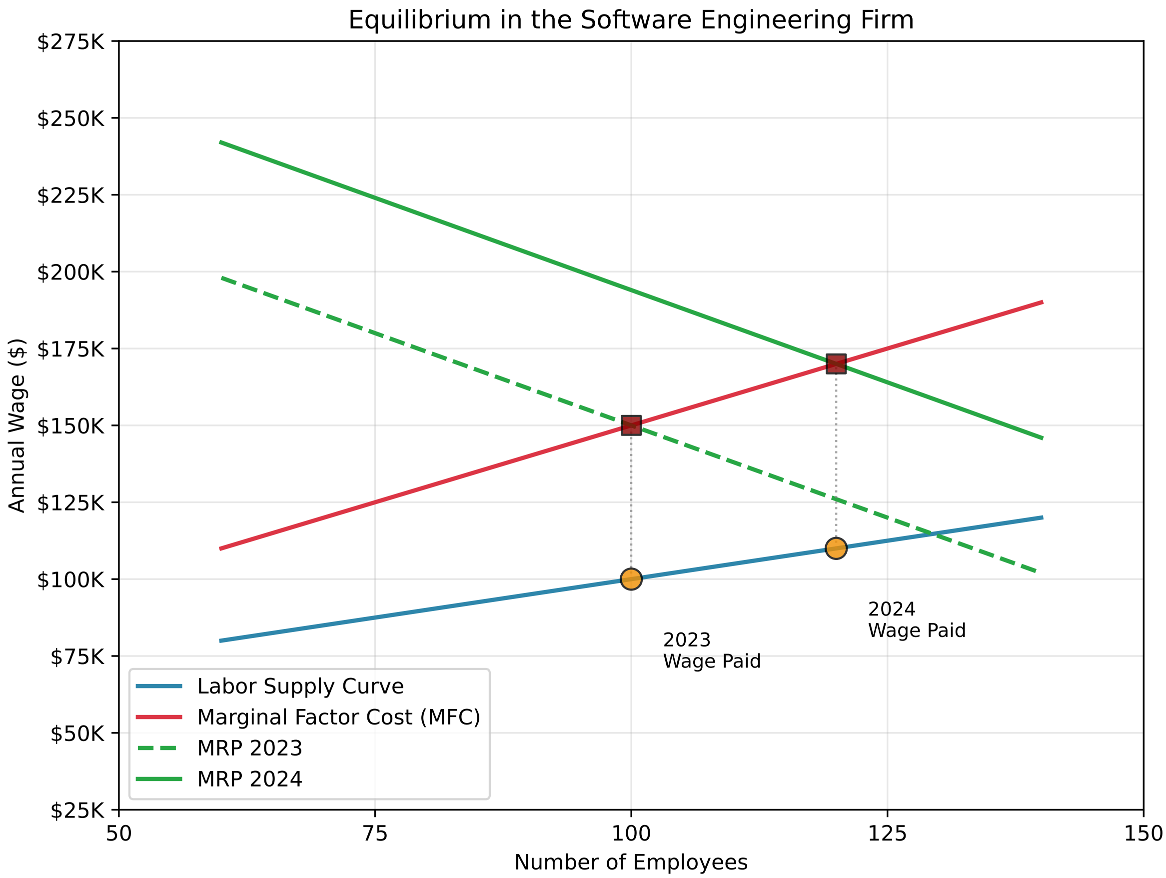


Figure 5.3: Equilibrium in the Software Engineering Firm

But this leads to an interesting possibility. If the marginal revenue product of labor curve shifted upwards, what is to say the labor supply curve didn't also shift between 2023 and 2024? For example, imagine that more workers moved into the local area, or that another local software engineering closed down. This time, let's see if we can draw two labor supply curves, one for 2023 and one for 2024, that also pass through the two equilibrium points.

Now in this second scenario, the labor supply curve we drew was purposely quite flat. Therefore, a small change in wages leads to a much larger change in employment. This implies the firm has less monopsony power than in the first scenario.

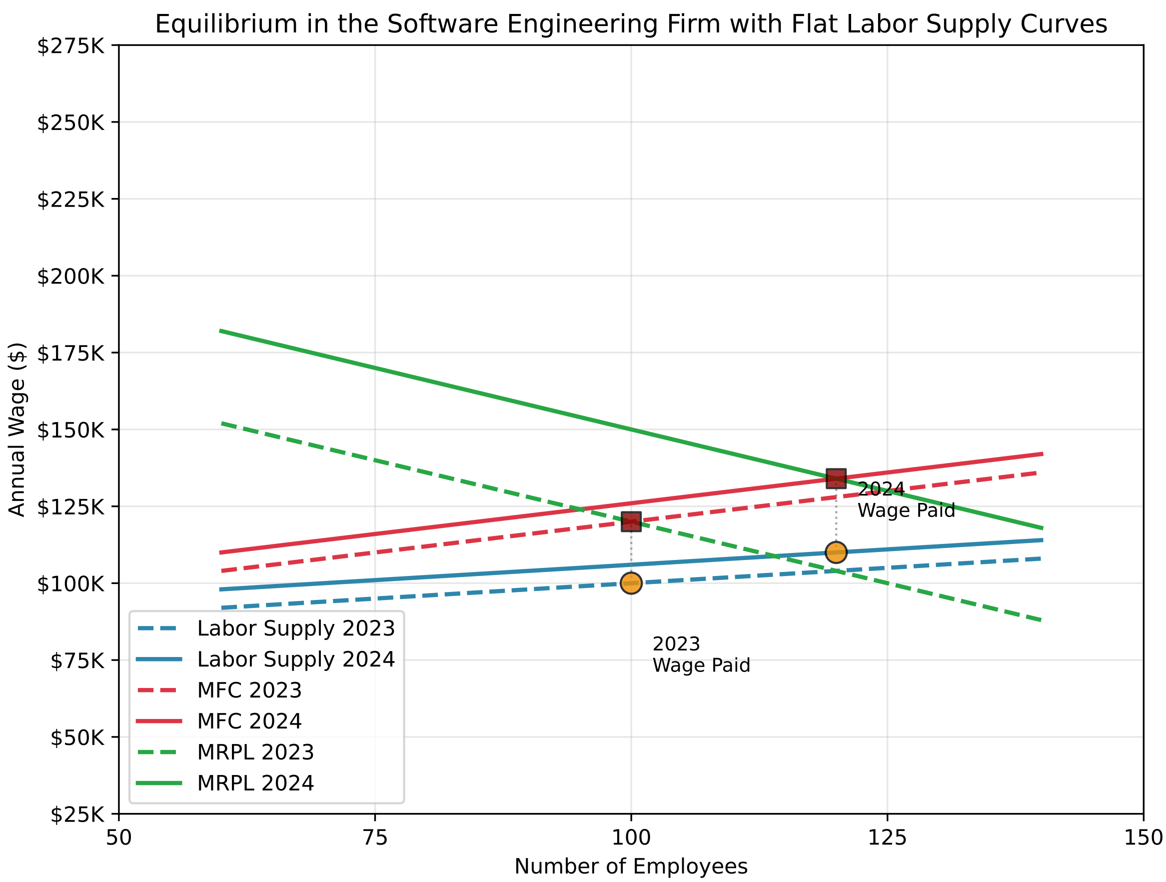


Figure 5.4: Equilibrium in the Software Engineering Firm with Flat Labor Supply Curves

But it did not need to be this way. For example, the plot below changes the labor supply curve to be much steeper. In this scenario, labor supply is shifting down instead of up. But the main point here is that these labor supply curves are also consistent with the observed data!

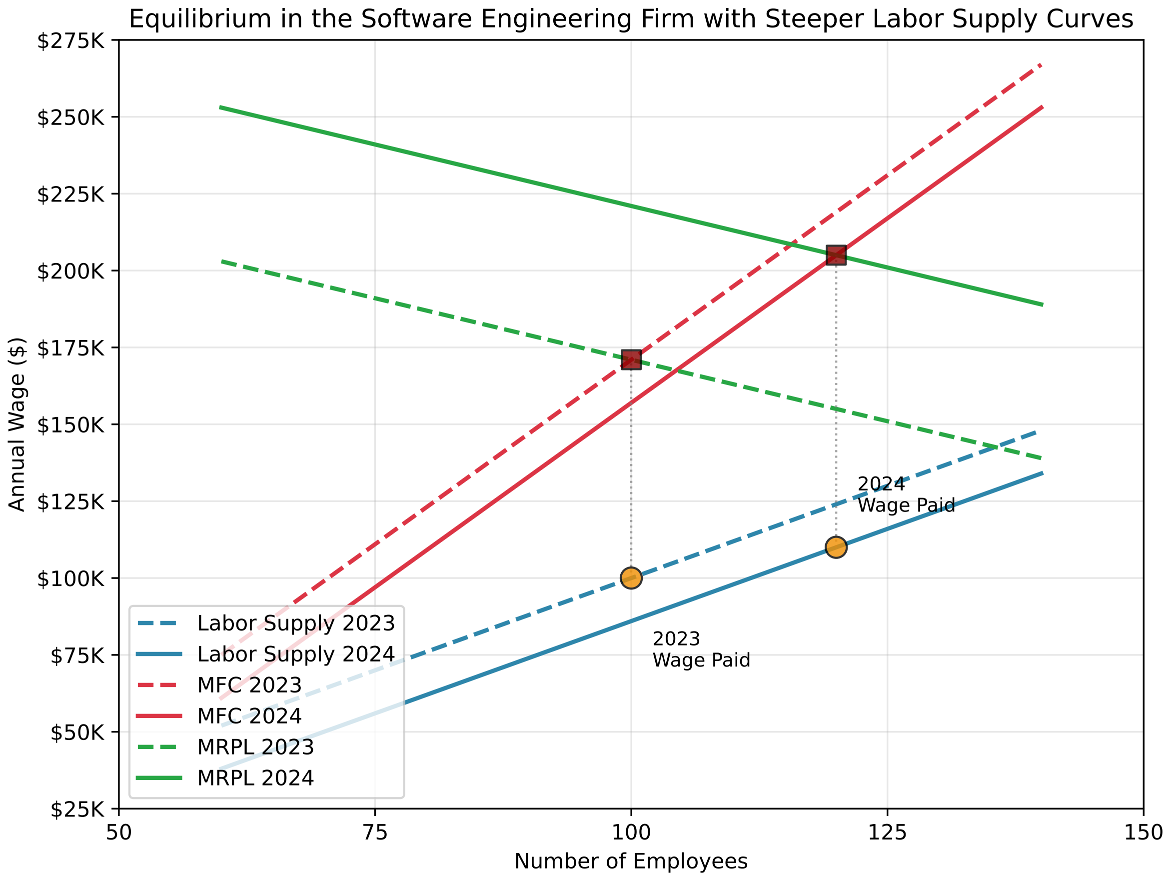


Figure 5.5: Equilibrium in the Software Engineering Firm with Steeper Labor Supply Curves

What has this exercise shown us? From two sets of employment and wages, we can't really draw any strong conclusions about the labor supply curve to the firm. We have drawn three different versions of supply curves all consistent with the data, but which have different implications for the elasticity of labor supply facing the firm. The problem is that when both MRPL and labor supply can shift, then there is no way to conclude what the labor supply curve looks like. This is a very general problem in economics when estimating supply and demand.

So what has the academic literature done to try to estimate the elasticity of labor supply to the firm? Imagine instead of a firm deciding to raise wages, we could experiment with the wages of a firm. For example, we could decrease the wage by 10% and see how many workers leave the firm. Why would this be helpful.

The reason is that now the change in wages is **exogenous** to the firm. We are not changing the wage because of shifts in the marginal revenue product of labor curve or supply curve. We are changing the wage because we are experimenting with the firm. If we did this for many firms, then we could get an average estimate of the elasticity of labor supply to the firm in the economy.

Now, this probably sounds like a pretty unusual thing to do. Will firms allow researchers to experiment with their wages? Probably not. In general, most studies therefore try what are sometimes referred to as quasi-experimental methods. These methods try to find situations where wages change for reasons that are not related to the firm's marginal revenue product of labor or labor supply curve.

Alan Manning's Handbook of Labor Economics chapter (Manning (2011)) on monopsony power provides a good overview of the empirical literature on estimating the elasticity of labor supply to the firm. The chapter discusses several different strategies that have been used to estimate the elasticity of labor supply to the firm. For example, Staiger, Spetz, and Phibbs (2010) use a natural experiment where Veterans Affairs hospitals in the United States changed the way they were paying nurses. Before the experiment, Veterans Affairs hospitals paid nurses based on a fixed salary schedule at the national level. However, in 1991, salaries were switched to being paid based on local labor-market surveys. In effect, this lead to an abrupt increase in the wages of nurses at VA hospitals, which Staiger, Spetz, and Phibbs (2010) use to estimate the elasticity of labor supply to the firm. Another prominent example comes from Falch (2010), who again uses a natural experiment in Norway which increased the wages for teachers in a subset of schools. The authors again used this natural experiment to estimate the elasticity of labor supply to the firm.

There are numerous other examples of natural experiments that have been used to estimate the elasticity of labor supply to the firm. The estimates vary from setting to setting, but in general, the qualitative results are similar: the elasticity of labor supply to the firm is far from infinite. In other words, firms do have some monopsony power in labor markets. A recent survey by Azar and Marinescu (2024) reports that empirical estimates to date suggest that the elasticity of labor supply to the firm falls between 1-4, with an associated wage markdown of approximately 15% to 50%, depending on the setting. This means that firms are paying workers 15% to 50% less than they would in a perfectly competitive labor market. The way these estimates are generally fomulated is by first estimating the elasticity of labor supply to the firm, and then using this estimate to calculate the wage markdown through a theoretical link between the two.

Perhaps the most convincing estimate of a labor supply elasticity to the firm comes from Dal Bó, Finan, and Rossi (2013). In their study, they had the ideal the experiment: truly randomized wages. In August 2011, the government of Mexico held a massive recruitment to hire workers to help with a public goods provision program. In total there were 106 recruitment sites in the study. For each recruitment site, the offered wage was randomized between 5,000 pesos per month and 3,750 pesos per month. This 33% increase in wages led to a 26% increase in the number of applicants, and then conditional on applying, a 35% increase in the probability an individual accepts the job. This implies an elasticity of labor supply to the firm of approximately 2.15, which is quite similar to the range of quasi-experimental estimates discussed above. There are a few additional experimental studies that have now been done. For example, Dube et al. (2020) study Mechanical Turk, a popular online labor market. In this online labor market, the authors are able to randomly vary wages and estimate the elasticity of labor supply to the firm, finding substantial scope for monopsony power. Lastly, Caldwell and Oehlsen (2023) use a field experiment to study the elasticity of labor supply to the firm among Uber drivers. The experiment offered drivers a 10-50% increase in wages, finding an elasticity of labor supply to the firm of approximately 1.5.

5.3 Firm-Specific Pay Premia

In Chapter 2, we considered the case of two cafes. These two cafes had different marginal revenue product of labor curves, but under the assumption of perfect competition, they paid the exact same wage. Why? Because under perfect competition, firms have no control over the wage. If they want any workers, they have to pay the market wage. So what happens in equilbrium? The firm with the higher marginal revenue product of labor hires more workers, and the firm with the lower marginal revenue product of labor hires fewer workers, but they both pay the same wage.

However, this might seem like a somewhat unrealistic assumption. At least anecdotally, you might think that firms pay widely different wages. Do giant firms like Apple and Google pay the same wages as small startups? It doesn't seem like that would be the case. However, in practice, it is a bit more complicated. In our simple example, we assumed all workers were the same. It doesn't matter who you hire, the marginal revenue product of that worker will be the same regardless. But in practice, this may not be the case. Apple may pay higher wages because they are hiring the most-skilled workers.

Abowd, Kramarz, and Margolis (1999) proposed a solution to this problem. To make things concrete, let's consider the case of Google and Apple. Imagine we want to understand whether Google or Apple pays higher wages. If we simply compare the average wage at Google and Apple, we might run into a number of issues. Although they are both tech firms, they may be hiring different types of workers. Maybe Google hires more high-wage engineers relative to Apple, which has more low-wage customer service workers. However, even conditional on the exact job the worker is performing, maybe Google is able to attract the highest-skilled workers. Therefore, even within occupation, maybe there are issues with just comparing across the two firms.

The solution proposed by Abowd, Kramarz, and Margolis (1999) is to compare the wages of workers as they transition across firms. So, for example, we have some workers that move from Apple to Google, and some workers that move from Google to Apple. Let's imagine in the data, that on average, we find that wages decrease by 5% when workers move from Google to Apple, and increase by 5% when workers move from Apple to Google. This suggests that Google pays a 5% wage premium relative to Apple. Now we can't argue that the difference in wages is due to worker quality, because we are only comparing changes in wages for the same worker.

This basic idea was expanded on in Card, Heining, and Kline (2013), who also proposed methods to test the assumptions underlying this approach. Since the publishing of Card, Heining, and Kline (2013), this approach has been used in a number of different countries to understand the importance of firm-specific pay premia. In the U.S., the standard deviation of log pay across firms has been consistently estimated to be around 0.2 (Kline (2024)). What does this mean exactly? Let's plot a normal distribution with a standard deviation of 0.2 and mean zero below. When comparing firm-specific pay premia, we often de-mean the estimates meaning the average firm pay is zero.

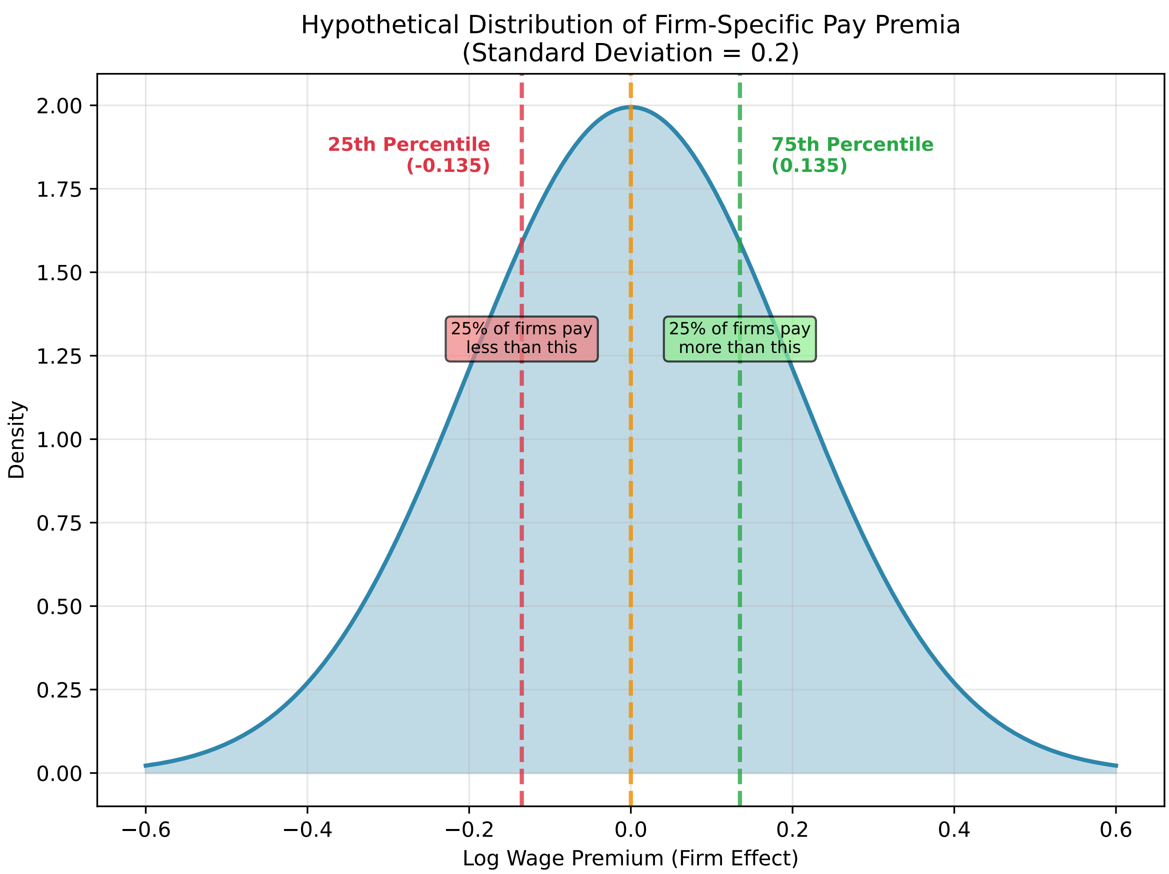


Figure 5.6: Hypothetical Distribution of Firm-Specific Pay Premia

Now if moving from an average firm to the 75th percentile firm raises your log wage by 0.135, this implies that your wage increases by approximately 14.3%. If moving to the 25th percentile firm decreases your log wage by 0.135, this implies that your wage decreases by approximately 14.5% ($exp(0.135) - 1 = 0.145$). This is a substantial difference in wages! Remember, this is a difference in wages for the same worker. So we can't necessarily attribute this to differences in the background skills of the workers.

This type of variation is difficult to reconcile with the perfectly competitive model. In a perfectly competitive model, why would there be firms that pay higher wages for the exact same worker? However, in a monopsonistic model, this is what we would expect. Places need to pay higher wages to attract more or better workers, giving rise to variation in wages across firms.

5.4 Institutional Boundaries to Mobility

Remember, why does monopsony power arise? Because workers are not perfectly mobile across firms. They don't just leave the firm the minute wages fall behind the market wage. In this section, we will discuss some institutional boundaries to mobility that can lead to monopsony power in labor markets.

In the United States, some workers are required to sign non-compete agreements. There is a sound economic reason for having non-compete agreements in certain settings. For example, imagine there is a biotechnology firm that is aiming to patent a new drug. They need to recruit scientists to work on this. They need to invest in machinery. They may need to train the scientists to use this equipment.

If they fear that a worker can leave the firm and immediately start working for a competitor, they may not find that it is worth investing in the first place. Non-compete agreements can help alleviate this hold-up problem. If the worker is not allowed to work for a competitor, then the firm can invest in training and equipment, knowing that the worker will not immediately leave and start working for a competitor.

However, Starr, Prescott, and Bishara (2021) find some striking findings about the prevalence of non-compete agreements in the United States using a survey of workers. While non-competes are generally discussed as most relevant for high-skilled workers, they find that for workers with high-school education or less, 20% report having signed a non-compete agreement. Non-compete agreements are non-negligible in industries with little theoretical justification for them. They are even prevalent in states in which they are illegal. For example, California and North Dakota both do no enforce non-compete agreements, but about 19% of the surveyed workers in these states report having signed a non-compete agreement.

One way to rationalize the prevalence of non-compete agreements is that they are also being used to restrict labor mobility. If a worker is not allowed to work for a competitor, then they will be less responsive to changes in wages at their own firm. This is one reason that the Federal Trade Commission proposed making the majority of non-compete agreements illegal in 2024.

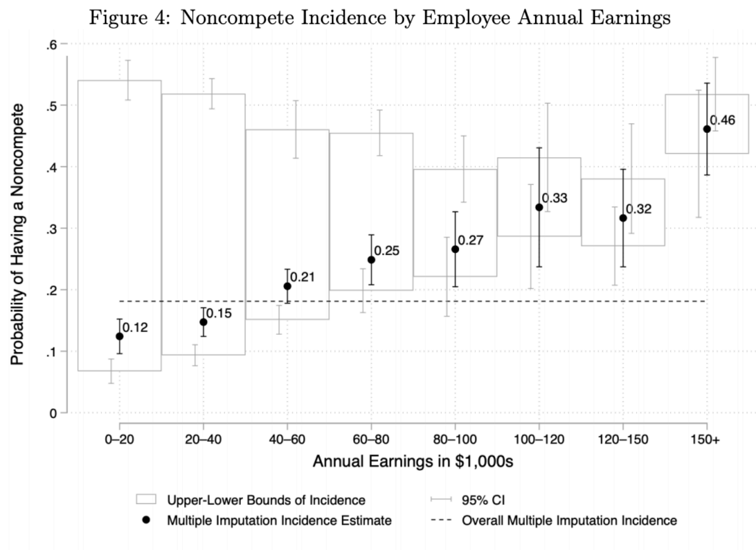


Figure 5.7: Share of Workers with Non-Compete Agreements by Education

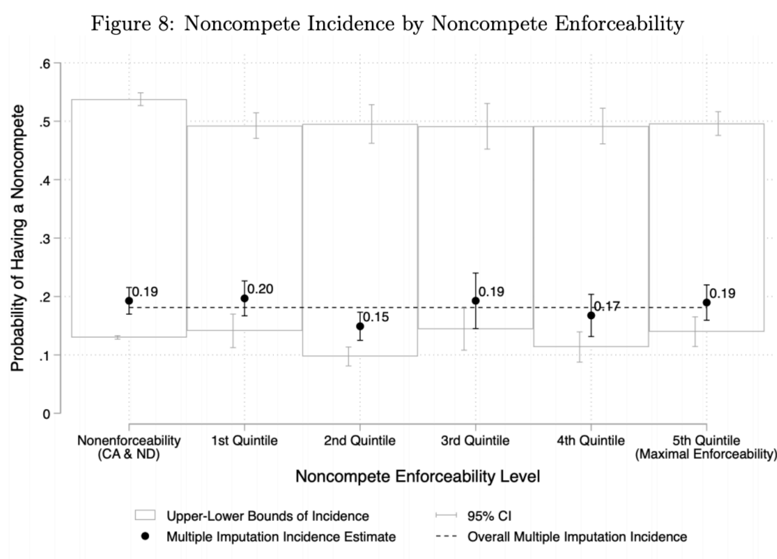


Figure 5.8: Share of Workers with Non-Compete Agreements by Industry

However, this proposal was blocked by federal courts in 2024, arguing the FTC exceeded its authority by proposing the ban on noncompetes.

5.5 Concentration in Labor Markets

We started Chapter 4 with a quotation from Adam Smith about collusion arising in settings with few employers. For a long time, economists have believed that imperfect competition is more likely to arise in settings with few firms: both on the labor-market side and the product-market side. This could happen through explicit collusion, as Adam Smith suggested, or through other channels that lead to firms having market power.

In the 1950s-1960s, economists started coming up with measures that quantified the idea of "small number of firms". These are referred to as concentration measures. One of the most influential measures is the Herfindahl-Hirschman Index (HHI), named after economists Orris C. Herfindahl and Albert O. Hirschman. These economists independently developed a measure aimed at measuring concentration. To compute the measure, you take the market share of each firm, square it, and then sum up all of these market shares.

To take a concrete example, let's consider the US phone market. For now, we will take the traditional approach and compute the HHI based on market shares in the product market. We are going to consider the market as the three largest firms: Verizon, T-Mobile and AT&T. In practice, one should always include all firms in the market, focusing on just three is for simplicity. In the telecommunications market, these three have over 90% of the market, so our calculation will be close to correct. Among these three firms, the market share of Verizon is about 37%, T-Mobile is about 33%, AT&T is about 30%. Therefore, the HHI is computed as follows:

$$\text{HHI} = 0.37^2 + 0.33^2 + 0.30^2 = 0.1369 + 0.1089 + 0.09 = 0.3358$$

Next, let's consider the two extreme cases of HHI. If there is only one firm in the market, then the HHI will be 1 because the market share of that firm is 100%, and $1^2 = 1$. Now let's assume there are many, many firms. Let's say there are 10,000 firms, each with a market share of 0.0001 (or 0.01%). In this case, the HHI will be:

$$\text{HHI} = 10,000 \times (0.0001)^2 = 10,000 \times 0.00000001 = 0.0001$$

In other words, basically zero. So the two extreme cases are between 0 and 1. Now, that we understand the procedure to compute the HHI, let's ask a follow-up question. Why does it make sense to measure concentration in this way?

The intution can be seen in the following example. Imagine there are three firms, but two have small shares (let's say 0.1 each) and one has a large share (0.8). In this case, the HHI will be:

$$\text{HHI} = 0.1^2 + 0.1^2 + 0.8^2 = 0.66$$

Next, let's consider the case where there are three firms, but they all have equal shares (0.33 each). In this case, the HHI will be:

$$\text{HHI} = 0.33^2 + 0.33^2 + 0.33^2 = 0.3267$$

So the HHI is higher when there are two firms with small shares and one firm with a large share, compared to the case where all three firms have equal shares. This intuitively seems like it would make sense. If you have three larger competitors, they may be more likely to compete fiercely with each other. If there is one large dominant firm, and two smaller firms, then it seems like the large firm may be more able to dominate the market and exercise market power. By squaring the market shares, the HHI gives more weight to larger firms.

However, it is just one measure of concentration that has been proposed. There are other measures of concentration that have been proposed, some of which are more theoretically driven. One reason the HHI is so widely studied, however, is due to its practical use in antitrust policy.

In the United States, the Department of Justice (DOJ) and Federal Trade Commission (FTC) use the HHI to assess market concentration when evaluating mergers and acquisitions. The DOJ and FTC have established thresholds for the HHI to determine whether a merger is likely to substantially lessen competition. In the past, these thresholds and the use of HHI generally have been applied to product markets. However, they can also apply to labor markets.

To test this, Azar, Marinescu, and Steinbaum (2022) measured HHI in labor markets across the United States. They defined a labor market as an occupation in a given commuting zone (a commuting zone is a geographic region defined by the U.S. Census Bureau that groups together counties based on commuting patterns). They then computed the HHI for each labor market, using the share of employment in each occupation in a given commuting zone. What they find is that once you look within a given region and occupation, labor markets are actually quite concentrated. The figure below shows the average HHI within different commuting zones in the United States (the average is taken over the occupations in the commuting zone).

Further, they find evidence that labor markets that experience increases in concetration experience declines in wages. This provides empirical evidence that monopsony power is higher in more concentrated labor markets.

One concern with using measures of concentration is that they may be associated with changes to the marginal revenue product of labor curve. For example, imagine there is increased foreign competition that means the price in an industry is plummeting. Since the price is lower, that

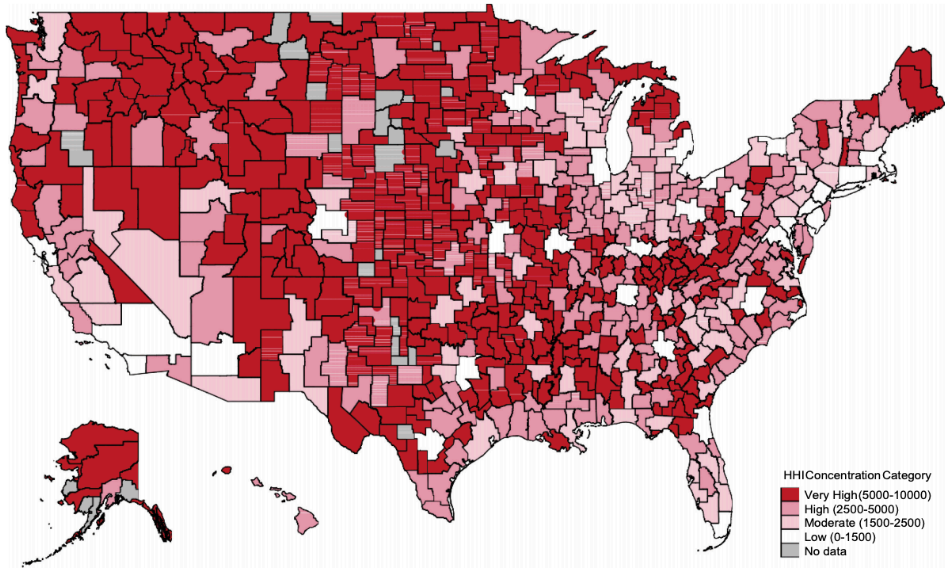


Figure 5.9: Concentration in US Labor Markets

means the marginal revenue brought by each additional worker is lower. If the decline in prices also leads firms to exit, then there will be fewer firms in the market, leading to a higher HHI.

One way to assess this concern is to look for events where this story is potentially less of a concern. One such event is perhaps a merger and acquisition (M&A). From a policy standpoint, these are the concentration-increasing events that are most relevant, as the DOJ and FTC can potentially block mergers that increase labor-market concentration. Therefore, it would be useful to understand whether mergers and acquisitions that generate concentration increases lead to declines in wages.

Prager and Schmitt (2021) and Arnold (2025) are two papers that study this question. Prager and Schmitt (2021) study the effects of mergers and acquisitions in the U.S. hospital industry, while Arnold (2025) study the effects of mergers and acquisitions across a wider range of industries. Both studies find qualitatively similar results.

For example, Arnold (2025) distinguishes between three different types of mergers. Low-impact mergers occur between two firms that are operating in different areas. Because labor-market competition is likely local in nature, these mergers have no impact on monopsony power. Indeed, Arnold (2025) finds a miniscule 0.5 percent decline in earnings in these types of events. In mergers that do increase concentration, the impact depends on the baseline level of concentration. For markets with relatively low levels of concentration, there is again a small and statistically insignificant decline in earnings of about 0.8 percent. Again, these medium-impact mergers likely have no impact on monopsony power, because these are not concentrated markets to begin with. However, in the high-impact mergers, which occur in high-concnetration markets, Arnold (2025) finds a decline in earnings of about 3.1 percent.

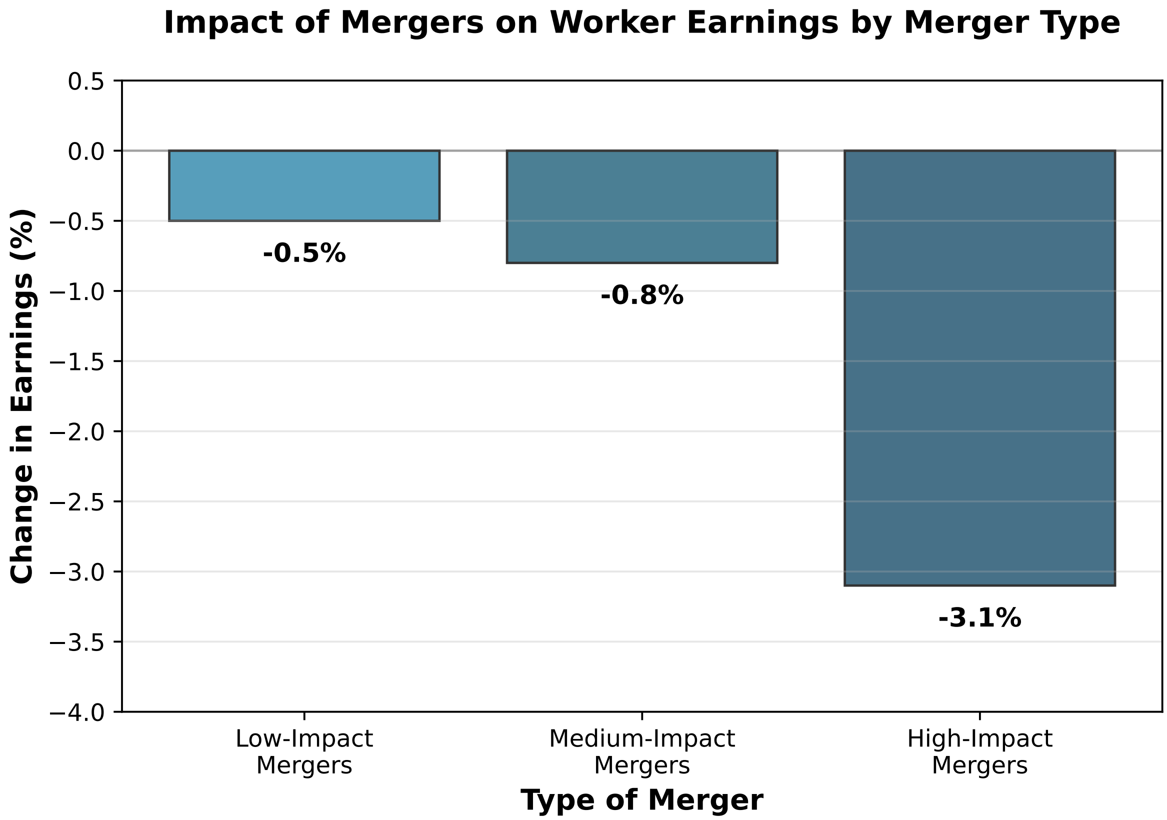


Figure 5.10: Impact of Different Types of Mergers on Worker Earnings

The fact that these large mergers lead to wage losses is a potential concern for antitrust authorities. However, traditionally antitrust authorities have been solely focused on product-market competition. This is changing. A recent complaint against Kroger and Albertsons, two large grocery chains in the United States, alleged that the merger would not only increase grocery prices, but also lower wages for grocery store workers. This is one of the first examples of labor-market concerns making their way into merger enforcement.

While the antitrust authorities are starting to consider labor markets in relation to mergers, there is a bit more history considering collusion in labor markets. Collusion is a notoriously difficult concept to study in economics. By its nature, firms have a strong incentive to keep it hidden. The industrial organization literature in economics has spent a lot of effort trying to detect collusion in product markets.

However, in some cases, antitrust authorities have uncovered collusion, giving economists an opportunity to study it empirically. This includes collusion in labor markets. One of the most famous cases of collusion in labor markets is the case of Silicon Valley tech firms. In 2010, the U.S. Department of Justice (DOJ) launched an investigation into several major tech firms, including Apple, Google, Intel, and Adobe, for allegedly colluding to suppress wages for tech workers. The investigation revealed that these firms had entered into "no-poach" agreements, where they agreed not to recruit or hire each other's employees. After the high-tech investigation, there was another similar case involving animation studios, including many you have likely heard about: Disney, Pixar, Lucasfilm, Blue Sky, and DreamWorks. These studios allegedly agreed not to poach each other's employees, which again suppressed wages for animators. The interesting thing is that by the time the lawsuit was settled, many of these animation studios had already been bought by Disney.

5.6 Monopsony and Inequality

As we discussed in Chapter 1, a large body of research in economics has been focused on what drove the rise in labor-market inequality in the United States. In this section, we will discuss how monopsony power can help explain the changes in labor-market inequality in the United States. To recap, since the 1980s, there has been a large increase in wage inequality in the United States. One way we discussed how to measure this is by looking at the ratio of the 90th percentile wage to the 10th percentile wage. The logic of this is the 90th percentile is a measure of how rich those with the highest incomes are (if you are above the 90th percentile, you are richer than 90% of the population), while the 10th percentile is a measure of how poor those with the lowest incomes are (if you are below the 10th percentile, you are poorer than 90% of the population). The ratio of these two numbers therefore gives a measure of inequality.

Sometimes, researchers instead study the 90/50 ratio and the 50/10 ratio. The 90/50 ratio is a measure of inequality among relatively higher-wage workers. It studies whether the richest workers are pulling away from the median worker. The 50/10 ratio is a measure of inequality among relatively lower-wage workers. It tells us whether the lowest-wage workers are falling behind the median worker.

Sometimes, instead of reporting the ratio, researchers will report the log of the ratio. For example, if the 90th percentile wage is \$150,000 and the 10th percentile wage is \$50,000, then the 90/50 ratio is 3. The log of this ratio is $\log(3) = 1.098$ (we are using the natural logarithm here).

So why do we take the log of the ratio? One is that small changes in the log ratio correspond to percent changes in the ratio. For example, imagine the 90th percentile wage increases to \$160,000 while the 10th percentile wage remains at \$50,000. The new 90/10 ratio is 3.2, and the log of this ratio is $\log(3.2) = 1.163$. The change in the log ratio is $1.163 - 1.098 = 0.065$, which corresponds to a percent change of about 6.5% in the original ratio (this is an approximation, but it is close). While we do not cover regression analysis in this book, using logs is very common in regression analysis in economics, as it allows one to interpret coefficients as elasticities. As you have probably noticed, elasticities come up a lot in economic theory.

For our purposes, one nice feature of the log ratio is that it is additive. This will allow us to clearly decompose changes in the 90/10 ratio into changes in the 90/50 ratio and the 50/10 ratio. We can use this information to understand how inequality has been changing over time in the United States. First, to see why this true, the log of the 90/10 ratio can be written as:

$$\log\left(\frac{W_{90}}{W_{10}}\right) = \log(W_{90}) - \log(W_{10})$$

This is using a property of logarithms: $\log(A/B) = \log(A) - \log(B)$. Next, we can add and subtract the log of the median wage (W_{50}) to the right-hand side:

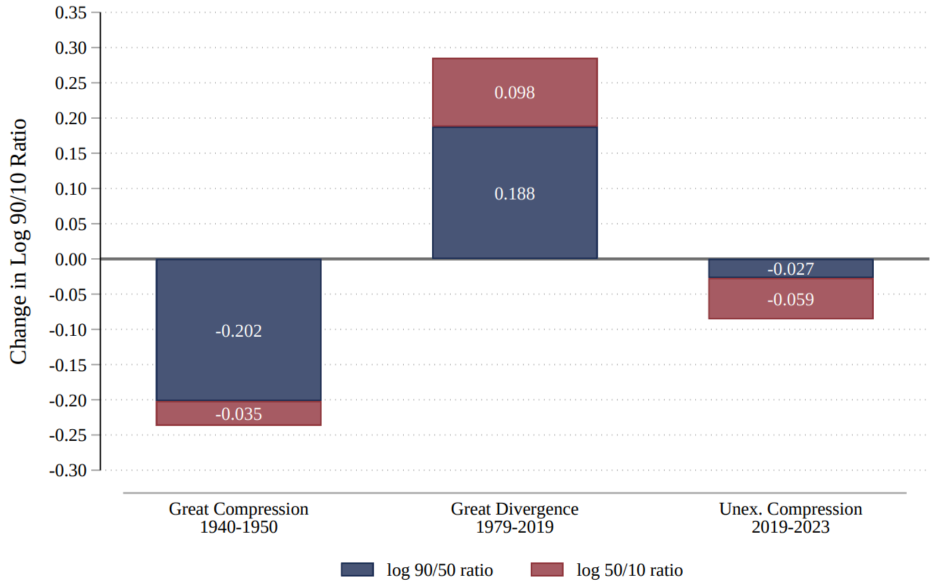
$$\log\left(\frac{W_{90}}{W_{10}}\right) = \underbrace{\log(W_{90}) - \log(W_{50})}_{\text{Upper Tail Inequality}} + \underbrace{\log(W_{50}) - \log(W_{10})}_{\text{Lower Tail Inequality}}$$

This equality shows that we can decompose the log of the 90/10 ratio into two parts: the log of the 90/50 ratio (which measures inequality among higher-wage workers) and the log of the 50/10 ratio (which measures inequality among lower-wage workers).

This decomposition is useful for understanding the dynamics of wage inequality. For example, if the log of the 90/50 ratio is increasing while the log of the 50/10 ratio is stable, it suggests that inequality is rising among higher-wage workers without a corresponding increase in inequality among lower-wage workers. For example, one claim is that rising CEO or more generally, manager pay, is a driver of inequality. However, if we see that the 50/10 ratio is also increasing, then this probably isn't the only story going on. We would need a story about why the lowest-wage workers are also falling behind the median worker.

In D. Autor, Dube, and McGrew (2023), the authors plot this decomposition for three different time periods. The first they refer to as the great compression (1940-1950). We found similar results in Chapter 1. During this period, inequality was falling. The majority of this decline in inequality was driven by a decline in the log(90/50) ratio. In other words, median workers were catching up to higher-wage workers. This is the classic story of the middle class growing in the post-war period.

Figure 2: Change in Log 90/10 Ratio during Three Episodes of Wage Structure Change



The second period is the great divergence (1979-2019). During this period, inequality was rising. We have already talked about this a bit in Chapter 3, arguing at least a portion of it was due to the falling real value of the minimum wage. However, we can also see that the log(90/50) ratio was rising. In other words, inequality was rising both among higher-wage workers and lower-wage workers during this period.

Now, for a long time, all we saw in the data was increasing inequality. But perhaps surprisingly, since 2019, in particular after COVID, inequality has been falling. In this short period between 2019 and 2023, the log(90/10) ratio has fallen by 0.086 (= -0.059 - 0.027). To compare to the great divergence, inequality rose by 0.282.

To put some of these numbers in perspective, let's assume for illustrative purposes that the 10th percentile wage in 1980 (adjusting for inflation) was \$10,000. The 90th percentile (adjusting for inflation) was \$100,000. These are just approximate numbers to get a sense of the math, but the real 90/10 ratio is not far off.

Now if the 90/10 ratio is 10, then the log of the 90/10 ratio is $\log(10) = 2.302$. If the $\log(90/10)$ ratio rose by 0.282 during the great divergence, then the new $\log(90/10)$ ratio would be 2.584. If we are to convert that two a ratio of wages, we would get: $exp(2.584) = 13.23$. exp is the inverse log function. So $exp(log(X)) = 1$, where X is some positive number. Therefore, if the 10th percentile wage remained at \$10,000, the 90th percentile wage would have risen to \$132,300. So workers at the 90th percentile are earning 132% more than workers at the 10th percentile, while previously they were earning 100% more.

Now let's consider what happens during the unexpected compression. The log(90/10) ratio fell by 0.086. Therefore, the new log(90/10) ratio would be 2.584 - 0.086 = 2.498. Converting this back to a ratio, we get $exp(2.498) = 12.15$. Therefore, if the 10th percentile wage remained at \$10,000, the 90th percentile wage would have fallen to \$121,500. So workers at the 90th percentile are now earning 121.5% more than workers at the 10th percentile, down from 132% more.

This is a substantial change! The great divergence took place over 40 years, while the unexpected compression took place over just 4 years. Still, even in this short amount of time, we have seen a substantial change in inequality. This isn't actually unprecedented. In the 1940s and 1950s, inequality fell quite rapidly as well. But it is perhaps surprising. The post-war period was a time of rapid economic growth, while the post-COVID period has been a time of economic uncertainty.

So what is driving this unexpected compression? The decomposition shows that the majority of the decline in inequality has been driven by a decline in the log(50/10) ratio. In other words, lower-wage workers have been catching up to median workers. D. Autor, Dube, and McGrew (2023) argue that the key factor in this unexpected compression has been an increase in competition in the low-wage labor market.

Hypothetically, this could be explained in a perfectly competitive model. For whatever reason, there has been an increase in the demand for low-wage workers. This has led to an increase in the wages for these workers, thereby decreasing inequality.

However, D. Autor, Dube, and McGrew (2023) argue that this is unlikely to be the full story. Instead, they argue that monopsony power is key in explaining this compression. The argument is that during this time, as demand for low-wage workers has increased, workers became more sensitive to wage differences across firms. In other words, if your employer was paying slightly lower than another employer, in this period, workers were much more likely to quit and move jobs. This increase in mobility has decreased the monopsony power of firms, leading to higher wages for low-wage workers. The perfectly competitive model cannot explain this, as in a perfectly competitive model, workers are always perfectly mobile across firms.

In this setting, the monopsony model offers a richer, more realistic explanation for the observed changes in inequality. What they find is that after the pandemic, low-wage workers became more sensitive to wage differences across firms. Remember, our key assumption in the monopsony model is that firms face an upward sloping labor supply curve. If they can decrease their wage a lot, but only lose a few workers, this implies they have significant monopsony power. However, if they can decrease their wage a little and lose a lot of workers, then they have less monopsony power. D. Autor, Dube, and McGrew (2023) find that after the pandemic, low-wage workers became more sensitive to wage differences across firms, implying that firms monopsony power decreased. Therefore, the post-pandemic labor market for low-wage workers had two important mechanisms at play: an increase in demand for low-wage workers and a decrease in monopsony power. These two forces led to a rapid rise in wages for low-wage workers, thereby decreasing inequality.

6 Human Capital

6.1 Human Capital

So far in this book we have generally considered workers skills as fixed. But individuals invest a lot into their skills. We call the accumulation of an individual's skills their human capital.

Human capital has always played a key role in Economics. One of the earliest discussions comes from Adam Smith, who recognized that investments in education, training, and health improve worker productivity. Modern growth theory also emphasizes the importance of human capital in explaining differences in income across countries. Below we consider data that was made available by Gapminder, an organization that collects data from around the world to make cross-country comparisons.

In Figure 6.1 we plot on the vertical axis the GDP per capita of a country. This is a common measure of the average wealth of a country. A higher GDP per capita implies the country produces more (in dollar terms) per person relative a lower GDP per capita country. Note that this scale is on the logarithmic scale. That is, each tick mark is double the previous tick mark. On the horizontal axis we plot the mean years of schooling for men 25 years and older. Figure 6.2 plots the same variables, but for women 25 years and older. The size of the bubble indicates the population of the country. The color indicates the region with blue indicating Africa, red indicating Asia, yellow indicating Europe, and green indicating the Americas.

The patterns in both of these figures is clear. Places with higher levels of schooling tend to have higher GDP per Capita. Of course we can't explain all of the variation with schooling alone, but the patterns are striking.

Schooling is often conceptualized as the primary way to increase the human capital of a citizenry. Since it is so tightly linked to growth across countries, it has been a key focus of economic policy for centuries. The importance of human capital has been understood for a very long time in Economic theory. For example, Alfred Marshall in his seminal work *Principles of Economics*, states "The most valuable of all capital is that invested in human beings."

What the Gapminder data allows us to do is show that countries that invest in education tend to be wealthier. What we can also ask is: "within a country, do individuals that invest more in education earn more as well?"

Gary Becker in his 1964 book *Human Capital* outlined the individual investment decision. He conceptualized human capital in the same way previous work had conceptualized investment

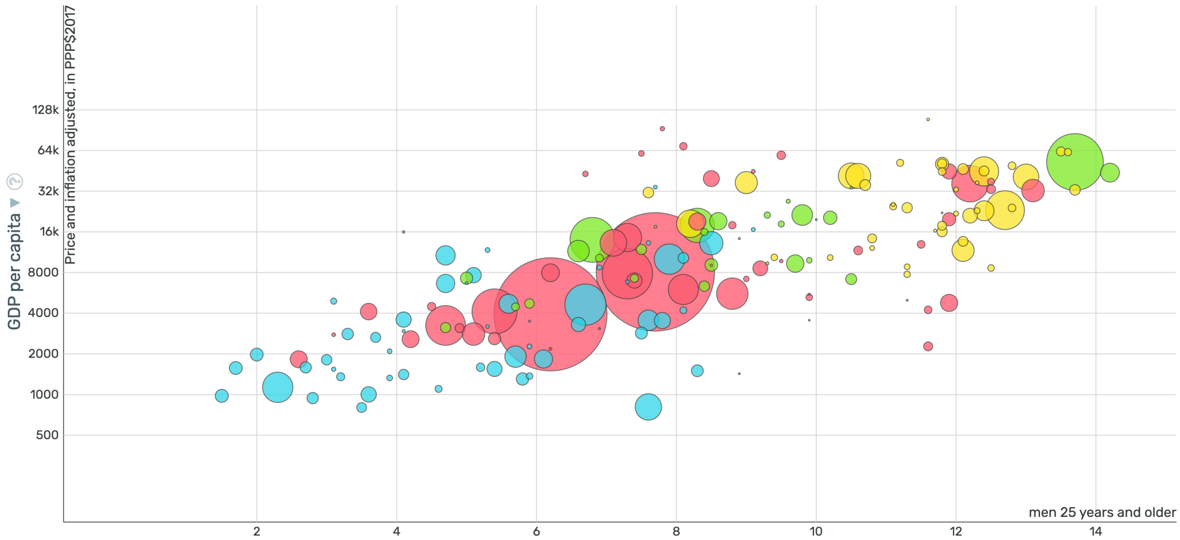


Figure 6.1: GDP Per Capita and Mean Schooling (Men)

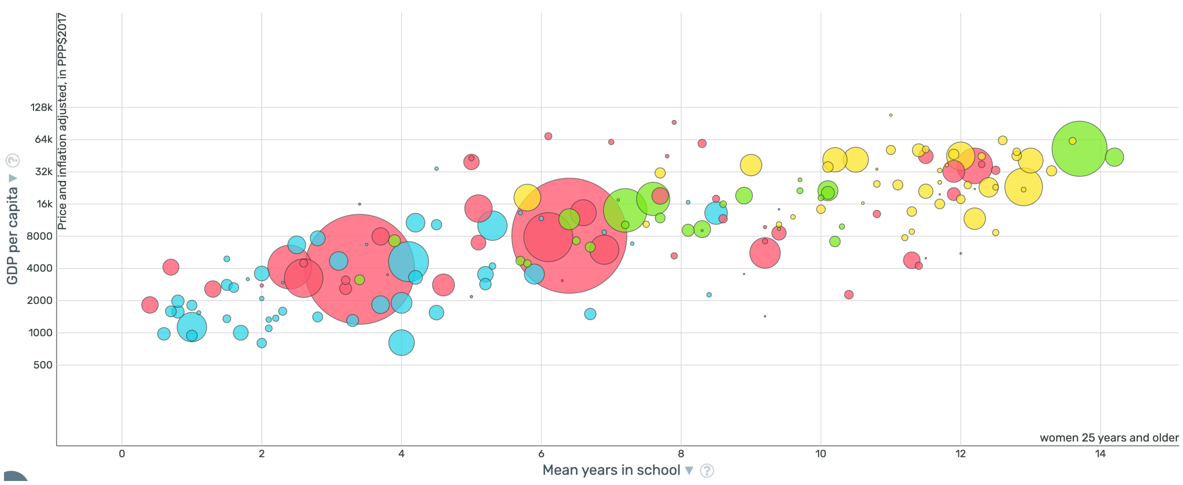


Figure 6.2: GDP Per Capita and Mean Schooling (Women)

in physical capital (i.e. better machines and technology.) The overall structure of the problem is actually relatively simple.

The investment decision for human capital is similar to other investment decisions. To be concrete, let's consider the decision to attend college or not. What does the decision depend on? Well, it depends first on how much you think this will increase your income in future periods. Let Y_t^1 be the income you would earn in period t if you attend college. Let Y_t^0 be the income you would earn in period t if you don't attend college. For example, let's start time at age 18 so $t = 0$ when a person is 18 years old. If the person is in college, then they are likely paying some amount for tuition and living expenses, and earning little current income. If they are not in college, they don't need to pay tuition and may be able to earn more by working in a full-time job. So initially, $Y_1^1 < Y_0^1$.

Of course we don't expect this to always be the case. Let's move forward 10 years, after you have graduated college. We will assume that college has given you training and opened doors that would have otherwise remained shut. So we expect $Y_{10}^1 > Y_{10}^0$. So how do we decide whether to invest in college or not? In one scenario, you start with lower amounts of income, but eventually gain more income in later periods. Becker suggests analyzing this decision in the same way as firms analyze investment decisions. This is done by computing the **net present value** of the investment.

The net present value (NPV) of the education decision can be calculated as:

$$NPV = \sum_{t=0}^{T} \frac{Y_t^1 - Y_t^0}{(1+r)^t}$$

where:

- Y_t^1 = income in period t with college education
- Y_t^0 = income in period t without college education

- r = discount rate (interest rate)

If $NPV > 0$, then the investment in education is worthwhile from a financial perspective. This is the exact same formulation as a firm would make in deciding whether to buy a new machine or not.

Now, why is the discount rate in the equation? The idea is that income today is more valuable than income tomorrow. Why? Because if you have income today, then you can invest it and get a rate of return equal to r. For example, if you have 100 dollars today, the interest rate is 5 percent, then you will have 105 dollars tomorrow. Therefore, having 100 dollars today is equivalent to having 105 dollars in a year from now.

To put it another way, say in a year from now, I have 250 dollars. How much money would I need to have today so that I can ensure that I have 250 dollars in a year? Well, this depends on the interest rate. If the interest rate is 5 percent, then if I have $\frac{250}{1.05} \approx 238$, I could put that money in the bank, collect my interest, and have 250 a year from now.

To see clearly how this equation works, let's take a more standard example of a physical investment. Imagine you are considering buying a machine for your factory. If you buy the machine, you expect your profits will be zero in the current year (because you need to pay for the machine which is expensive), but then for the next two years you expect your profits to be 100 dollars per year. If you don't buy the machine, you can continue business as usual. You expect to earn 20 dollars for the next three years. To keep this simple, let's assume we are only considering the next three years, after which you plan to shut down the business.

The question is, what is more valuable, an income stream of (0, 100, 100) or an income stream of (20, 20, 20). To answer this, we need to know the interest rate. Let's assume the interest rate is 10 percent. To compute the NPV of buying the machine, we calculate the sum total of the discounted income streams:

For buying the machine, I receive zero today, 100 a year from now, and 100 two years from now. The value of a 100 dollars is equivalent to $\frac{100}{1.1} \approx 90.91$ today. Why? Because if I had 90.91 today and invested it at 10 percent, in a year I would have 100 dollars. The value of 100 dollars two years from now is equivalent to $\frac{100}{1.1^2} \approx 82.64$ today. Again, if I had 82.64 today and invested it at 10 percent, in one year I would have 90.91. If I invest that again at 10 percent, in two years I would have 100 dollars. Therefore, the total NPV of buying the machine is:

$$NPV_{buy} = \frac{0}{1.1^0} + \frac{100}{1.1^1} + \frac{100}{1.1^2} \approx 0 + 90.91 + 82.64 = 173.55$$

Now we can do the exact same thing for the other income stream. If I don't buy the machine, I receive 20 dollars today, 20 dollars a year from now, and 20 dollars two years from now. The NPV of this income stream is:

$$NPV_{no\_buy} = \frac{20}{1.1^0} + \frac{20}{1.1^1} + \frac{20}{1.1^2} \approx 20 + 18.18 + 16.53 = 54.71$$

So clearly in this case it is more valuable to buy the machine. Now, instead of conceptualizing this decision as buying a machine, Becker conceptualizes this as investing in human capital. The same logic applies. You give up some income today (paying tuition and not working full-time) in exchange for higher income in the future. If the NPV of investing in education is greater than the NPV of not investing, then it is worthwhile to invest in education.

For our purposes, we are going to focus much more on the term $Y_t^1 - Y_t^0$ in Becker's theory, rather than studying how choices vary by discount rates. For much of the discussion, we won't actually need to keep track of the time period. Of course the impact of education on your earnings may vary over the career. For now let's just consider the average across all periods as: $Y^1 - Y^0$. In other words, we will think about $Y^1 - Y^0$ as the average increase in income from attending college.

An enormous literature in labor economics has sought to estimate $Y^1 - Y^0$. It is notoriously difficult as we will discuss in the next section. The key issue is that what we are conceptualizing is a world in which you go to college vs. a world in which you don't. But for any individual, they can only do one of those two things. So when we think about the data, we will need to compare across people who go to college vs. people who do not.

But that again raises the fundamental issue of identification: what if people who attend college are different than people who do not attend college in unobservable ways? Then, how can we be sure education is driving income differences?

Lastly, this framework does not leave a lot of room for different types of education. If you get certain degrees you may get different future income streams. We will also consider returns to different college majors to see if the type of education makes a difference.

6.2 Mincer Equation

One of the most famous uses of Gary Becker's theory on human capital investment is the Mincer Equation, derived by Jacob Mincer. The goal of the Mincer Equation is to try and model the earnings of workers using a relatively simple list of variables which are inspired by Gary Becker's theoretical formulation (which in practice is much more involved than what was presented in the previous section). The Mincer Equation is presented below:

$$\ln(w) = \alpha + \beta_1 S + \beta_2 E + \beta_3 E^2 + \varepsilon$$

Where:

- w = earnings (wage or income)
- S = years of schooling/education
- E = years of work experience (often approximated as age - schooling - 6)
- α = intercept term
- β_1 = return to education (approximately the percent increase in earnings per additional year of schooling)
- β_2, β_3 = parameters capturing the relationship between experience and earnings
- ε = error term

This is probably one of the most famous and analyzed equations in all of labor economics. Our main focus will be in understanding the parameter β_1, which is the return to an additional year of schooling. This has been a parameter that researchers have long been interested in. Before we address this parameter, let's first understand the other parameters.

The Mincer Equation has two main variables that are used to explain wages: schooling and experience. Both of these reflect investments in human capital. The basic idea is that the more schooling and individual has, the higher their productivity will be, yielding higher earnings. However, individuals also accumulate human capital through work experience.

Work experience is often modeled as a quadratic. The basic idea is that initially, you learn a lot in your new job. These new schools will allow you to be more productive and earn more. But we don't think this return to experience continues forever. Eventually, you stop accumulating new skills on your job and so the return to experience starts to decline. Having a squared term allows us to capture this relationship.

To see this in action, let's ignore schooling for a moment and just consider experience. The equation becomes:

$$\ln(w) = \alpha + \beta_2 E + \beta_3 E^2 + \varepsilon$$

Let's choose some arbitrary values for the parameters and plot this relationship. We'll set $\alpha = 10$, $\beta_2 = 0.02$, and $\beta_3 = -0.0005$. Now we can plot the relationship between earnings and experience.

The Mincer Equation assumes a shape for how earnings change with respect to experience. But we don't need to necessarily believe this shape is correct. We can look in the data and see the shape for ourselves.

To do this, we are going to use data from the American Community Survey (ACS). Now, for this particular graph, we could have also used the Current Population Survey (CPS) data. The CPS and ACS are both surveys run by the U.S. Census Bureau that collect information on individuals' demographics, employment, and income. The CPS is conducted monthly and is primarily used to measure labor force statistics, while the ACS is conducted annually and provides more detailed information on social, economic, housing, and demographic characteristics. Additionally, the sample size for the ACS is larger than the CPS. While the two overlap considerably, each has its own strengths and weaknesses.

In the plots below, we will show the relationship between experience and two different measures of earnings using ACS data from 2023. The ACS does not actually keep track of a person's work experience. To construct this variable, we are going to use a common rule-of-thumb:

$$Experience = Age - YearsOfSchooling - 6$$

This assumes that a person starts school at age 6 and goes straight through school without any breaks. While this is not perfect, it is a common approximation used in the literature.

In Figure 6.3, we show the relationship between experience and annual wage income. However, the Mincer Equation we used above modeled log wages, therefore in Figure 6.4 we show the relationship between experience and log annual wage income. We plotted this for experience values between 0 and 40 years. Once you get to very high years of experience, the sample size starts to get quite small, so the estimates become noisy. The size of the markers in this graph are proportional to the number of observations. As you can see, they are all roughly the same size, indicating we have a similar sample size for various experience levels.

In both of these figures, we see rapid earnings growth at the beginning of a person's career, which eventually slows down. This type of relationship is consistent with Mincer's equation.

Now the key parameter we are actually interested in is the return to schooling, β_1. This parameter tells us how much an additional year of schooling increases a person's earnings. So let's next turn to this parameter. Again in the ACS we can study how earnings depend on the number of years of schooling.

As in the experience figures, the size of the dots is proportional to the number of individuals in the sample with that level of schooling. In the U.S. there are compulsory schooling laws that mandate children stay in school until a certain age, which can vary across state from a minimum of 16 to a maximum of 18. This may be one reason that we have very few individuals with low levels of schooling. Until we hit 12 years of schooling, we have very few observations.

If we consider the part of the figure where we have ample data (12 or more years of schooling), we can see a stark increase in earnings for each additional year of school.

Now let's get to an estimate of the Mincer Equation. Generally, this equation would be estimated by Ordinary Least Squares (OLS). This chooses the coefficients β_1, β_2 and β_3 that best fit the data. A key focus of your econometrics courses will be to understand what the

Figure 6.3: Experience and Annual Wage Income

Figure 6.4: Experience and Log Annual Wage Income

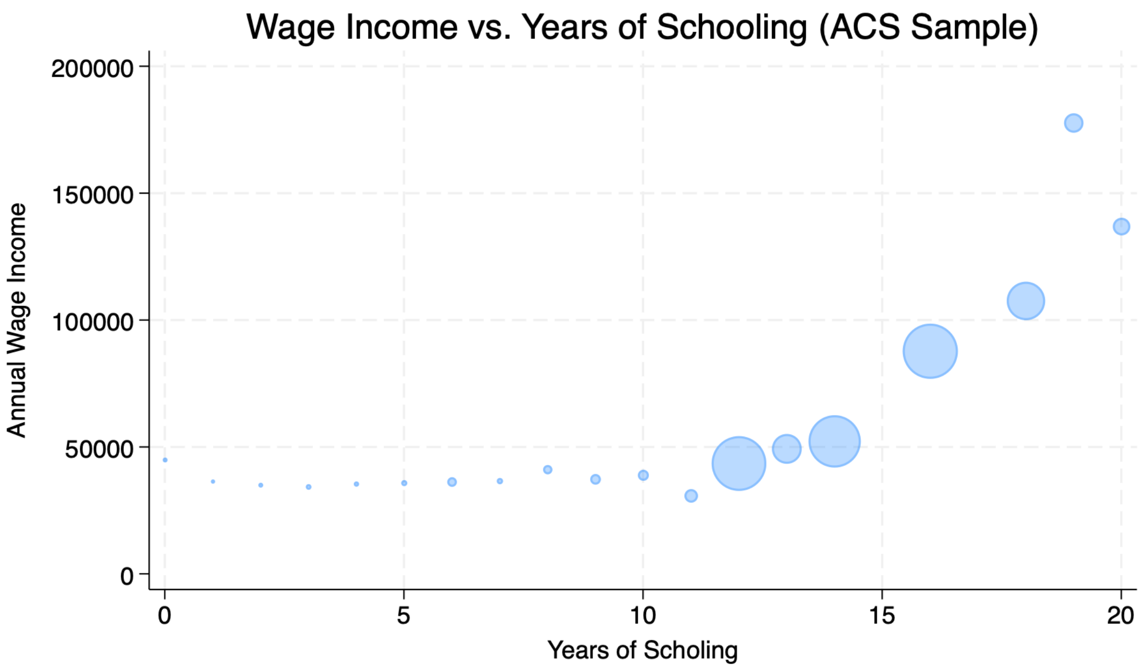


Figure 6.5: Years of Schooling and Annual Wage Income

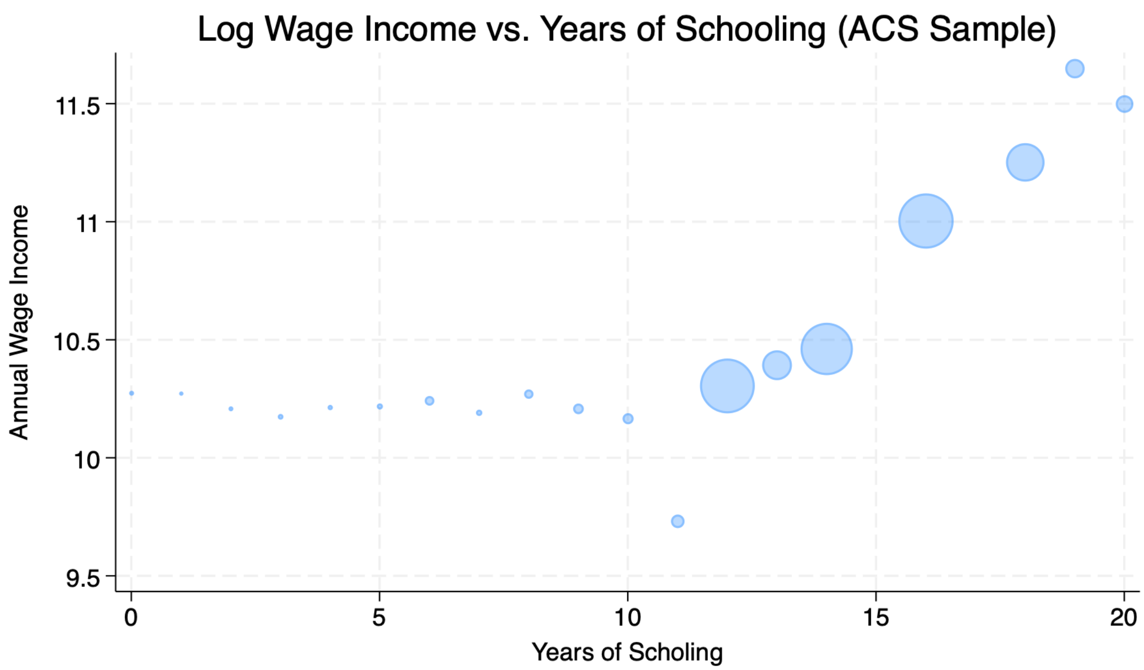


Figure 6.6: Years of Schooling and Log Annual Wage Income

statistical definition of "best-fit" means. In our case, we are going to take estimates and try to understand how to interpret them.

A typical estimated Mincer equation might look like:

$$\ln(w) = 7.069 + 0.173 \cdot S + 0.100 \cdot E - 0.002 \cdot E^2 + \varepsilon$$

These are the numbers that I retrieved when estimating the parameters using the 2023 ACS sample restricted to individuals with non-missing income between the ages 18-65.

What this equation is trying to tell us is how to predict an individual's log wage from their schooling in years and overall experience. For example, imagine a person graduated college ($S = 16$) and has spent ten years in the workforce ($E = 10$). Their predicted log wage is equal to:

$$\ln(w) = 7.069 + 0.173 \times 16 + 0.100 \times 10 - 0.002 \times 10^2$$

$$\ln(w) = 7.069 + 2.768 + 1.000 - 0.200 = 10.637$$

Now imagine we consider another person who has the same experience, but has 17 years of schooling (i.e. they went to graduate school for one year). Their predicted log wage is equal to:

$$\ln(w) = 7.069 + 0.173 \times 17 + 0.100 \times 10 - 0.002 \times 10^2$$

$$\ln(w) = 7.069 + 2.941 + 1.000 - 0.200 = 10.810$$

What is the difference in log wages between these two individuals?

$$\Delta \ln(w) = 10.810 - 10.637 = 0.173$$

So the coefficient on schooling, $\beta_1 = 0.173$, tells us that an additional year of schooling is associated with a 0.173 increase in log wages. But what does this mean in terms of actual wages? Well, there is a simple formula that allows us to convert changes in log wages to percent changes in wages. The formula is:

$$\%\Delta w \approx 100 \times exp(\beta_1) - 1$$

For this course, you don't need to memorize this formula or understand how it is derived. The point of this exercise is to understand how to interpret the coefficient on schooling. Since you are probably less familiar with log wages, we want to convert this into a more interpretable number: percent changes in wages.

Using the formula above, we can compute the percent change in wages from an additional year of schooling as:

$$\%\Delta w \approx 100 \times exp(0.173) - 1 \approx 18.9\%$$

So the interpretation of the coefficient on schooling is that an additional year of schooling is associated with an 18.9 percent increase in wages.

Now, so far what we have done is show that there is a strong correlation between wages and schooling. But we have not yet established that there is a causal impact of schooling on wages. There are many reasons why people with more schooling may earn more. For example, people who attend college may come from wealthier families and have better connections that help them get higher paying jobs. Or, people who attend college may be more motivated which could help later in finding jobs and being promoted.

Establishing the causal impact of education on earnings is a central question in labor economics. There have been many different strategies attempted to answer this question. We will discuss briefly here some of the more prominent strategies used in the literature, but only briefly. Instead our focus in this chapter will be on other aspects of human capital choice, which includes how individuals predict their own returns to education, and how returns vary by the type of education, such as college major or college quality.

6.3 Identification Strategies

There have been many different strategies used in the literature to try and estimate the causal impact of education on earnings. Below we briefly describe some of the more prominent strategies.

Twins Studies: Twin studies are a staple of many academic literatures. The idea is that twins share many of the same background characteristics. They come from the same family, have similar genetics (or the same genetics in the case of identical twins), grow up in the same environment. Therefore, if they get different levels of education, maybe we can attribute differences in their earnings to the differences in education. Ashenfelter and Rouse (1998) use this strategy on a sample of identical twins. They take advantage of a large twin festival held in Twinsburg each year in order to get a large sample of 700 identical twins. They find in their twin sample that the return to an additional year of schooling is around 9 percent. So that is quite a bit lower than our estimate using the 2023 ACS data. But it goes in line with our intuition. Individuals that attend more schooling may be different in unobservable ways that also help them earn more. By looking at twins, we can control for some of these unobservable differences.

Quarter of Birth: In the U.S., places often use cutoffs for determining when a child will start school. For a long time this was December 31st. For example, take a child is turning 6 on December 31st. This child would be assigned to start the first grade in the following year. However, take another child that was born on January 1st. This child would not be assigned to start first grade until the year after that. Therefore, even though these two children are only one day apart in age, they will start school a full year apart.

Now, Angrist and Krueger (1991) realized that this might interact with the compulsory schooling laws in the U.S. In most states, children are required to stay in school until they turn 16. Therefore, if you are born on December 31st, you would be required to stay in school for a full year longer than a child born on January 1st. The reason is simply because you started school a year earlier.

To put concrete numbers into this idea, imagine a child born on December 1st. This child would start first grade at age 6 on the following September. Therefore, this child would turn 16 they would have experienced about 9 1/2 years of schooling. Now, for the child born on January 1st. This child would start first grade at age 7 on the following September. Therefore, this child would turn 16 they would have experienced about 8 1/2 years of schooling.

So there are a few things to keep in mind with this design. First, this is really only relevant for individuals who are planning to drop out as soon as possible. If you are planning to go to college, then this design won't impact you. It doesn't matter if you start school early vs. late, as you plan to continue your education anyways. However, for individuals at the margin of dropping out, this design can have a big impact. Angrist and Krueger (1991) finds that spending an additional year in school due to compulsory schooling laws increases earnings about 7.5 percent. So pretty close to our estimate from the twin study.

These are two clever designs that have been used in the literature to try and estimate the causal impact of education on earnings. There are many more designs that have been used as well. In general much of the literature finds relatively similar estimates of the return to an additional year of schooling, generally in the range of 7-10 percent.

6.4 Regression Discontinuity Design

So far we have considered the return to an additional year of schooling. But surely the impact of a college education, for example, may vary by the major. Maybe different types of colleges have very different returns. In this section we will first introduce a new research design that has proven to be extremely useful in the economics of education: the **regression discontinuity design**. Then we will use this design to study various questions in understanding how different education choices impact earnings.

To understand the idea behind a regression discontinuity design, we are going to go through the first use of the design, which comes from Thistlethwaite and Campbell (1960). In their paper, they were interested in studying the effect of receiving a scholarship on various student outcomes, such as whether the individual goes on to purse graduate school or whether the individual goes on to win more scholarships. So this is a slightly different question than we have been considering so far, but it sets us up well to understand the methodological approach. At the college they were studying, a subset of students were given merit-based scholarships based on their results on a standardized test. If a student scored above a certain cutoff, they would receive the scholarship. If they scored below the cutoff, they would not receive the scholarship.

Now let's again consider how one might tackle this problem: if we had data on individual students, maybe we could compare the outcomes for students that received the scholarship vs. those that did not. But again, we run into a problem we have already seen: students that received the scholarship are likely different than students that did not receive the scholarship in unobservable ways. For example, maybe students that received the scholarship are more motivated. Due to this inherent motivation, they will be more likely to go on to graduate school or win more scholarships. Therefore, simply comparing the two groups will not give us a causal estimate of the impact of receiving the scholarship.

The idea in Thistlethwaite and Campbell (1960) is relatively simple. Because they had information on the score the students received on the standardized test, they could compare students that scored just above the cutoff to students that scored just below the cutoff. The idea is that students that scored just above vs. just below the cutoff are likely very similar. They likely have similar levels of motivation, similar family backgrounds, and similar innate abilities. Small differences in a test score are likely due to random chance. Maybe someone guess correct a few more times than another student and got a scholarship. Therefore, if we compare students that scored just above vs. just below the cutoff, we can get a causal estimate of the impact of receiving the scholarship.

This idea, regression discontinuity design, did not catch on for quite a while. However, recently, more and more researchers have been taking advantage of this relatively simple idea. The plot below shows the number of papers in top economics journals that use a regression discontinuity design. As you can see, there was very little interest in this design until around 2000, after which there has been an explosion of research using this design.

Now the regression discontinuity design has become influential for a variety of reasons. First, the basic requires that some treatment of interest is determined by a cutoff rule. This ends up being a relatively common in various policy settings. For example, elections also have cutoff rules. Imagine you are interested in the effect of having a Democratic vs. Republican mayor on city outcomes. You could compare cities that just barely elected a Democratic mayor vs. cities that just barely elected a Republican mayor. The idea is that these cities are likely very similar, and the difference in election outcomes is likely due to random chance (i.e. a few more people showing up to vote). Therefore, you can use this design to estimate the causal impact of having a Democratic vs. Republican mayor.

It was also later clarified that you don't need to have a strict cutoff rule. As long as the probability of receiving treatment jumps at a certain cutoff, you can still use this design. This

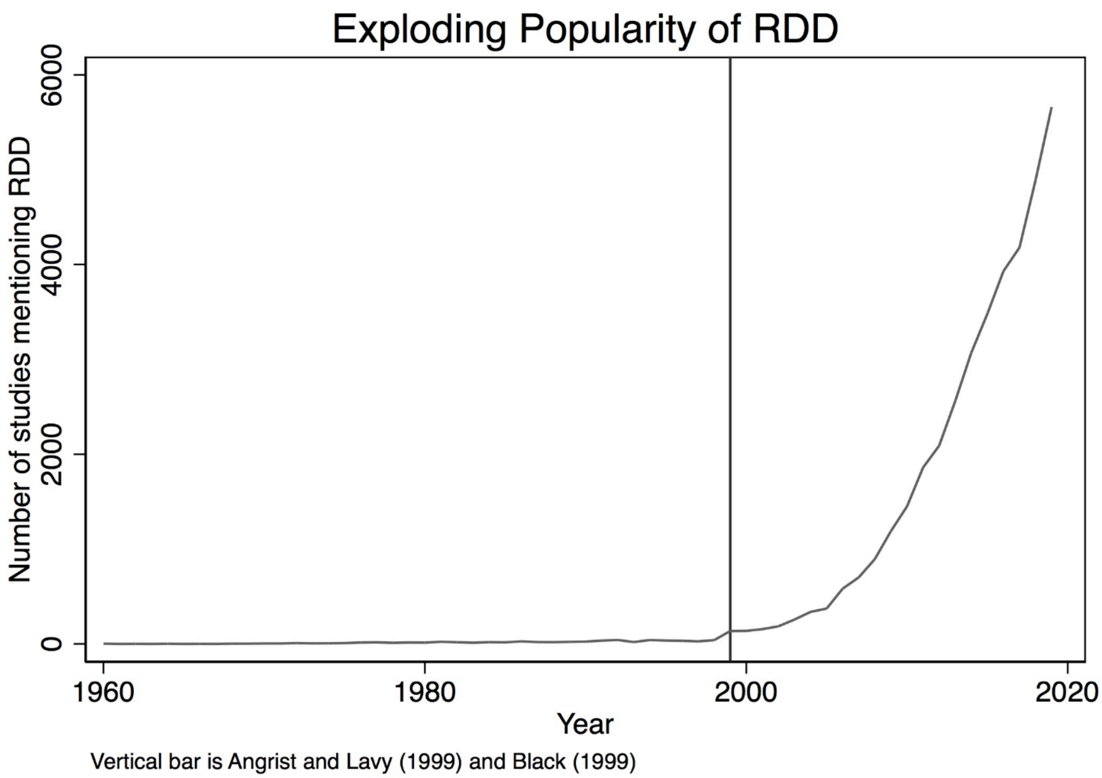


Figure 6.7: Source: Causal Inference the Mixtape

is called a fuzzy regression discontinuity design. For example, imagine a scholarship program that gives scholarships to students that score above a certain cutoff on a test, but also gives scholarships to some students that score below the cutoff (maybe due to financial need). In this case, you can still use the regression discontinuity design as long as the probability of receiving the scholarship jumps at the cutoff. This is something you can easily check in the data, so is easily verifiable.

To see the power of regression discontinuity designs, we are going to go through two recent papers that have used this idea. The first is similar to the original Thistlethwaite and Campbell paper. Instead of a financial scholarship at a single college, we are going to consider merit-based scholarships in Colombia. As we will see these scholarships have a big impact on college attendance and college completion, so we can use the regression discontinuity design to estimate the impact of college attendance on earnings. The second example we will consider is how different college majors impact earnings. In particular, we will study the causal impact of earning an economics degree.

Our first study comes from Londoño-Vélez, Rodríguez, and Sánchez (2020). While most of our discussion has focused on the U.S., the setting for this paper is the higher education system in Colombia. In Colombia, there is a large merit-based scholarship program called "Ser Pilo Paga" (Hard Work Pays Off). This program was started in 2014 and provided full scholarships to high-achieving students from low-income backgrounds to attend top universities in Colombia. The goal of the program was to increase access to higher education for talented students who might not otherwise be able to afford it. The scholarships covered full tuition to students who attend a four or five-year degree program at any university with High Quality Accreditation. Therefore, it was a very generous scholarship program, although restricted only to a certain subset of universities.

The key feature of this program that allowed to use a regression discontinuity design is that scholarships were awarded based on a standardized test score. In Colombia, when students exit high school, they take a national standardized test called the "SABER 11" exam. This is similar to the SAT or ACT exams in the U.S., but more widely used. The test is taken even by students that don't plan to attend college. In order to be eligible for the Ser Pilo Paga scholarship, students needed to score above a certain cutoff on the SABER 11 exam.

The plot below shows the probability of receiving the scholarship based on the SABER 11 exam score. As you can see, there is a big jump in the probability of receiving the scholarship at the cutoff. However, you might also notice that it is not a perfect jump from 0 percent to 100 percent. Instead, it jumps from 0 to about 60 percent (0.6 in the graph). The reason is because passing the cutoff implies you are eligible for the scholarship, not that you are required to take it. Some of these individuals may choose not to attend college. If they do attend college, they may not be attending a High Quality Accredited university for which the scholarship is valid. This is therefore an example of a fuzzy regression discontinuity design. The treatment here is receiving a scholarship. Because it does not jump from 0 to 1, we have a fuzzy design.

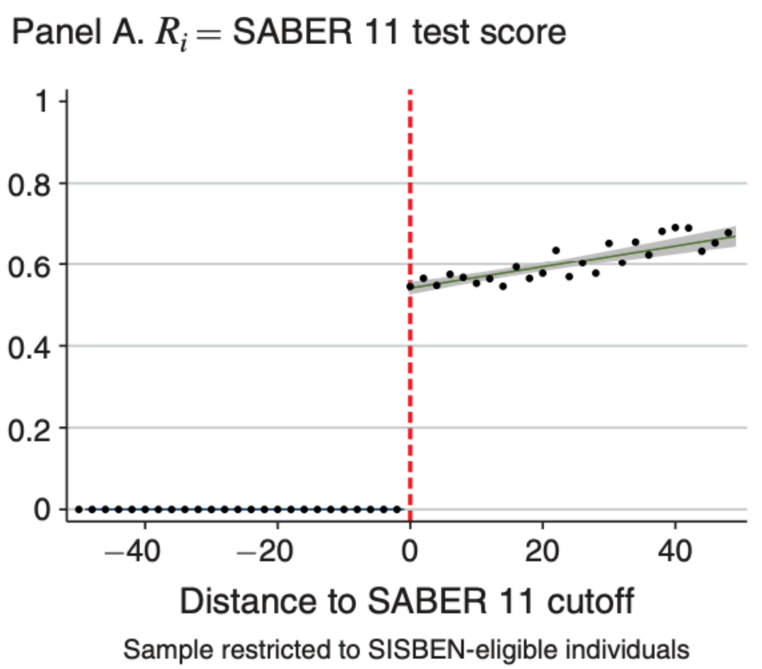


Figure 6.8: Probability of Receiving Scholarship by SABER 11 Score

Now, the key assumption of a regression discontinuity design is that individuals just above vs. just below the cutoff are similar. This is true if students cannot precisely manipulate their test scores. The definition of precisely manipulate has a statistical definition. However, the intuition is the following: imagine a subset of students were given access to the electronic records where their SABER 11 test score was held.

If they wanted the scholarship, they could change their score to be just above the cutoff. Therefore, for all these students that are given the ability to cheat, they can precisely manipulate where their score will fall relative to the cutoff.

Now, how will this manifest in the data? If we look at the distribution of test scores, then we would see an unusual spike in test scores right at the cutoff. This is because the students who could cheat would all change their scores to be just above the cutoff so there would be "more than expected" students just above the cutoff.

Well, this sounds like a pretty convoluted story, right? And that suggests that being able to precisely manipulate test scores is unlikely. Studying long hours, working hard, these are ways to improve your score, but not precisely manipulate it. But the great thing about the regression discontinuity design is that we can check for this in the data.

The plot below shows the distribution of SABER 11 test scores. The vertical dashed line indicates the cutoff for receiving the scholarship. As you can see, there is no unusual spike in test scores right at the cutoff. This implies that students are not able to precisely manipulate their test scores. This is a valid regression discontinuity design.

Now let's get to some results. What is the impact of receiving financial assistance on college attendance and future earnings. In the plot below, we see the impact of receiving the scholarship on college attendance. The vertical dashed line indicates the cutoff for receiving the scholarship. As you can see, there is a big jump in college attendance at the cutoff. Receiving the scholarship increases college attendance by about 40 percentage points. This is a huge impact! These students at the cutoff are very high achieving students, in the top decile of test scores. Yet, receiving the scholarship increases their college attendance by 32 percentage points. This shows that in this setting the financial constraints play an enormous role in determining college attendance.

In a follow-up study, Londoño-Vélez et al. (2025) study the impact of the financial aid on future earnings. The literature on financial aid often considers two potential impacts. First, if financial aid is given to students that would have attended college anyways then the impact on future earnings may be small. This is not to discount the importance of financial aid for equity reasons, but if the goal is to increase future earnings, then giving aid to students that would have attended college anyways may not be the most effective way to do this. From what we've seen above, this seems unlikely to be the case here, but we can also directly study the impact of being eligible for the scholarship on future earnings. This is one of the goals of Londoño-Vélez et al. (2025).

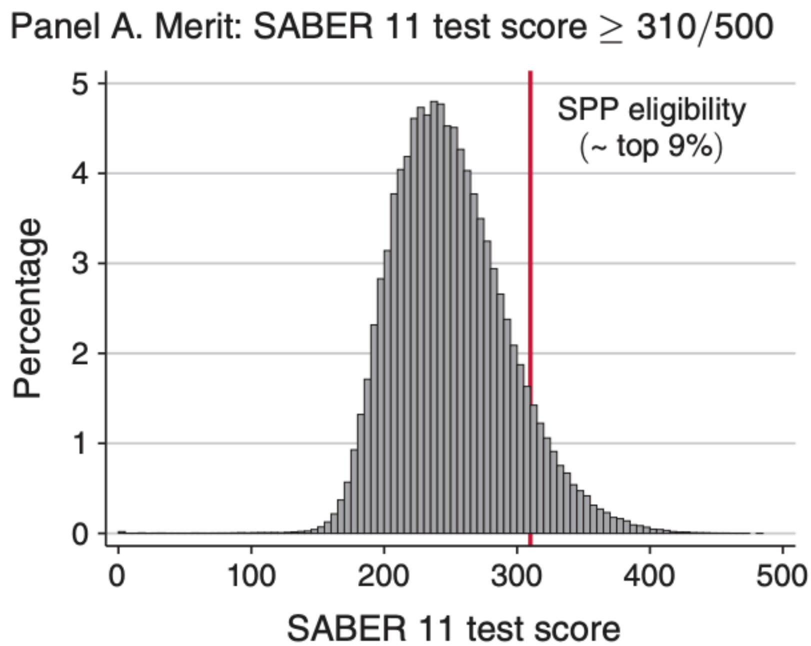


Figure 6.9: Distribution of SABER 11 Test Scores

Below, we show the plot of the impact of receiving the scholarship on future earnings. The vertical dashed line indicates the cutoff for receiving the scholarship. As you can see, there is a big jump in earnings at the cutoff. The y-axis is denoted in multiples of the monthly minimum wage in Colombia. If you receive zero formal earnings, you would be coded as having zero multiples of the minimum wage, so this is why averages can be well below one.

As you can see in Figure 6.11, individuals just below the cutoff earning about 1.0 times the monthly minimum wage, while individuals just above the cutoff earn about 1.2 times the monthly minimum wage. Therefore, one way to interpret this is that being eligible for the scholarship increases monthly earnings by 0.2 times the monthly minimum wage. Now, recall that not everyone even takes up the scholarship or attends college due to being eligible. So this isn't exactly the impact of attending college on earnings. To scale this up to get the impact of attending college on earnings, we need to divide this estimate by the increase in college attendance at the cutoff. In this context, we would then get an increase of about 0.35 times the monthly minimum wage. Since the average earnings for individuals just below the cutoff was about 1 monthly minimum wage, this represents a 35 percent increase in earnings from attending college.

Next, we are going to turn to our second example which brings us back to the U.S. and even within the UC system. In many universities, there are certain majors that have requirements to get into the major. At UC Santa Cruz, in order to get into any of the Economics majors,

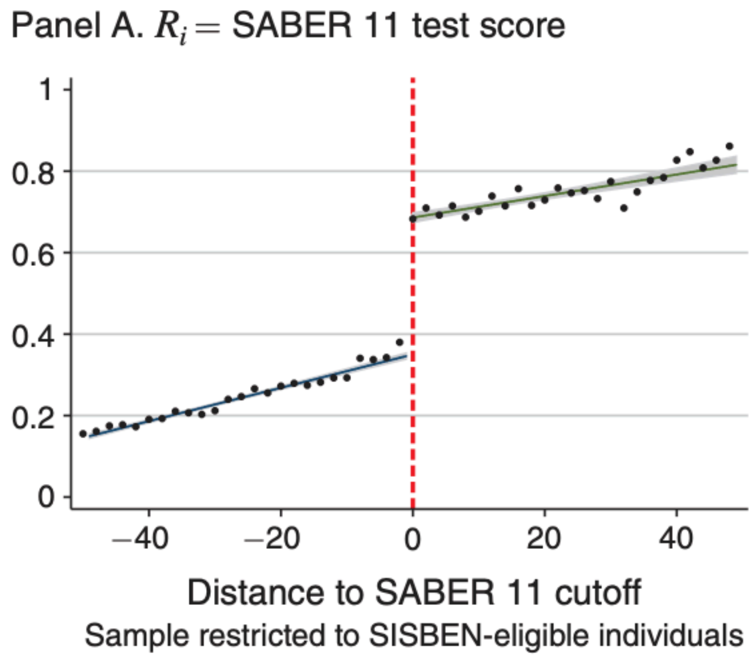


Figure 6.10: Impact of Scholarship on College Attendance

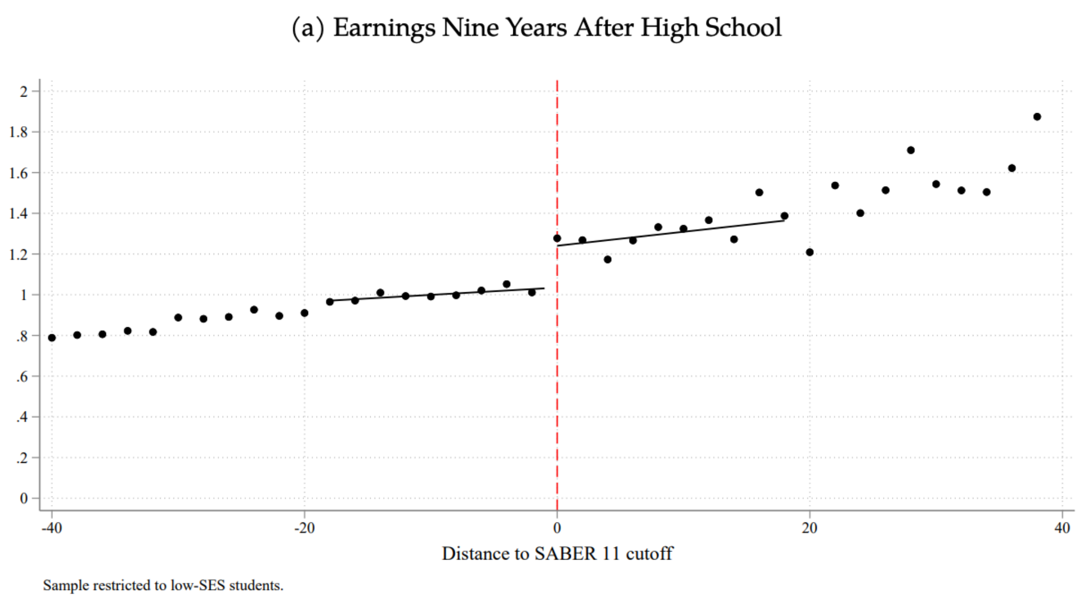


Figure 6.11: Impact of Scholarship on Future Earnings

you must receive a combined GPA of 2.8 in the prerequisite courses (Econ 1 and Econ 2). Therefore, again, whether you receive an Economics degree depends partially on a cutoff rule. Therefore, we can use regression discontinuity design to estimate the causal impact of receiving an Economics degree on future earnings.

This is exactly what Bleemer and Mehta (2022) does. As always, the first thing to check in a regression discontinuity design is does the threshold rule have an effect. The plot below shows the probability of receiving an Economics degree based on the combined GPA in Econ 1 and Econ 2. The vertical dashed line indicates the cutoff GPA of 2.8. As you can see, there is a big jump in the probability of receiving an Economics degree at the cutoff.

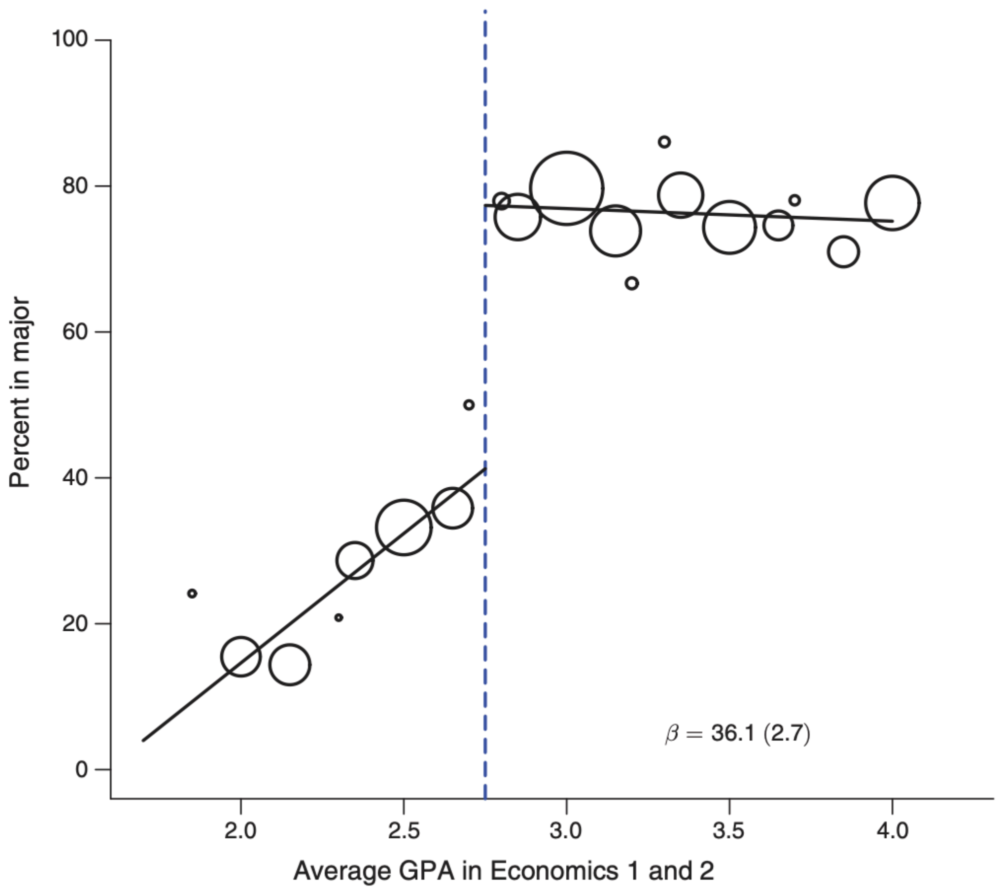


FIGURE 1. THE EFFECT OF THE UCSC ECONOMICS GPA THRESHOLD ON MAJORING IN ECONOMICS

Figure 6.12: Probability of Receiving Economics Degree by GPA

This is again a fuzzy regression discontinuity design. Why, because not everyone that passes the cutoff will receive an Economics degree, and some of the individuals that do not pass the cutoff may still receive an Economics degree. Individuals to the left of the cutoff can petition to be admitted into the major, and some of them are successful. Additionally, some individuals that pass the cutoff may decide to switch to a different major later on. Therefore, we have a fuzzy design. Regardless, we can see that the threshold is influential on whether an individual receives an Economics degree. If you fall just below, your probability of receiving an Economics degree is about 40 percent. If you fall just above, your probability of receiving an Economics degree is about 80 percent. Therefore, passing the cutoff increases your probability of receiving an Economics degree by about 40 percentage points.

Now, again, what is the key assumption of a regression discontinuity design? That individuals just above vs. just below the cutoff are similar. This will be true as long as individuals cannot precisely manipulate their GPA in these courses. While we can check for this in the data, it is a bit harder than for our prior example with SABER 11. This is because SABER 11 takes on many many values. We get a clear continuous distribution of test scores. However, GPA only takes on a few discrete values. Looking for spikes in a distribution with only a few discrete values is a bit harder.

Therefore, another way to check for this is to look at other characteristics of individuals just above vs. just below the cutoff. Our real goal is to argue that individuals just below vs. just above the cutoff are similar in all characteristics except for receiving an Economics degree. In Bleemer and Mehta (2022), the authors have a rich set of characteristics to study. For example, one compelling characteristic is the SAT score of individuals. If individuals just above vs. just below the cutoff have similar SAT scores, that is evidence that they are similar in terms of overall academic ability.

As can be seen in the graph, there is no clear break around the cutoff. It is true that individuals that had higher SAT scores tend to have higher GPAs in Econ 1 and Econ 2. But there is no unusual jump in SAT scores at the cutoff. This suggests that individuals just above vs. just below the cutoff are similar in terms of academic ability, at least measured by the SAT. The authors check many other characteristics as well and find similar results. Therefore, we can be more confident that this is a valid regression discontinuity design.

So let's get to the results. Below we plot earnings by GPA in Econ 1 and Econ 2. The vertical dashed line indicates the cutoff GPA of 2.8. These earnings are for the years 2017 and 2018, which implies that the students in the sample are between 23 and 28 years old. Therefore, this should be interpreted as an estimate of the change in earnings early in a person's career.

Individuals just below the cutoff earn about $47,000 per year, while individuals just above the cutoff earn about $55,000 per year. Therefore, being eligible for an Economics degree increases earnings by about $8,000. Again, recall that not everyone that passes the cutoff will receive an Economics degree, and some of the individuals that do not pass the cutoff may still receive an Economics degree. Therefore, if we scale this impact by the increase in probability of receiving an Economics degree at the cutoff (40 percentage points), we get that receiving an Economics degree increases earnings by about $20,000 per year. This is what the term IV is referring to in the figure (although we have just approximated the numbers, the exact effect of a Economics degree is $22,123).

(d) SAT Score

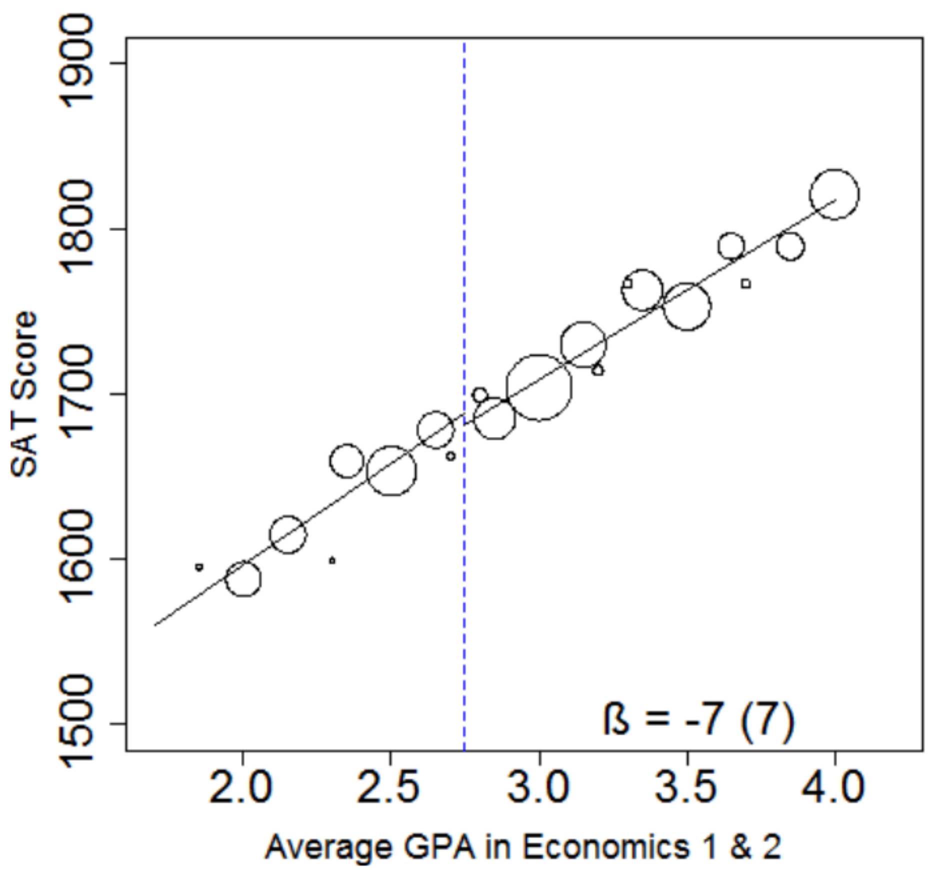


Figure 6.13: SAT Score by GPA

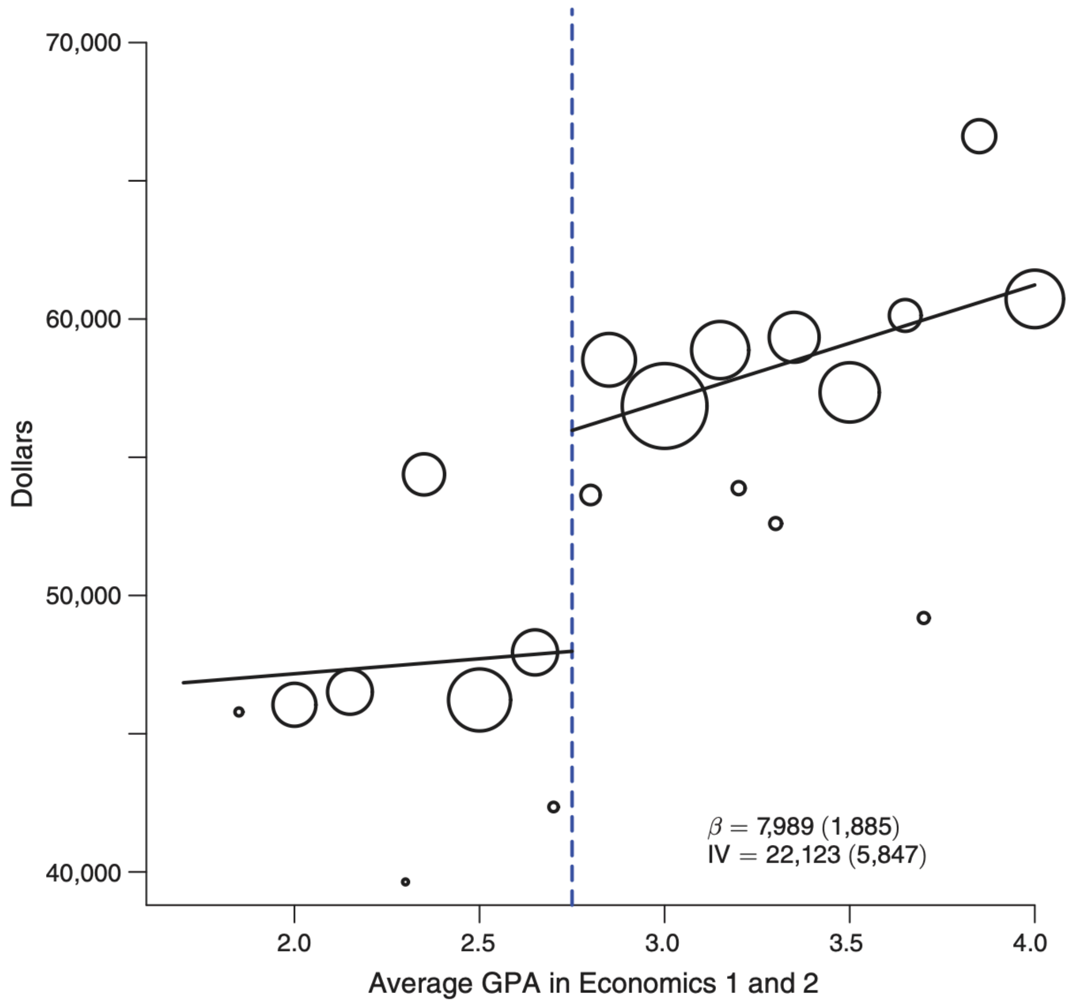


Figure 6.14: Earnings by GPA

Now, this is an enormous effect. Remember, we have been consider the impact of an additional year of schooling on earnings. Typical estimates are in the range of 7-10 percent. Here, receiving an Economics degree relative to another college decrease increases earnings early in your career by over 40 percent!

And the amazing thing that the regression discontinuity design allows us to do is argue that this is causal. However, one nice feature of this study is that the authors also consider what the observational data would suggest. For an observational study, one could take all the students at UC Santa Cruz and compute earnings by major. Then to estimate the return to an Economics degree, one could compare the average earnings of Economics majors to the average earnings of non-Economics majors. Or to perhaps be more realistic, you could compare Economics majors to other popular social science majors, such as Psychology or Sociology.

When the authors do this, they estimate that Economics majors earn about $20,000 more than other social science majors. This is remarkably close to the causal estimate of $22,123 they get from the regression discontinuity design. Now, to emphasize, just because the observational estimate is close to the causal estimate in this particular case does not mean that we didn't need the regression discontinuity design. If you had just shown someone the observational estimate, there would be a number of counterarguments. Maybe Economics majors tend to have higher GPAs and would earn more anyways. Maybe Economics majors are interested in different careers, but even if they were unable to get an Economics degree, they would have still pursued those careers. These arguments are all valid for observational studies. The power of the regression discontinuity design is that it allows us to get around these concerns and estimate a causal impact. So in this context, there is a clear causal impacto of receiving an Economics degree on earnings.

7 Technology: A Task Approach

7.1 The Luddites

In the early 19th century, the textile industry was one of the most important industries in England. One in four workers were employed in the industry, making it a key source of employment. But the early 19th century was also a time of dramatic technological change. Driven by the industrial revolution, owners of factories were continually introducing more and more machinery. Not everyone was excited about this. Textile workers were worried the new mechanized weavers would replace them in the long run. According to legend, one weaver in Leicester grew so enraged by a new mechanical loom that he smashed it in a fit of fury. His name, reportedly, was Ned Ludd.

Figure 7.1: 1812 Cartoon of Ned Ludd

Whether or not Ned Ludd actually existed, his story spread across England and he became a folk hero to workers anxious about the advance of machinery. A political movement emerged with Ludd as its symbolic figurehead. The so-called "Luddites" organized nighttime raids on factories and destroyed the machines they saw as threats to their livelihoods. In one dramatic incident at Rawfolds Mill, a group of 100 armed Luddites clashed with local soldiers. The violence triggered a harsh government crackdown and eventually brought the movement to an end.

Today, the term *luddite* is often used pejoratively to describe someone opposed to new technology. Economists have typically argued that even labor-displacing innovations benefit society in the long run. In the case of the Industrial Revolution, the initial disruption to jobs eventually gave way to widespread economic growth and rising living standards. If you had told a textile worker in 1800 that by the year 2000, employment in English textile mills would be virtually nonexistent, they might have assumed disaster had struck. But instead, employment shifted to other sectors, often ones with higher wages and better working conditions.

This will be the first in two chapters on how technology has shaped the labor market over time. As we have seen with the luddites debates on the impact of technology on labor markets has been around for a long time. There are still two basic views on technology. First, there is a view you may often hear in Silicon Valley, the view that new technology leads to new prosperity, automatically. This certainly has some appeal when looking at the long sweep of history. Real incomes have risen dramatically over time and we owe a lot of this to new technology. But even if technology does lead to growth in aggregate, there is nothing that implies that these gains will be shared equally. This will be the topic of this chapter. How has technology impacted inequality in the 20th century.

7.2 A Brief History of Computing

There are many places that one could start when talking about the impact of technology on labor markets. The industrial revolution is an obvious place since it fundamentally changed the occupational structure of economies, shifting individuals from farms into cities and factories. There are a large range of inventions during this time that had enormous impact on workers, including the steam engine, the expansion of railways, automatic sewing machines, new industrial processes, among others. There are volumes of books on these topics and their impact on not only workers, but history more generally.

Electrification is another potential place to begin. The National Academy of Engineering calls electrificaton "the greatest engineering achievement of the 20th century". Electrification brough tremendous changes to the economy, leading to further innovations and growth which impacted all facets of life, includng the labor market.

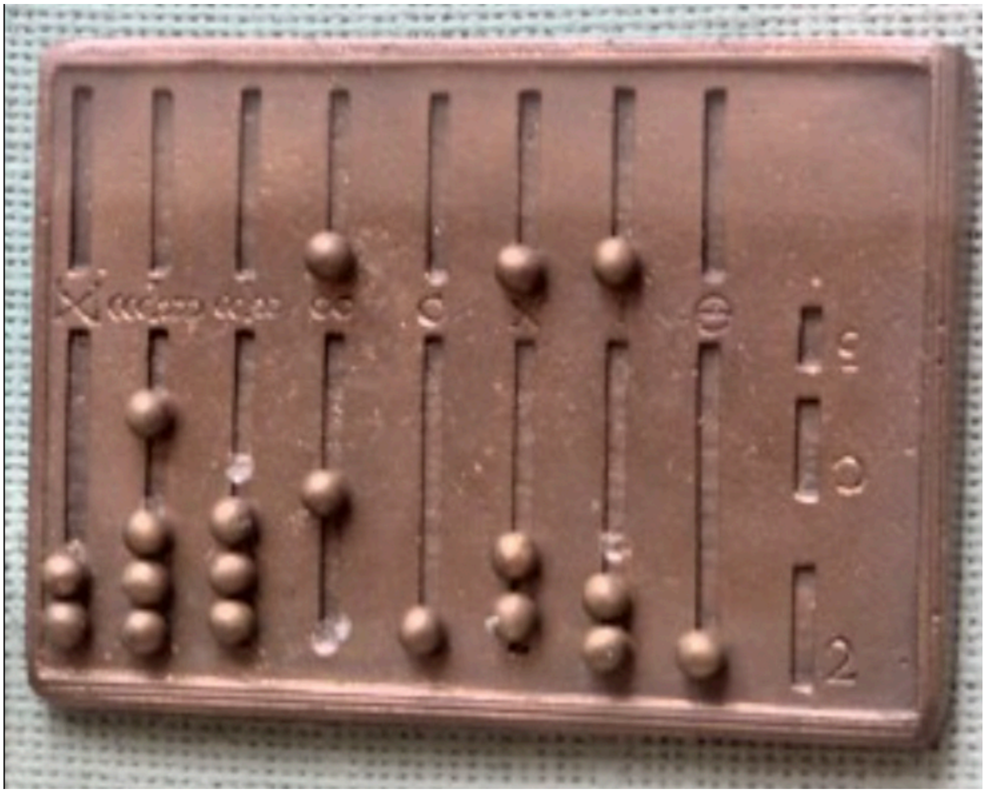


Figure 7.2: Roman Abacus

Modern labor economics, however, has had much more focus on more recent technological advancements. And one has garnered more attention than others because of its now ubiquitous use in nearly all industries: the computer. A computer is at its core an information processing system. One of the simplest historical information processing systems is the abacus. Throughout human civilization, a constant need for societies everywhere was to collect taxes. Collecting taxes often involves adding large sums together to figure out the total tax owed or the total taxed paid in a given jurisdiction. The abacus was a tool that has been around for millenia that has aided workers in keeping track of large sums. It is, at its core, an information processing system. The data inputs are often beads on a series of rods, which can be moved to represent numbers. By manipulating the beads according to simple rules, users could quickly perform addition, subtraction, multiplication, and division. Many of these tools were used in a time before the zero was invented, making these even more difficult!

There were a number of conceptual breakthroughs that occurred between the ancient times of using abacuses and the modern computer. First, Charles Babbage invented the idea of a programmable computer. Calculators and abacus prior to this time were hardwired to do certain operations. They could do basically one task and they performed that task no matter what the inputs where. A programmable computer is different. It can do several tasks, you just need to give it information on what it should do. This is the birth of programming (a set of instructions for a computer to follow so that it accomplishes a given task). Ada Lovelace (daughter of notable English poet Lord Byron) was the first to identify that Babbage's computer had applications beyond calculating numbers. For her insights into the use of Babbage's machine she is often credited as the first programmer.

Babbage's idea was revolutionary, but the true power of computation was not realized until the conceptual work of Alan Turing. It seemed intuitive at the time that very different tasks would need entirely different machines. To take two examples from this era, let's consider the calculator and a tabulating machine. Mechanical calculators are much like calculators we have today. The exact mathematical operation could depend on the settings (i.e. be programmable like Babbage envisioned), but its scope was limited to certain mathematical operations: often addition, subtraction, multiplaction, and division.

Figure 7.3: Herman Hollerith's Tabulating Machine

The tabulating machine was invented by Herman Hollerith in order to aid the U.S. Census in its data-gathering mission. The way it worked is that Census workers encoded individuals responses on punch cards. The tabulating machine would read the holes in the cards and count or sort the data. This would allow the Census to quickly count up the number of individuals, and because the cards were also sorted, one could count the number of females, number of individuals of different ages, etc.

But try to put yourself into the shoes of people at the time. Does it seem intuitive that you could build a single machine that both does mathematical calculations and counts and sorts data? A machine where you could input the data and then perform complex mathematical operations. The two machines don't appear to have much in common. Maybe you think it seems possible to somehow combine the two in some way, but what if you also want to write text on the machine. That seems like it would take even more hardware to add to the machine.

Alan Turing's insight was the idea of a universal machine. He showed that you can construct a machine that can simulate the behavior of any other machine. This is referred to now as a universal Turing machine. The machine just requires the correct instructions. The insight was purely theoretical. But it set the stage for the foundation of general-use computers. A key idea here is the difference between hardware and software. The hardware is the universal machine, the software tells the universal machine what to do. With one set of software it is a calculator. With another it is a tabulating machine. Today, our personal computers, phones, tables, these are universal turing machines.

Since Turing's insight, there have been many more revolutionary insights into computing, including the design of the architecture of the first modern computer by John von Neuman, it's practical implementation in the invention of the EINAC, the invention of programming languages, microprocessors, and finally, the personal computer, the focus of our next section.

7.3 The Computer

The widespread use of personal computers began in the U.S. in the 1980s. Now, try to imagine you are an accountant before the rise of personal-use computers. Let's say you have a relatively straightforward task for the day. You need to calculate the payroll for all the employees. This would involve adding up the hours for each employee, multiplying by the wage and then adding it to the payroll ledger. It's a simple task, but could take an extremely long time depending on the number of workers in the firm. And you might make some errors. Those errors might be costly to correct if the paychecks go out with the wrong information.

Now imagine your firm purchases the new Apple computers which come with a spreadsheet. You've got a column for wages and a column for hours. Computing payroll adds up to a few keystrokes on the computer. If you're the accountant you might have two very different reactions. You might be relieved. What used to be an extremely tedious error-prone process has become simple. You can focus your effort on less tedious, more conceptually difficult tasks. Your productivity is going to skyrocket.

But you might have an entirely different reaction. You might be extremely worried. Do they need all these accountants now that computers can do these operations so quickly and with fewer errors. This isn't a productivity boost, it's a replacement. In 1980, it was not at all obvious which of these two scenarios would play out. But to fully understand the picture of how the wage structure evolved, we have to add another type of worker to our example: the support staff.

7.4 Routine vs. Non-Routine Tasks

To understand the impact of computers on the labor market, we are going to use a framework that is referred to as a "task model", which was first proposed in 2003 in an article by David Autor, Frank Levy, and Richard J. Murname. Before the task model, economists typically thought of workers as being high-skilled or low-skilled, high-productivity or low-productivity. High-skill workers were able to produce more output than low-skilled workers. But this framework had many shortcomings. In reality, workers are not directly producing output. If you think of basically any firm there are a wide range of tasks that must be accomplished in order to produce some output. For example, to produce a phone, you need a team of engineers to design the phone. You need a team to source the inputs for the phone. You need a team and machinery to assemble the parts into a phone. Finally, you need a team to market and distribute the phone. You can think of each of these steps as a task that must be accomplished. These tasks can be performed by different types of workers or they can be performed by machines. In this section, we will think about what types of tasks the computers can perform, and how this will impact different types of workers.

Until recently, there was basically a single dominant paradigm for how to get a computer to do something. You needed to write a computer program that told the computer exactly what to do. You specify each step carefully in code. We are going to refer to tasks that can be accomplished by following a set of rules as *routine tasks*. Routine tasks can further be broken down into two types: cognitive and manual. Examples of cognitive routine tasks include record keeping, performing calculations, and simple customer service interactions, like taking money out of an ATM. Examples of manual routine tasks include things like sorting items or assembling items on a production line.

In contrast to routine tasks, we have *non-routine tasks*. Non-routine tasks are tasks that would be very difficult by following a set of rules, often because the task may vary greatly depending on the setting. For example, take a truck driver. While most people can drive a truck relatively easily, imagine trying to write a set of rules for a machine to follow. You would need to figure out all the potential situations the truck would find itself in and come up with a solution for each individual one. While humans can react quickly and figure out a solution, it would be difficult to codify this in a set of rules. Non-routine tasks can also be broken down into two types: cognitive and manual. Examples of non-routine cognitive tasks include things like medical diagnoses, legal writing, managing a team. Examples of non-routine manual tasks include things like cleaning a house, driving a truck, or cooking a meal.

Until very recently, computers were not very good at performing non-routine tasks. The machine-learning breakthrough has changed this. To understand why, let's consider a very simple example. Imagine we are watching someone throw a baseball and we want to predict where the ball will land. This is relatively easy to figure out by writing a traditional computer program. The code would be Newton's laws of motion. You can write a program that takes the initial position, velocity, and angle of the throw and then uses these rules to predict where the ball will land. We know the rules of motion, so we can write a program that will work very well. This will be a simple program.

Now, imagine we didn't know Newton's laws of motion. Instead, we had a large dataset of baseball throws. A machine-learning approach would use this data to figure out how baseballs generally behave. The machine-learning algorithm would look for patterns in the data and use these patterns to make predictions. This is a very different approach than writing a traditional program. We don't need to understand the laws of physics, we just need a lot of data.

The machine-learning approach is very powerful because it can be used in situations where we don't know the underlying rules. For example, let's take the self-driving car example. It would be extremely difficult to write a traditional program that could handle all the potential situations a car might find itself in. But with enough data, a machine-learning algorithm can learn to recognize patterns and make decisions based on those patterns. Writing human language turns out to be similar. You could hypothetically write a chatbot that follows a set of rules to respond to questions. But then you would need to come up with rules for every possible question. Machine-learning approaches have proven to be much more tractable in this setting.

While conceptually machine learning has been around since the middle of the 20th century, it had very limited impact until recently. Therefore, for this section in which we seek to understand changes in the labor market from 1980 to 2010, we will focus on the traditional computer programming approach. For traditional computers, they are very good at performing routine tasks, both cognitive and manual. They are not very good at performing non-routine tasks, either cognitive or manual. In a later chapter, we will return to the machine-learning revolution and how it has changed the landscape of what computers can do.

7.5 The Demand for Tasks

To understand how computers will impact the labor market, let's return to our example of the accountant. For an accountant, the work is entirely cognitive. Some of the tasks are likely routine, like calculating payroll. Other tasks are likely non-routine, like researching tax codes, advising management and organizing clerical staff. One source of information on what occupations do is the Dictionary of Occupational Titles (DOT). The DOT was a massive effort by the U.S. Department of Labor to classify occupations and describe what tasks are performed in each occupation. The DOT was first published in 1939 and was updated several times until it was replaced by the O*NET database in the 1990s. Let's look at the DOT description for accountants in the 1965 version, before the introduction of computers into the workplace.

> Work activities in this group primarily involve the application of the principles of accounting, cost analysis, and statistical analysis to problems of fiscal management, auditing, and the like. Typically, workers are engaged in devising accounting systems and procedures; appraising assets and evaluating costing methods, investment programs, and monetary risks and rates; and preparing statistical tabulations and diagrams and financial reports, statements, and schedules for use by management.
>
> — *Dictionary of Occupational Titles*, 1965

Now let's think about how the introduction of computers will impact the demand for accountants. To be very concrete, let's pick out one task mentioned in this prior description: "preparing statistical tabulations and diagrams and financial reports, statements, and schedules for use by management."

Performing this task is actually a mix of routine and non-routine tasks. First, someone needs to actually collect the data. This is often not done by the accountant themselves, but rather by support staff. Once the data is collected, the accountant needs to perform the statistical tabulation. But the accountant needs to further decide how to interpret and present the results. This last part is highly non-routine. It could require judgment, creativity, and an understanding of the audience.

Now, what does a computer do well? It can perform the statistical tabulation very quickly and accurately. This means that the demand for accountants to perform this routine task will fall. But the non-routine part of the task, interpreting and presenting the results, is not something a computer can do well (at least until very recently).

In this example, there are two forces at work. First, some of the tasks the accountant performs are routine and can be automated by computers. This will reduce the demand for accountants to perform these tasks. However, the accountant has become much more productive. The accountant can now focus on the non-routine tasks that require judgment and creativity, and not on spending hours and hours tracking down small calculation errors. If this increase in productivity is large enough, it could actually lead to an increase in the demand for accountants overall. If you are a firm deciding whether to hire an accountant, you might be willing to pay more for an accountant who can use computers to quickly prepare financial reports. Maybe the computer has made the accountant so productive that you want to hire accountants not just for payroll, but also to help with budgeting, forecasting, tax planning, etc.

But what about the support staff of the accountant? The support staff may have been performing routine tasks, like data entry. Before computers, keeping track of data was a huge task. But with spreadsheets and databases, much of this work can be automated. You may need someone to enter the data initially, but keeping track of it and updating it is much easier. Therefore, the demand for support staff may fall.

At the end of the day, what happened is an empirical question. We can probably come up with alternative stories that make sense, so let's see what actually happened in the data. To do so, we are going to take advantage of the fact that the US Census has been collecting data on the occupations of workers for a long time. In particular, we are going to look at the 1980 Census, which was the last Census before computers were widely used in the workplace, and compare it to the 1990 Census. We will compare accountants and auditors. These workers tend to be highly educated and as described above, perform a number of non-routine cognitive tasks. We will compare them to bookkeeping, accounting, and auditing clerks. These workers tend to perform more routine cognitive tasks.

The figure below shows the change in employment and wages for these two occupations between 1980 and 2000, the period in which computers were introduced into the workforce. To compute these changes, I utilized the IPUMS USA database, which provides microdata for a 5% sample of the US Census. I identified individuals in the 1980 and 2000 Censuses who reported their occupation as either "Accountants and Auditors" or "Bookkeeping, Accounting, and Auditing Clerks".

To compute changes in employment, I calculated the fraction of the total workforce in each year that reported being in each occupation. I then computed the percentage change in this fraction between 1980 and 2000. The figure below shows the results.

These are some pretty striking results. The employment of accountants and auditors increased by 35% between 1980 and 2000. In contrast, the employment of bookkeeping, accounting, and auditing clerks decreased by 27%. This is consistent with our theory that computers have increased the productivity of accountants, leading to higher demand for their services, while reducing the demand for routine clerical tasks which accountants use as inputs and can often be performed by computers. To be clear, it is not as if the raw number of accountants is much larger than the number of clerks. In 2000, there are still many clerks. About 0.8 percent of all workers are in these bookkeeping, accounting, and auditing clerk roles. This is actually similar to the total number of accountants and auditors.

The key here is the growth rates vary wildly. In 1980, about 0.5 percent of all workers were accountants and auditors. By 1990, this had grown to about 0.67 percent. In contrast, bookkeeping, accounting, and auditing clerks made up about 1.0 percent of all workers in 1980, but this had fallen to about 0.8 percent by 2000.

7.6 Job Polarization

The relative decline in the demand for clerks is notable. One natural question is where might these workers go? One possibility is that they move to occupations that are not as exposed to automation by computers. For example, cashiers are likely not as exposed to automation.

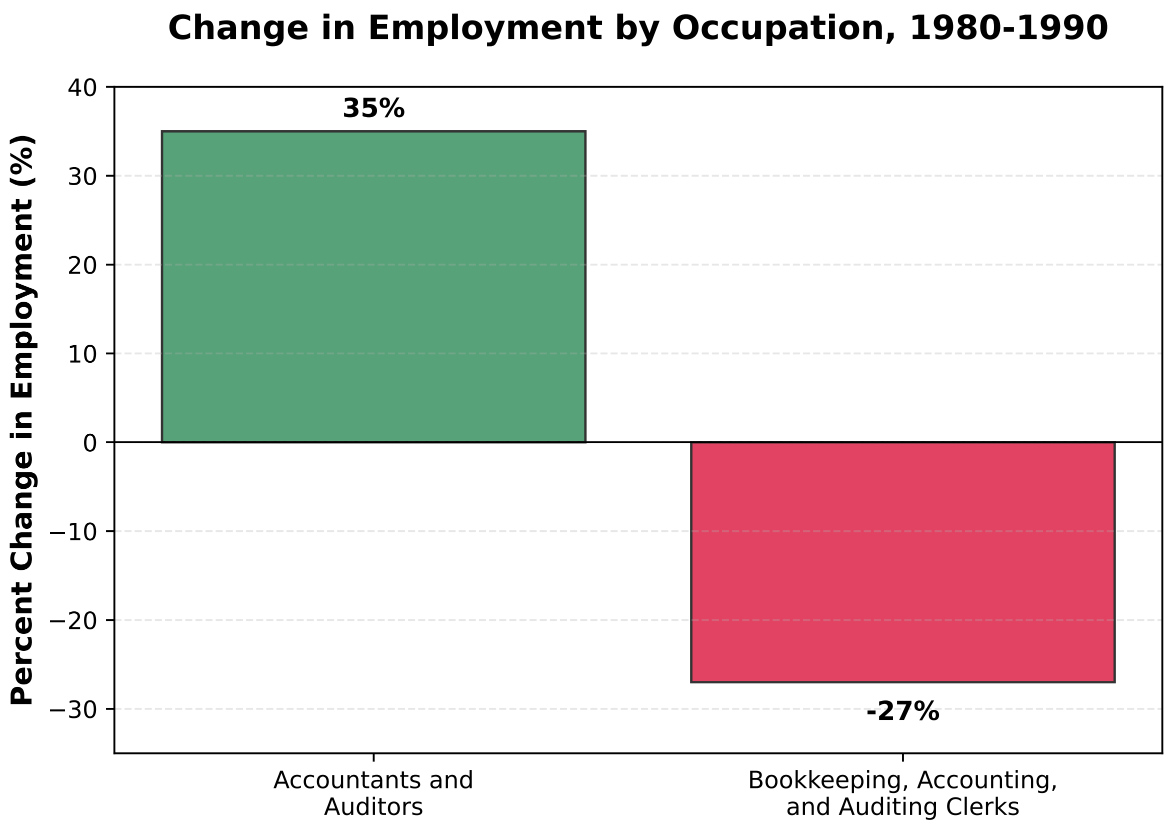


Figure 7.4: Change in Employment by Occupation, 1980-1990

While it was easier to monitor revenue and sales with computers, the actual task of checking out customers was not easily automated. In the figure below, we show the change in employment for cashiers between 1980 and 2000, plotted alongside the changes for accountants and clerks.

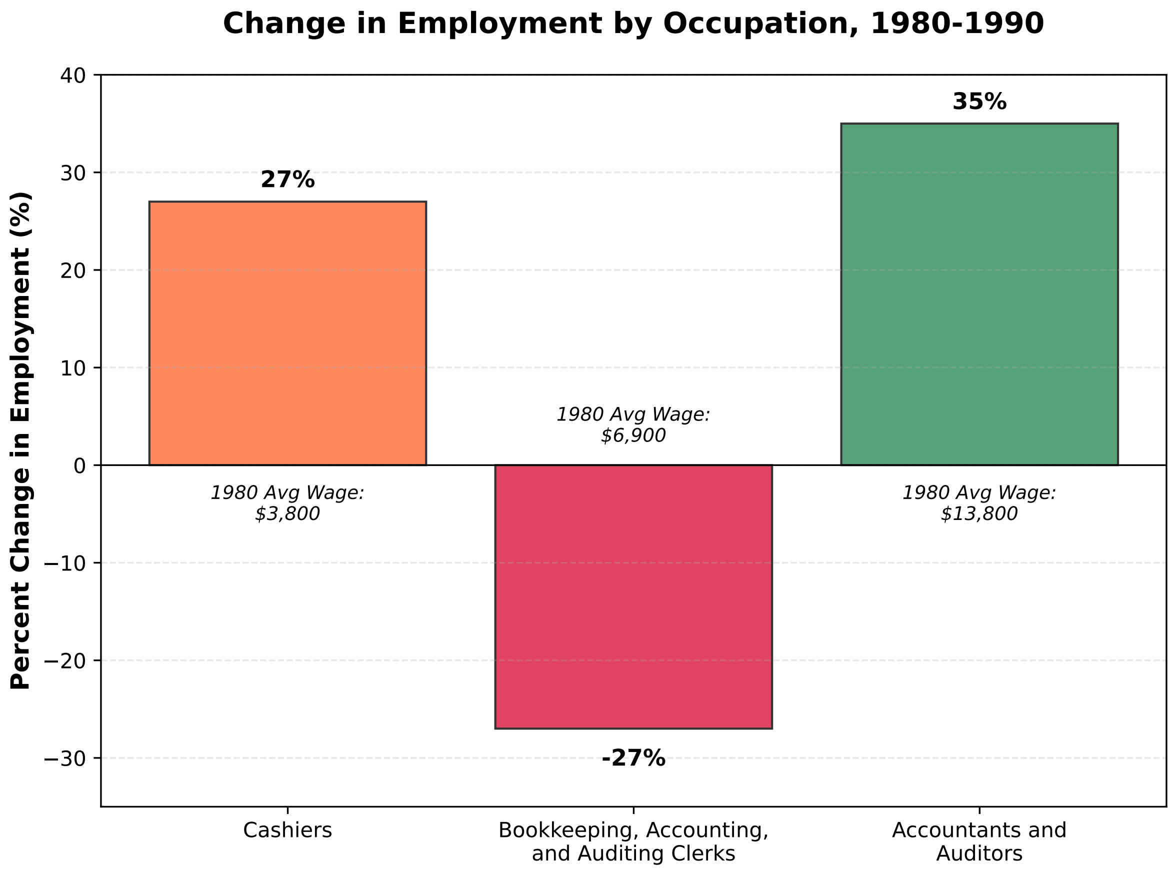


Figure 7.5: Change in Employment by Occupation, 1980-1990

As can be seen in the figure, cashiers saw a 27% increase in employment between 1980 and 1990, similar to our increase for accountants. As you may have noticed in the figure, we also ordered these occupations by their average wage level in 1980. Cashiers were the lowest paid of the three, clerks were in the middle, and accountants were the highest paid. Therefore, from 1980 to 1990, we see growth in employment at both the low-wage end (cashiers) and the high-wage end (accountants), while we see a decline in employment in the middle (clerks). This pattern is referred to as *job polarization*. Technology has displaced middle-skill workers.

So far we have three occupations: accountants, clerks, and cashiers. Let's map these back to our task framework. Accountants generally perform non-routine cognitive tasks. Clerks generally perform routine cognitive tasks. Cashiers generally perform non-routine manual tasks. This begs the question: what about routine manual tasks? Are there occupations that perform routine manual tasks that have also seen declines in employment? Let's assume the accountant is an accountant for a firm that produces clothes, and as part of the production process they have workers who operate sewing machines. This is a routine manual task (replacing sewers with machines is a long tradition, with the original luddite movement being a reaction to the mechanized looms). The figure below shows the change in employment for sewing machine operators between 1980 and 2000, alongside our other three occupations.

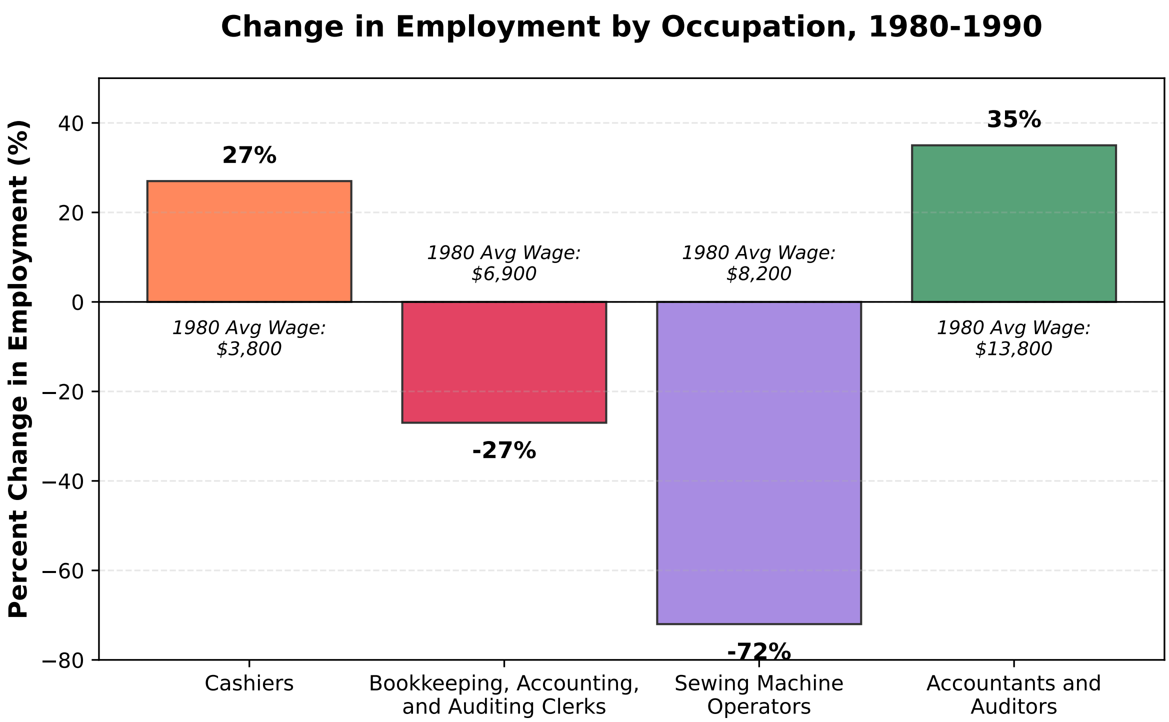


Figure 7.6: Change in Employment by Occupation, 1980-1990

So in this figure, we see that sewing machine operators saw an enormous 72% decline in employment between 1980 and 2000. This is consistent with our theory that routine tasks are being displaced during this period. To summarize, we considered four occupations: - Accountants and Auditors: non-routine cognitive tasks, saw a 35% increase in employment - Bookkeeping, Accounting, and Auditing Clerks: routine cognitive tasks, saw a 27% decrease in employment - Cashiers: non-routine manual tasks, saw a 27% increase in employment - Sewing Machine Operators: routine manual tasks, saw a 72% decrease in employment

Because the routine tasks (both cognitive and manual) had wages in the middle of the wage distribution (relative to accountants and cashiers), we see a pattern of job polarization: growth in low-wage and high-wage occupations, and decline in middle-wage occupations.

Now, we've done this for a few set of occupations. This is useful as it gives us a concrete way to tell a story about how technology may or may not impact these occupations, but we should be careful not to extrapolate too much from a few examples. Our goal next is to understand whether this pattern of job polarization is a general phenomenon or just holds for these few occupations we happened to pick.

One way to do this is to look across all occupations and rank them in terms of their average wage in 1980. Then, we can compute the change in employment for each occupation between 1980 and 2000. If job polarization is a general phenomenon, we should see that occupations at the low end and high end of the wage distribution saw increases in employment, while those in the middle saw decreases. The figure below (D. H. Autor and Dorn (2013)) shows this relationship. The horizontal axis is the rank of the occupation in terms of its average wage in 1980. The vertical axis is the change in employment between 1980 and 2000. So to be concrete, the average wage for accountants in 1980 put them at just above the 70th percentile of this distribution. So 70 percent of occupations had lower average wages than accountants. Accounting clerks were around the 23rd percentile, while cashiers were around the 6th percentile. The figure shows a smoothed version of the relationship between employment changes and the average wage of the occupation. Smoothed in this context means that we are not plotting the raw data for every single occupation, instead we are plotting a smoothed curve that captures the general trend.

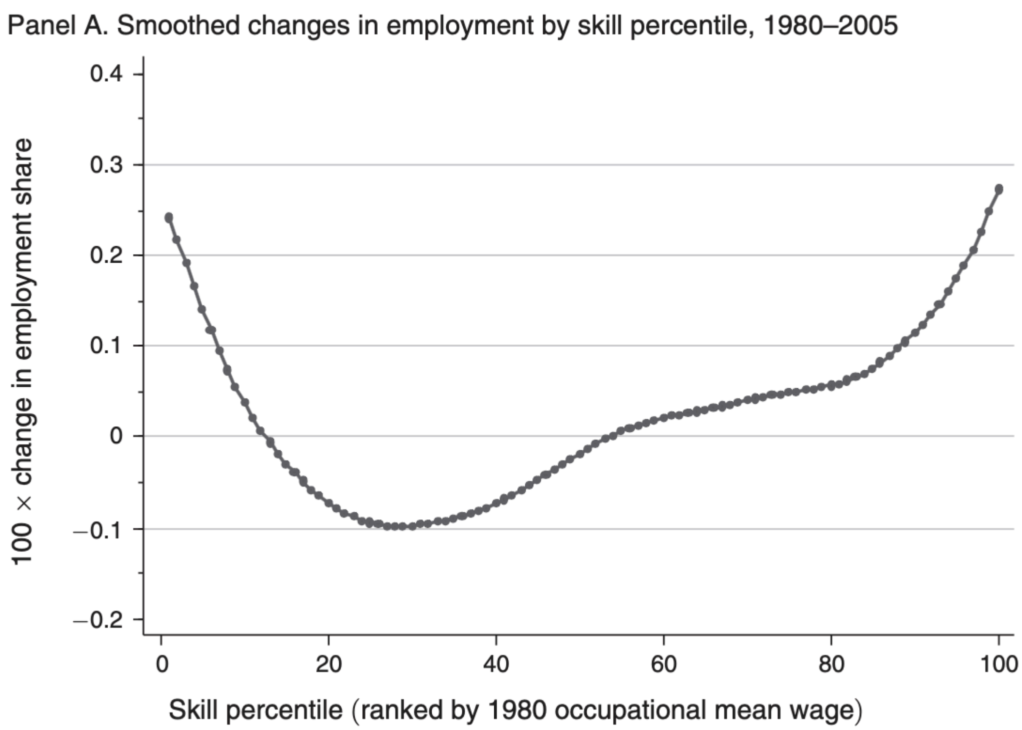


Figure 7.7: Employment growth across occupations by initial wage level

As we can see in the figure, occupations at the low end of the wage distribution (left side of the figure) saw large increases in employment. Many of these low-wage occupations are in the service sector, things like food preparation, cleaning, and waiting tables. Occupations at the high end of the wage distribution (right side of the figure) also saw increases in employment. Many of these high-wage occupations are in professional and technical fields, like management, law, and medicine. In contrast, occupations in the middle of the wage distribution saw decreases in employment. Many of these middle-wage occupations are in clerical and administrative support roles.

One remarkable aspect of this story is that it is not at all specific to the United States. For example, Acemoglu and Autor (2011) present results for a number of countries. To make things easy-to-visualize, they break occupations into three groups: low-wage (bottom third of the distribution), middle-wage (between 33rd percentile and 66th percentile), and high-wage (top third of the distribution). They then plot the change in employment for each group over time. The figure below shows the results for a number of countries.

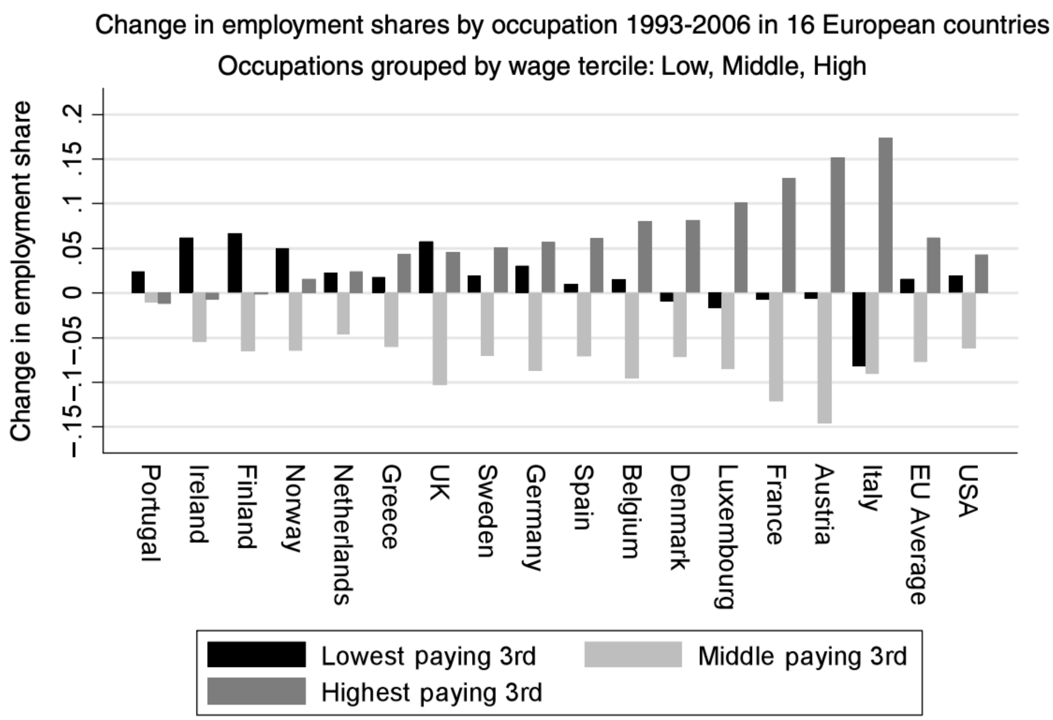


Figure 7.8: Job polarization patterns across multiple countries

If we zoom in on the right-most part of the figure, we see the USA, where we know lowest paying jobs and highest paying jobs have grown, while middle-paying jobs have shrunk. But we see this same pattern across many countries. The most stark pattern is the persistent increase in growth of high-wage jobs in many countries, and the persistent decline in middle-wage jobs. The low-wage jobs have more mixed results, but many countries show increases in low-wage jobs as well. Overall for the EU, the pattern is remarkably similar to the US.

7.7 Job Polarization and Tasks

Let's revisit our original argument: computers are good at performing routine tasks. For our accountant, clerk and cashier example, we argued that the accountants specialize in non-routine cognitive tasks, clerks specialize in routine cognitive tasks like tracking data, and cashiers specialize in non-routine manual tasks. Now, to show that job polarization is really driven by technology, it would be helpful if we could show that the occupations that are declining in employment are the ones that specialize in routine tasks.

There are a couple of different ways to do this. One way is to use the Dictionary of Occupational Titles (DOT) to classify occupations by the types of tasks they perform. The DOT has a number of variables that describe the tasks performed in each occupation. This is the approach taken by D. H. Autor, Levy, and Murnane (2003), which first introduced the task framework.

Another option is to use the O*NET database, which replaced the DOT in the 1990s. The O*NET database has a number of variables that describe the tasks performed in each occupation. One advantage of the O*NET database is that it is more detailed than the DOT. It has over 200 variables that describe the tasks performed in each occupation.

However, arguably the simplest is to just use broad categories that seem plausible. Acemoglu and Autor (2011) use a simple classification of occupations into four categories: routine cognitive, routine manual, non-routine cognitive, and non-routine manual. They argue that professional, managerial, and technical occupations are non-routine cognitive. This group contains many detailed occupations, specific types of engineers, lawyers, doctors, managers, etc.

The second group is clerical, sales and administrative support occupations. These occupations specialize in routine cognitive tasks. This group contains occupations like secretaries, bookkeepers, and data entry clerks. The third group is service occupations. These occupations specialize in non-routine manual tasks. This group contains occupations like food service workers, janitors, and personal care aides. Lastly, the fourth group is production, craft, repair and operative occupations. These occupations specialize in routine manual tasks. This group contains occupations like assemblers, machine operators, and construction workers.

The table below summarizes this classification as well as the prediction of how employment should change in each group if computers are driving changes in employment shares. If computers can replace routine tasks, then we should see (relative) employment declines in the routine cognitive and routine manual groups. In contrast, we should see (relative) employment increases in the non-routine cognitive and non-routine manual groups.

Table 7.1: Task Classification and Predicted Employment Changes

Group Label	Example Occupations	Routine/Non-Routine	Cognitive/Manual	Predicted Change
Professional, Managerial, and Technical	Managers, Engineers, Lawyers, Doctors	Non-Routine	Cognitive	↑ *Increase*
Clerical, Sales and Administrative Support	Bookkeepers, Secretaries, Data Entry	Routine	Cognitive	↓ *Decrease*
Service Occupations	Food Service, Janitors, Personal Care	Non-Routine	Manual	↑ *Increase*
Production, Craft, Repair and Operative	Assemblers, Machine Operators, Laborers	Routine	Manual	↓ *Decrease*

So let's see what this looks like in the data. The figure below (Acemoglu and Autor (2011)) shows the change in employment shares for each of these four groups between 1959 and 2007. The vertical axis is the change in employment share, which is the fraction of all workers in that group.

Let's first look at the production and operators group, which tend to perform routine manual tasks. This group has seen an incredibly large decline in employment share, falling from about 40 percent of all workers in 1959 to just over 20 percent in 2007. Clerical and sales support occupations, which are also exposed to automation, saw a rise in the earlier part of the period, but declined throughout the 1990s and 2000s.

In contrast, non-routine cognitive occupations have seen a steady increase in employment share, rising from about 20 percent of all workers in 1959 to about 35 percent in 2007. Non-routine manual occupations have also seen an increase, though it is more modest, rising from about 14 percent in 1959 to about 18 percent in 2007.

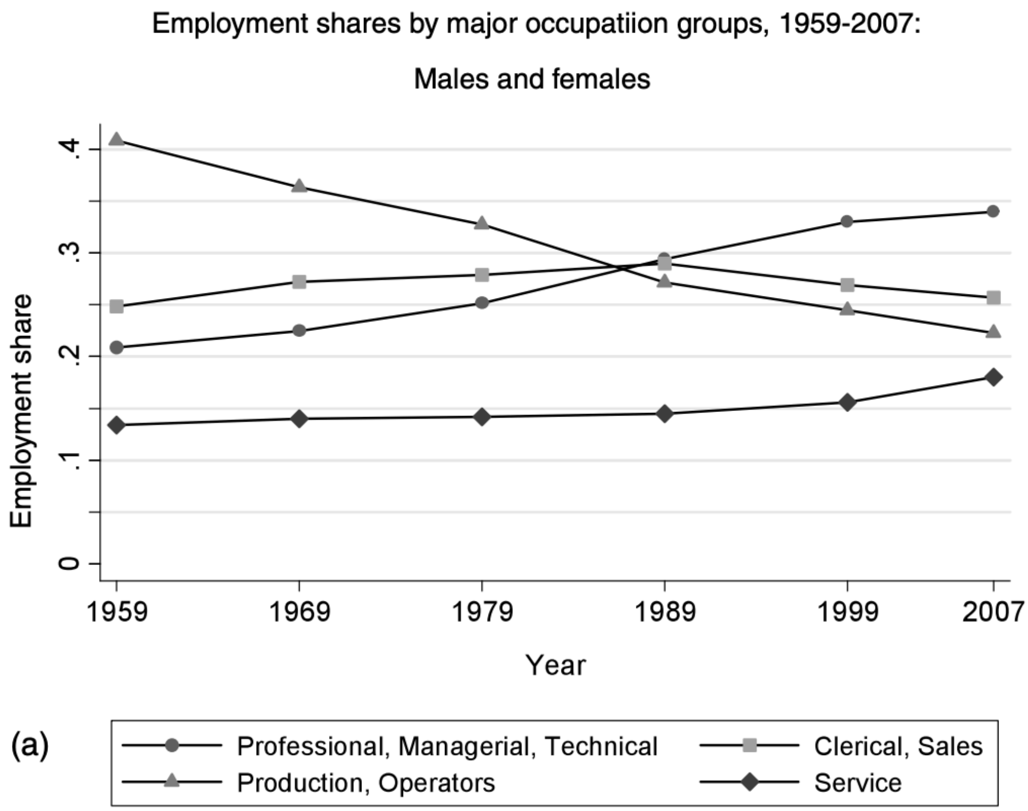


Figure 7.9: Change in Employment Shares by Task Type, 1959-2007

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Part II

Datasets

8 The Current Population Survey (CPS)

8.1 Brief Introduction

Modern labor economics started with the advent of large-scale surveys of individuals and households which began shortly after World War II. One of the most important, widely used, and to this day, influential surveys is undoubtedly the Current Population Survey (CPS).

8.2 The Current Population Survey (CPS)

The current population survey (CPS) is a monthly survey of about 60,000 households conducted by the U.S. Census Bureau and the Bureau of Labor Statistics. The CPS is the primary source of information on the labor force characteristics of the U.S. population, as well as the source for unemployment rates. To discuss the CPS, we are going to compute the unemployment rate and construct a measure of median personal income.

The CPS is what is referred to as a panel survey, implying it follows individuals over time. It has a somewhat complex design, where households are interviewed for four consecutive months, then not interviewed for the next eight months, and then interviewed again for four consecutive months. For example Figure 1 below illustrates the design for a household that was first interviewed in January 2024.

The 4th and 8th interview are referred to as outgoing rotation groups. These groups are often given extra consideration in analysis because in these interviews, respondents are asked additional questions about work and income. So if you are doing an analysis that requires an individual's income, you may need to restrict your sample to just the outgoing rotation groups. If you are more interested in labor force status, then you don't need to restrict your sample.

The variable MISH indicates the month-in-sample of the individual. For example, if MISH=1, then the individual is in their first month of being interviewed. If MISH=4, then the individual is in their fourth month of being interviewed and is part of an outgoing rotation group. If MISH=5, then the individual is in their fifth month of being interviewed and so on. If you want to restrict to outgoing rotation group, you need to restrict to MISH=4 or MISH=8.

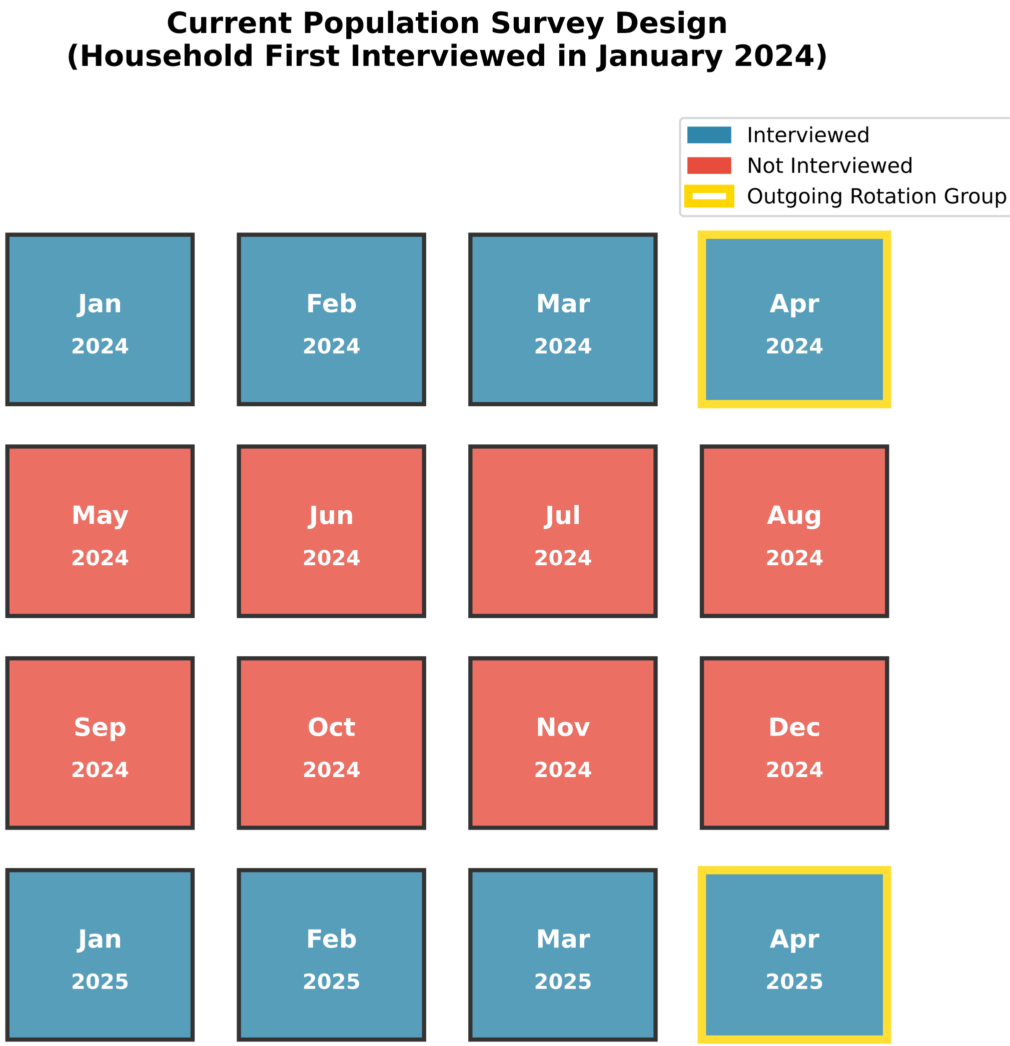


Figure 8.1: Current Population Survey Design

8.3 Downloading the CPS

There are two main ways to access the CPS data. The first is directly through the Bureau of Labor Statistics website. The second is through the IPUMS-CPS project, which provides harmonized CPS data. This is much easier for a number of reasons. First, the IPUMS-CPS project (IPUMS stands for Integrated Public Use Microdata Series). IPUMS provides a user-friendly interface to select variables and download data. You can also select the samples you want, implying you can select information from multiple years and download it all in one file. The raw data files provided by the Bureau of Labor Statistics require much more data cleaning and processing. Second, the IPUMS-CPS project provides harmonized variables that are consistent over time, which is especially useful if you are working with data from multiple years. Lastly, the IPUMS-CPS project provides detailed documentation and support for users. For example, if you are confused about what a variable in the CPS means, you can often google the variable name and find a relevant IPUMS-CPS documentation page that explains the variable in detail. For example, when you submit an extract request, it comes with a codebook that provides detailed information about each variable, including its coding scheme.

8.4 Computing the Unemployment Rate

For our first foray into the CPS, we are going to compute the unemployment rate. We are going to do this for August 2025 data (see course website for link to the data file `cps_aug_25.csv`).

Let's first load the data and take a look at the variables that are included.

Table 8.1: CPS Dataset Variables (Total: 24 variables)

0	1	2	3
YEAR	STATEFIP	AGE	UHRSWORKT
SERIAL	PERNUM	SEX	WHYUNEMP
MONTH	WTFINL	EMPSTAT	EMPSAME
HWTFINL	CPSIDP	LABFORCE	EDUC
CPSID	CPSIDV	OCC	UNION
MISH	EARNWEEK2	CLASSWKR	UHRSWORKORG

Let's compute the unemployment rate among these group of workers. The unemployment rate is defined as the number of unemployed individuals divided by the labor force. Remember from Chapter 1 that if you are not currently employed and not actively looking for work, then you are not part of the labor force. Before we compute the unemployment rate, let's first see how many individuals are in the labor force. We can do this by looking at the variable `LABFORCE`, which indicates whether an individual is in the labor force or not.

Table 8.2: Distribution of Labor Force Status (LABFORCE Variable)

Labor Force Status	Description	Count	Percentage
0	NIU (Not in Universe)	16167	16.54%
1	No, not in labor force	32988	33.75%
2	Yes, in labor force	48585	49.71%

In the CPS, variables are generally coded numerically. To understand what the numbers represent, however, you need to use the codebook. This is where the labels for the numbers come from. The first thing you might notice is that a larger share of people are recorded as "Not in Universe". This implies the labor force status is not applicable for these individuals. This can happen for two reasons, basically. First, the CPS does ask about individuals who are under 16 years old, and these individuals are not part of the labor force. For these individuals, often the parents will fill out the survey on their behalf. For our purposes, let's restrict to individuals that are 16 years old or older. We can do this by restricting to individuals with `AGE>=16`. This restriction drops most of the 16,167 individuals who are "Not in Universe", but there remain a few left. In this case, these individuals should have been asked this question, but for some reason, they were not. This can happen for a variety of reasons, such as the individual refusing to answer the question or the interviewer making a mistake. For our purposes, we are going to drop these individuals from our analysis.

Table 8.3: Distribution of Labor Force Status (Ages 16+)

Labor Force Status	Description	Count	Percentage
1	No, not in labor force	31929	39.72%
2	Yes, in labor force	48448	60.28%

So in our relevant sample, just above 60% are in the labor force and just below 40% are not in the labor force. This is a pretty standard estimate of this statistic. To compute the unemployment rate, we need to know how many individuals among those in the labor force are not currently employed. This information is contained in the variable `EMPSTAT`, which includes a more thorough breakdown that indicates whether an individual is employed, unemployed, or not in the labor force. Let's take a look at the distribution of this variable among our remaining sample.

Table 8.4: Distribution of Employment Status (EMPSTAT Variable)

Employment Status	Description	Count	Percentage
10	At work	44348	55.17%
12	Has job, not at work last week	2069	2.57%

Employment Status	Description	Count	Percentage
21	Unemployed, experienced worker	1815	2.26%
22	Unemployed, new worker	216	0.27%
32	Not in labor force, unable to work	3704	4.61%
34	Not in labor force, other	9190	11.43%
36	Not in labor force, retired	19035	23.68%

The Not In Labor Force categories match what we saw before (19035+9190+3704=31929). The employed categories (10 and 12) sum to 46,417, while the unemployed categories (21 and 22) sum to 2,031. The unemployment rate is then computed as:

$$\begin{aligned} \text{Unemployment Rate} &= \frac{\text{Number of Unemployed}}{\text{Labor Force}} && (8.1)\\ &= \frac{2031}{2031 + 46417} && (8.2)\\ &= 4.4\% && (8.3) \end{aligned}$$

You might see a slightly different unemployment rate reported in the news or on the BLS website. This is because the BLS uses a slightly different sample and also applies weights to the data to make it representative of the U.S. population. The weights are provided in the CPS data ('WTFINL'). The basic reason behind the weights is that the CPS purposely oversamples some populations. For example, imagine you want to understand the age distribution of a University, but also wanted to understand what fraction of seniors at a University have already found a job. You may want to oversample seniors to get a better estimate of this statistic. However, if you oversample seniors, then your sample will not be representative of the overall population. If you take the average age, then it will be biased upwards because seniors are older. To correct for this, you would apply weights to the data so that seniors are weighted less and other groups are weighted more. This is the basic idea behind weights in survey data. Weights are often provided in survey data to make the sample representative of the population. For our purposes, the weights will not actually make too much of a difference and would complicate the analysis, so we will ignore them for now.

⚠ AI for Data Analysis

AI is undoubtedly an extremely useful tool for data analysis. Much of the code writtent to clean and organize data is fairly simple and has been performed thousands of times before by many different people. This makes it ideal for AI to help generate code snippets and explanations.

However, AI tools can also make mistakes. For example, when constructing the frequency tables above, the AI I used (Claude) provided suggestions for the labels of the different numeric codes in the CPS. While most were correct, a few were incorrect. If I had not cross-referenced with the official documentation, I would have classified a large number of workers that were not in the labor force as not in universe. This would lead to big issues if your goal was to study labor force participation. Additionally, the AI sometimes made simple mathematical mistakes (such as incorrectly summing up a list of numbers).

8.5 Occupation

In this course, one important focus will be on occupations. In particular, when we study how technological changes impact the labor market, our predictions will depend crucially on the types of tasks performed in different occupations. Therefore, let's dig into the occupation variable in the CPS.

The occupation variable, by necessity, has gone through some changes over time. When the CPS was created in the 1940s, there simply weren't occupations like "Software Developer" or "Data Scientist". Therefore, the CPS has periodically updated the occupation codes to reflect changes in the labor market. The occupation variable in the CPS is called `OCC`. Since we are using the 2025 data, we are going to use the classification system that was introduced in 2018.

Let's see what the five most common occupations are in our sample.

Table 8.5: Top 5 Most Common Occupations Among Employed Workers

Occupation Code	Occupation Title	Count	Percentage
440	Managers, all other	1707	3.68%
3255	Registered Nurses	1035	2.23%
9130	Unknown Occupation	1003	2.16%
2310	Elementary and middle school teachers	959	2.07%
4700	First-line supervisors of retail sales workers	893	1.92%

Depending on your analysis and what you are interested in, you may need to either aggregate data from many more time periods or group some occupations together. For the most popular occupation (Managers, all other), we have 1,707 observations in August 2025.

8.6 Earnings

Next, we are going to studying earnings. As discussed before, earnings information is primarily studied using the outgoing rotation group (i.e. MISH=4 or MISH=8). There are a few things about earnings data that are important to note.

First, because earnings information is only asked in the outgoing rotation group, many people have a numeric code that indicates that they were not asked this question. The variable we will be using in `EARNWEEK2`, which indicates the total earnings in the past week. If you use earlier versions of the CPS, we could have used `EARNWEEK`, which is similar. The only difference between these two variables is that `EARNWEEK2` rounds numbers to preserve confidentiality of the respondent. This was introduced in April 2023. If you are using data from before this date, you should just use `EARNWEEK`.

If you go to the codebook for `EARNWEEK2`, you will see that there are a number of numeric codes that indicate the respondent was not asked this question. For example, the codebook reports that `EARNWEEK2=999999.99`, indicates that the person is not in universe for this question.

Table 8.6: Distribution of EARNWEEK2 = 999999.99 (Not in Universe)

EARNWEEK2 Value	Count	Percentage
999999.99	70372	87.55%
All Other Values	10005	12.45%

The individuals with missing earnings information are either individuals who are not working or individuals not in the outgoing rotation group. That is why the fraction with earnings information appears relatively low. Not dropping individuals with missing earnings information from the sample would be extremely problematic. Because the missing data is stored as a numeric variable, imagine we tried to compute the average weekly earnings. If we failed to drop the missing individuals, then the average would be around 900,000 dollars per week. This is obviously not correct. So we need to drop these individuals from our sample. Then, let's compute some summary statistics for weekly earnings.

Table 8.7: Weekly Earnings Summary Statistics (Excluding Missing Values)

Statistic	Value
Mean Weekly Earnings	$1,464.58
Median Weekly Earnings	$1,076.00
10th Percentile	$390.00
90th Percentile	$2,700.00
Observations	10,005

This table gives us a sense of the distribution of weekly earnings among workers in our sample. The median weekly earnings is $1,076.00. Assuming a 52-week year, this implies median annual earnings of about $55,952. The average is quite a bit higher at $1,464.58. This is common in earnings data, which tends to be right-skewed. When there are some really large value, this will pull the average up, while the median doesn't depend on the magnitude of the values, just their rank.

To see this clearly, imagine we had only 5 workers with weekly earnings of \$500, \$500, \$1000, \$1,500, and \$1,500. In this example, both the mean and median are \$1,000. Now imagine the 5th worker sees an increase in their weekly earnings, growing all the way to \$10,000. This has no impact on the median, which remains \$1,000. However, the mean increases to \$2,700.

To see the entire distribution of weekly earnings, let's plot a histogram.

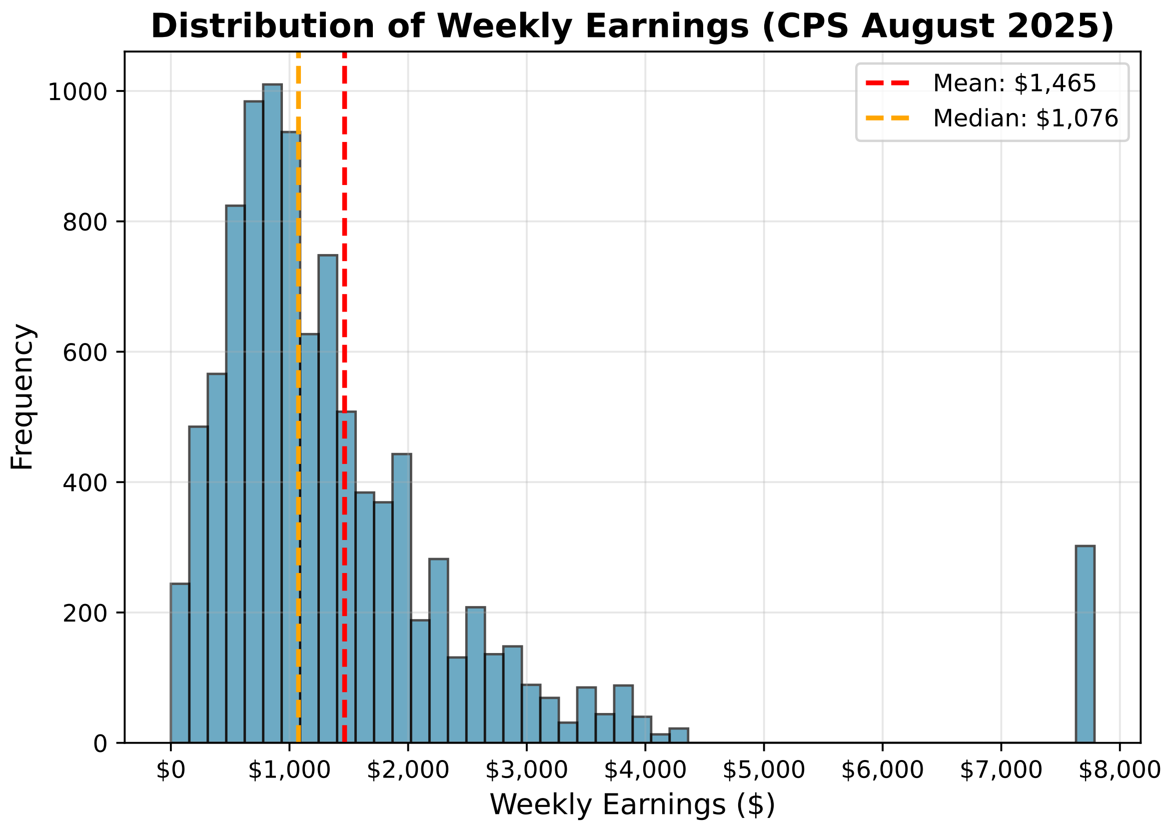


Figure 8.2: Distribution of Weekly Earnings

This distribution has a long right tail, which is typical of earnings data. The mean is pulled up by the high earners, while the median is more representative of a typical worker's earnings. One thing you might notice is that there seems to be a spike just below \$8,000 a week. It seems unlikely that so many people would have exactly that amount of eanrings. Additionally, it is surprising that there is basically no one with earnings between \$5,000 and \$7,000. What is going on here?

If you look carefully in the data, you will find that there are a non-negligible fraction of individuals with weekly earnings reported as exactly equal to $7,785.88. But wasn't this supposed to be a rounded earnings variable? The reason this is not rounded is because these individuals earnings have been top-coded. For earnings above a certain amount, there are concerns that the individual would be identifiable based on their earnings. Therefore, to preserve confidentiality, the CPS top-codes earnings at a certain level. For 2025, the Census top-codes the top 3 percent of earners and sets the value of their weekly earnings to the average earnings of the top 3 percent of earners.

If you look even more closely at the data, you'll find that the largest value that is not top-coded is $4,300. So if you make more than $4,300 a week, that implies you are in the top 3 percent of earners. Among individuals that make more than $4,300 a week, the average is $7,785.88. This is why there is a spike at this value because anyone with earnings above $4,300 is recorded as having earnings of $7,785.88.

Top-coding has changed over time. The threshold has changed, and in prior years, it was static. In other words, instead of top-coding the top 3 percent of earners, the CPS would top-code anyone making more than a certain amount.

8.7 Conclusion

The Current Population Survey is one of the most important and influential surveys in labor economics. It remains important today as the source for up-to-date unemployment numbers. However, using the CPS for research requires one to be careful. Without a thorough understanding of how variables are coded, the structure of the survey, and changes to the survey over time, it is easy to make mistakes that can lead to incorrect conclusions.

9 The American Community Survey (ACS)

9.1 Brief Introduction

The American Community Survey (ACS) is a nationwide survey conducted by the U.S. Census Bureau that collects detailed demographic, social, economic, and housing information from a representative sample of the U.S. population. It overlaps considerably with the Current Population Survey (CPS), although there are some important differences.

9.2 Downloading the ACS

IPUMS USA (same organization that produces IPUMS CPS) provides harmonized versions of the ACS. There are a couple different types of data that you can download from IPUMS USA. For example, you can download Decennial census microdata from 1790 to 2010 or the American Community Survey data from 2000 to present. For this chapter, we are going to download the most recent version of the ACS data from 2023.

Again, like the CPS, there are many variables available in the ACS. For this chapter, we will focus on a subset of variables related to health insurance coverage, labor force participation, and demographics. However, to stress, the ACS can (and has) been used to study a wide variety of topics. If you are interested in exploring the ACS data further, I encourage you to visit the IPUMS USA website. This will allow you to see the massive time span that the ACS and Census data cover as well as the wide variety of variables that are available.

9.3 Retirement Policies

There are a number of policies in the United States that are tied to age and may impact retirement decisions. The one we are going to focus on is the availability of publicly provided health insurance through Medicare, which begins at age 65. Since health insurance is often tied to employment in the U.S., gaining access to Medicare may influence individuals' decisions about whether to remain in the labor force or retire.

However, there are two other important age-based policies. First, at age 62, individuals become eligible to receive Social Security retirement benefits, although these benefits are reduced compared to waiting until full retirement age. At full retirement age (which is 67), individuals can receive their full Social Security benefits. Therefore, the availability of Social Security may impact when individuals choose to retire.

Since we are focusing on retirement, we are going to limit our sample to individuals aged 55-75.

9.4 Key Variables

The dataset contains several binary indicator variables. These are variables that take on either a value of zero (condition is false) or one (condition is true). For example, let's first take a look at the variable indicating whether an individual has health insurance coverage. The name of this variable is `has_health_insurance`. Note: this does not come directly from IPUMS USA, but rather was created by me while cleaning the data. In particular, as part of the cleaning process I dropped all individuals for which this variable was missing and recoded the variable so that it is equal to 1 if the individual has any health insurance coverage and 0 otherwise.

	Has Health Insurance	Count	Percentage
0	0	37395	3.87
1	1	927643	96.13

So in the data, most individuals do have health insurance coverage. Many of of these individuals likely have private heath insurance through their employer, while others may have public health insurance through programs like Medicare. It is also possible that some individuals younger than 65 may have public health insurance through Medicaid.

As a quick aside, it is generally very useful to code binary indicator variables as 0/1 variables rather than using string labels like "yes" and "no". This makes it much easier to work with these variables in statistical software. Additionally, you want the variables to be coded as 0/1, not some other numbers like 1/2 or -1/1. There are many reasons for this. One simple reason is that if you want the proporition of individuals with health insurance, you can just take the mean of the variable when it is coded as 0/1. For example, imagine we have three individuals with health insurance coded as follows:

Individual	has_health_insurance
1	1
2	0
3	1

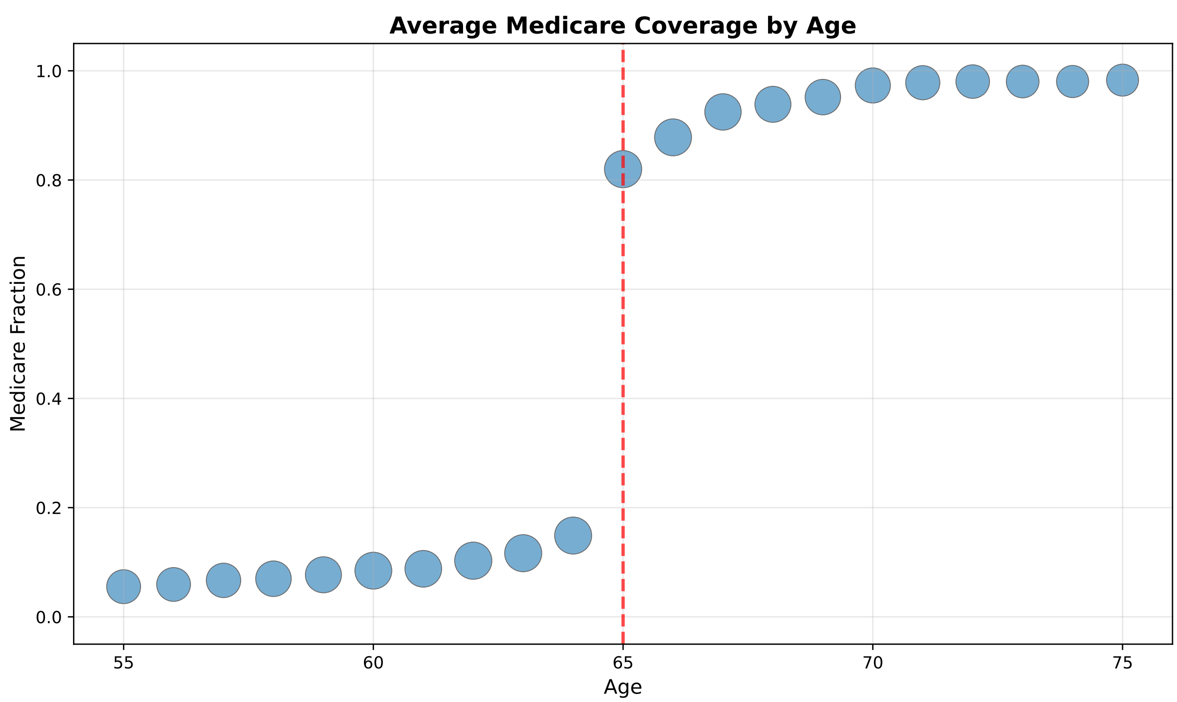
It is easy to see that 2 out of 3 individuals have health insurance, or 66.67%. We can also calculate this by taking the mean of the variable: (1 + 0 + 1) / 3 = 0.6667. This will be useful later when we construct graphs. If we want the fraction in the labor force by age, we can just take the mean of the `in_labor_force` variable. Now, if the variable were coded as "yes" and "no", or as 1 and 2, we would not be able to do this.

Since we are going to be focused on retirement decisions due to Medicare, let's look directly at this variable as well. This variable is called `has_medicare` and is again a binary indicator variable equal to 1 if the individual has Medicare coverage and 0 otherwise.

	Has Medicare	Count	Percentage
0	0	462326	47.91
1	1	502712	52.09

9.5 Regression Discontinuity Design

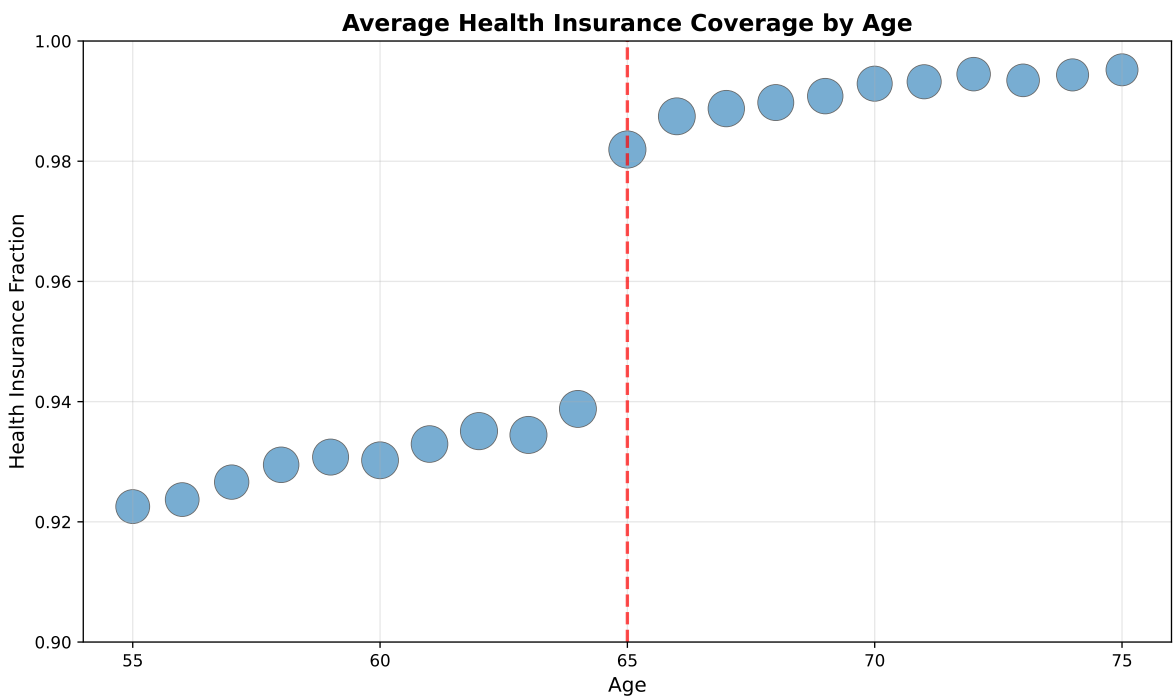
All of these policies are determined by cutoff rules, making them potential candidates for regression discontinuity analysis, which we studied in Chapter 6: Human Capital. First, let's see how the probability of having Medicare coverage changes at age 65. This will be our "treatment" variable. We will then study how this change in Medicare coverage affects labor force participation.



So clearly, turning 65 leads to a large increase in the probability of having Medicare coverage. However, it doesn't go from zero to 100 percent. There are some rare cases that individuals under 65 may have Medicare coverage (for example, individuals with certain disabilities). There is also potential for measurement error. People may be reporting that they are covered by Medicare when they are actually on Medicaid or insurance program. This could explain why we see a non-negligible fraction of individuals under 65 with Medicare coverage.

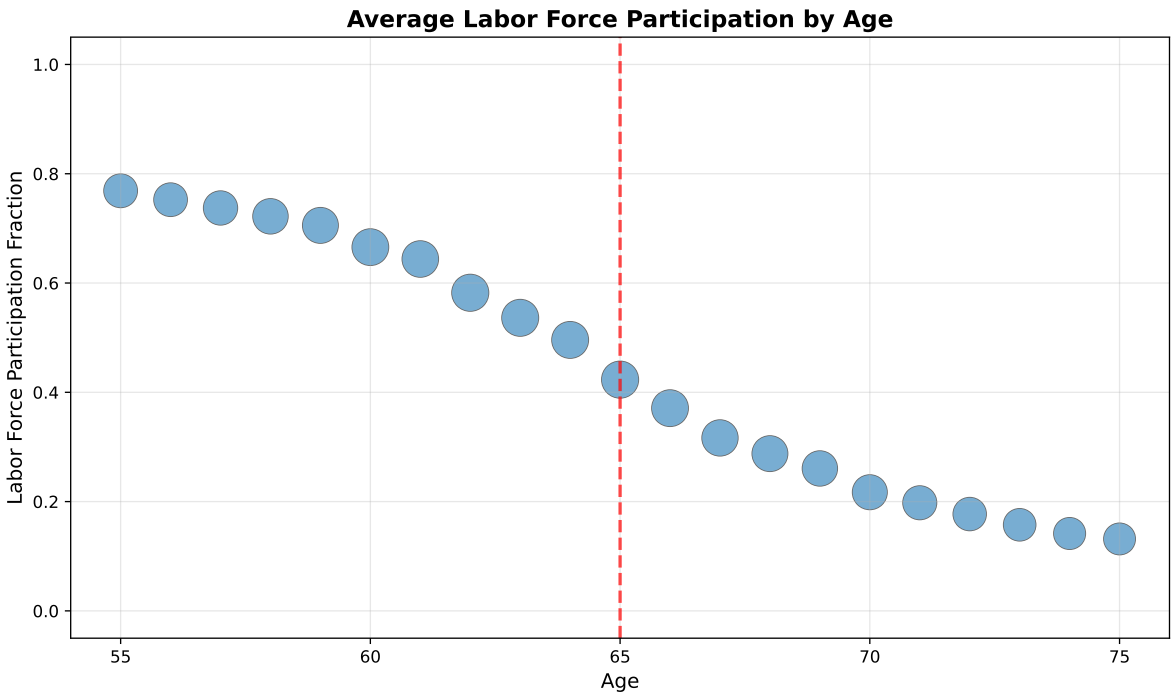
However, it is clear there is an enormous jump in the fraction of individuals with Medicare coverage at age 65. For 65 and older, more than 80 percent of individuals have Medicare coverage. This makes this a fuzzy regression discontinuity design. Crossing the threshold has a large impact on the probability of treatment (having Medicare), but it is not a perfect jump from zero to one.

Now, for contet, many people do have health insurance through private options, generally provided by employers. So let's see if this enormous jump in Medicare coverage at age 65 is associated with increased coverage more generally. Therefore, we are going to replace the `has_medicare` variable with the `has_health_insurance` variable in the graph above. This variable indicates whether the individual has any health insurance coverage, whether through Medicare, private insurance, Medicaid, or other sources.



Again we see a noticeable jump in the fraction of individuals with health insurance coverage at age 65. However, we have to be careful about the scale on the y-axis. As we can see, the fraction of individuals with health insurance before age 65 is already quite high, at 94 percent. However, there is still a noticeable increase in health insurance coverage at age 65 due to the availability of Medicare. The fraction of individuals with health care coverage increases to nearly 98 percent at age 65 and older.

Finally, let's see if this increase in health insurance coverage at age 65 is associated with changes in labor force participation. The variable we will use to measure labor force participation is called `in_labor_force`, which is again a binary indicator variable equal to 1 if the individual is in the labor force (either employed or actively looking for work) and 0 otherwise.



So perhaps surprisingly, we don't see a large drop in labor force participation at age 65. There is a gradual decline in labor force participation as individuals age, but there is no clear discontinuity at age 65. This suggests that gaining access to Medicare does not have a large impact on individuals' decisions to remain in the labor force or retire.

10 O*NET

10.1 Brief Introduction

The goal of O*NET (Occupational Information Network) is to provide comprehensive information about different occupations in the US. It replaced the Dictionary of Occupational Titles which was paper-based. It includes information on the tasks that individuals perform in occupations, the skills one needs to perform the occupation, details on the work activities performed, among other information.

In order to get this information on occupations, O*NET generally sends survey to workers who are current performing the occupation (referred to as incumbents). They are asked various questions about different tasks and skills and asked to rate them on a scale from 1 (not important) to 5 (very important). Some questions are given to all occupations while others are specific. For example, when asking about tasks, it would be unreasonable to ask about certain tasks for certain occupations. Therefore, experts use prior information (just as information from the Dictionary of Occcuapational Titles) to come up with a set of tasks that would be reasonable to ask about.

10.2 Interacting with O*NET

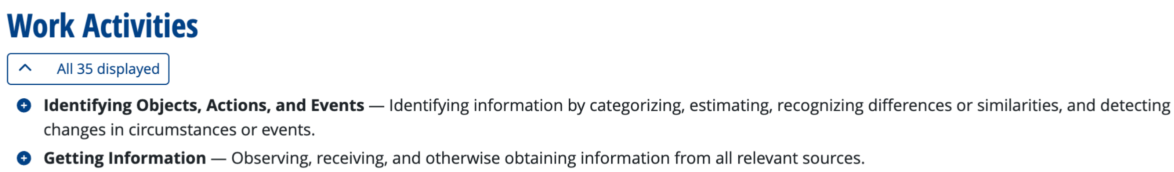
How one interacts with O*NET likely depends on the user. For example, a job seeker will generally use O*NET online. This is a website that allows users to search for specific occupations. The information will then be clearly presented. Researchers, generally, will tend to prefer to use the raw data downloads. This will allow one to perform statistical analysis on the data, while considering a wide array of occupations at once.

The raw data, however, can be quite complicated. Unlike when we studied the CPS or ACS, we don't get one table of information. Instead O*NET comes as a large list of files that can be merged together depending on the analysis that you are trying to perform. We are going to go through the structure of a few of these in this chapter, but there is much more to learn about the O*NET.

10.3 Work Activities

We are going to start quite broad. As we will show later, O*NET will allow us to drill down much more narrowly to get into exactyl what a person is doing in an occupation. But first we will start with relatively broad **Work Activities**.

As a case study to begin, we are going to look at Security Guards. If you go to O*NET Online and search for security guards, you will find under the tab Work Activities, a number of different activities that are performed by Security Guards. Below I have pasted the two work activities that appeared highest up in the list.


Work Activities

All 35 displayed

- **Identifying Objects, Actions, and Events** — Identifying information by categorizing, estimating, recognizing differences or similarities, and detecting changes in circumstances or events.
- **Getting Information** — Observing, receiving, and otherwise obtaining information from all relevant sources.

Figure 10.1: Work Activites (First 2) for Security Guards

We are going to focus on the second work activity **Getting Information**. O*NET also makes available the questionnaire that it sends to workers. Let's go look at exactly how this question is asked.

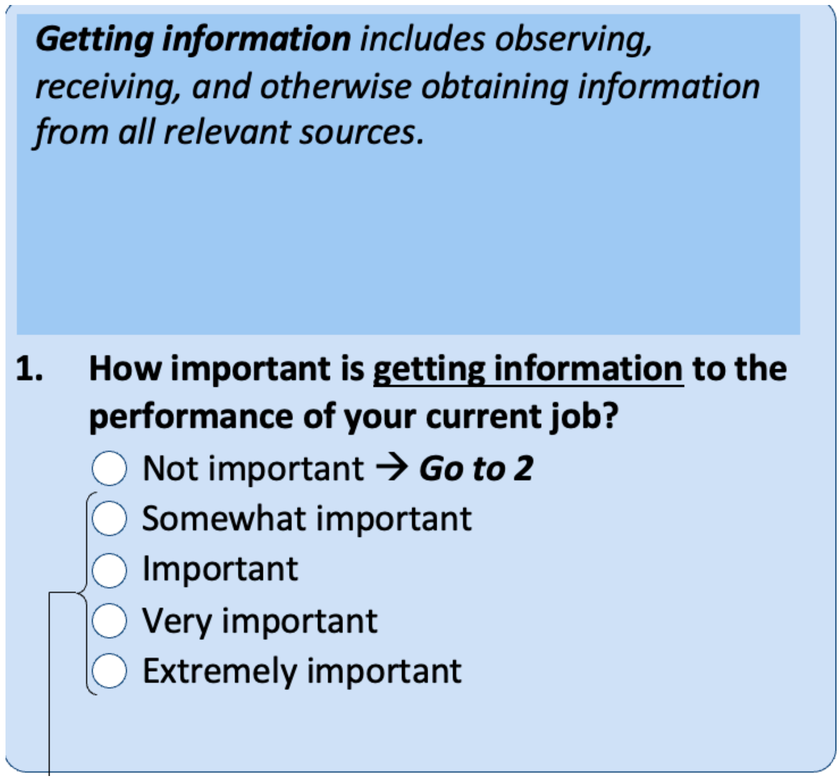


Figure 10.2: How Important is Getting Information

When we turn to the raw data, we will see that this information will be coded as a numeric variable. If you select not important, than that will be represented by a 1. If you select 5 that will be treated as extremely important. This format for scoring is extremely common in survey questions and is often referred to as a Likert scale (named after its inventor Rensis Likert).

Now, if you choose anything besides "Not Important", you will then be asked another question.

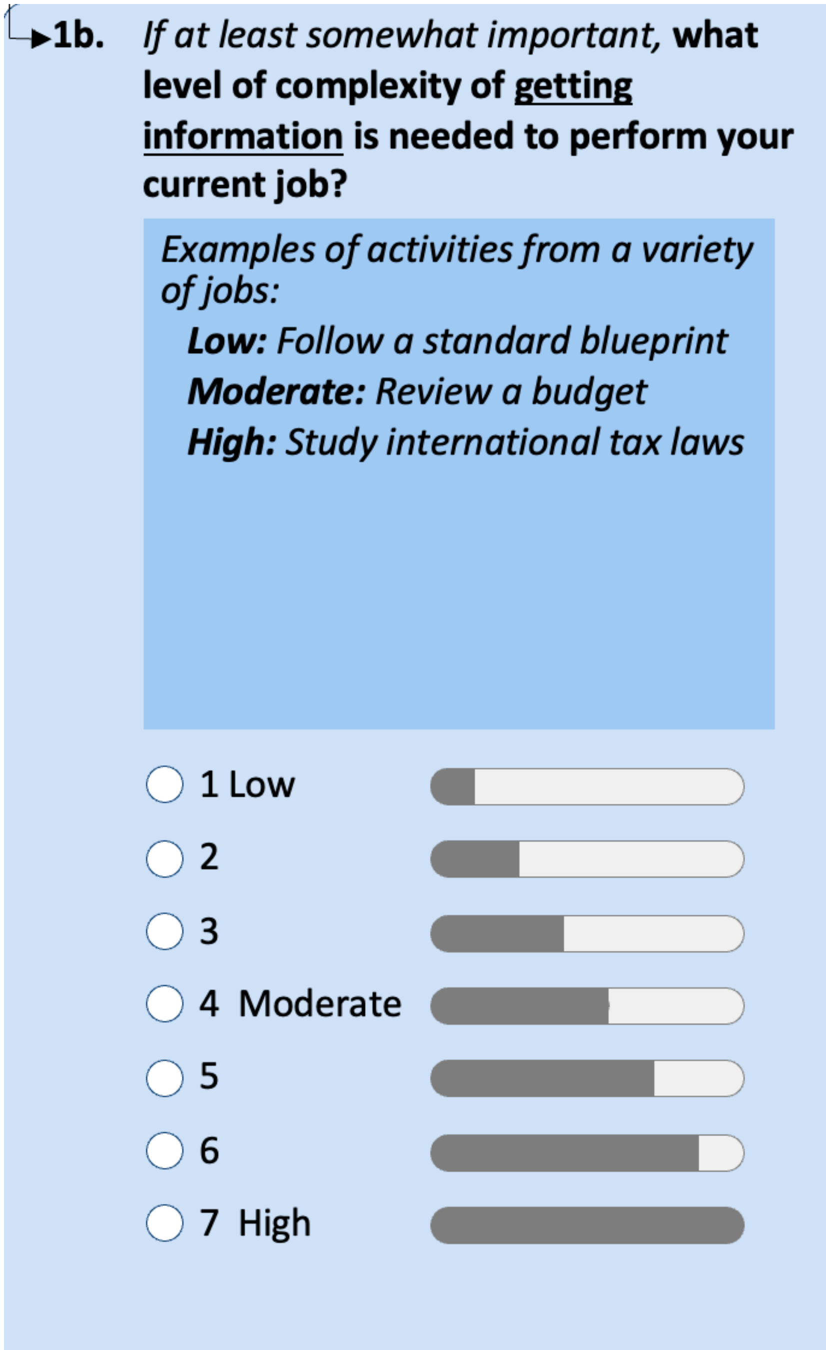


Figure 10.3: Complexity of Work Acitivity

This is referred to as the **Level** question. This is also coded as a numeric variable in the raw data, with higher numbers indicating a higher level of complexity. So when we go to the raw data next, we should keep these questions in mind. Unlike the online data, the raw data files will allow us to see exactly how individuals answered these questions.

The information below comes from a file named “Work Activities”. The way I downloaded the data is through going to the O*NET database website here: https://www.onetcenter.org/database.html. This website allows you to donwload the entire O*NET databse. Like mentioned before, this is a lot of files. You may not need all of these files. You can download individuals files by going to O*NET online and searching O*NET data tab at the top of the site.

Let’s get a list of the different variables in the Work Activities file.

O*NET-SOC Code	Element ID	Element Name
Scale ID	Data Value	N
Standard Error	Lower CI Bound	Upper CI Bound
Recommend Suppress	Not Relevant	Date
Domain Source		

We are going to discuss a few of these. The O*NET-SOC Code is goint to be how we identify occupations in the data. For example, the O*NET-SOC code for security guards is 33-9032.00. Each row in the data is a different question. The Element ID tells us which question we are referencing and Element Name gives us a description of that question. For example, if we look at the first element of Element ID an Element Name in the dataset we get:

```
Element ID: 4.A.1.a.1
Element Name: Getting Information
```

So the code associated with our Getting Information question is 4.A.1.a.1. However, there are actually two Getting Information questions. One is about the importance of getting information. The other is about the level of complexity of getting information. These questions actually have the same Element ID, but whether you are considering the importance or level question is given by the variable Scaled ID. This variable is either equal to “IM” for the importance questions or “LV” for the level questions. Let’s get a frequency table of IM vs. LV in the dataset.

```
Scale ID
IM    36654
LV    36654
Name: count, dtype: int64
```

So in total we have 36,654 rows of data that correspond to importance questions, and 36,654 rows of data that correspond to level questions. Before we continue, let’s see how many unique occupations there are in the data:

Metric	Count
Number of Unique Occupations	894

Now, this data structure isn't quite as straightforward as the previous datasets we covered, where the unit-of-observation was a person. In this dataset, the unit-of-observation is a question for a specific occupation. The variables O*NET-SOC Code, Element ID and Scale ID together make up the unit-of-observation. **Data Value** is going to be our key variable of interest. This tells us how people are answering the Work Activities Question. But to understand how to use it, let's focus on a single question first. Let's restrict our data to the importance version of the Getting Information question. In code, we are going to restrict to rows of the data in which

`Element ID == "4.A.1.a.1" & Scale ID == "IM"`

How you will restrict to these individuals depends a bit on what program you are using. Below we include Python, Stata and R code snippets that would perform this operation.

Python:

```python
wa_filtered = wa[(wa['Element ID'] == '4.A.1.a.1') & (wa['Scale ID'] == 'IM')]
```

Stata:

```stata
keep if ElementID == "4.A.1.a.1" & ScaleID == "IM"
```

R:

```r
wa_filtered <- wa %>%
  filter(`Element ID` == "4.A.1.a.1" & `Scale ID` == "IM")
```

You will note that sometimes the exact names of the variables have changed a bit. Stata does not allow names to have spaces (generally it is a good idea not to have spaces in variables names. Spaces make the code unnecessarily more complicated).

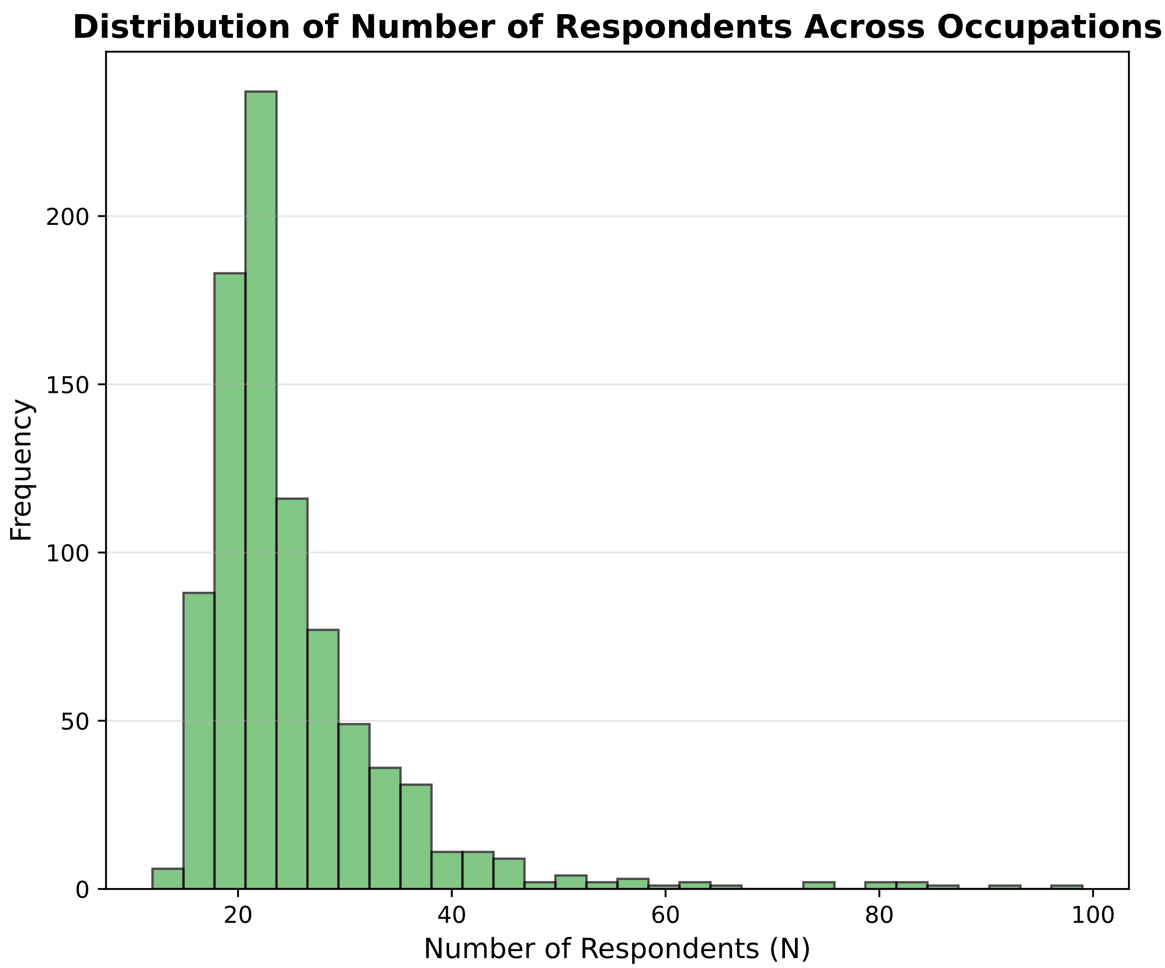
Now that we've restricted to our question of interest, let's make a summary statistics table that shows some key percentiles of the Data Value variable (i.e. the answer to how important getting information is for the job).

Statistic	Value
Mean	4.22
10th Percentile	3.71

75th Percentile	4.48
Number of Observations	894

Reminder, this question asked on a scale of 1 to 5, how important it is to get information in this job. As you can see the average across all occupation is 4.22, which indicates getting information on average is very important in many jobs. Even the 10th percentile, getting information is around 3.7, indicating it is still close to very important.

One question we should ask ourselves is: how many people are actually answering this question. This is given by the variable N in the dataset. Each value tells us how many incumbent workers responded to this question. Let's create a histogram of N across occupations so we can better understand the source of our information.



So the first thing to note is that there are not a tremendous number of respondents per occupation. Most occupations have around 20. This is why O*NET also provides standard errors and confidence bounds for questions. While we will not discuss these statistics in the class, they are ways to understand the uncertainty around an answer. For example, if an occupation reports a 4.5, but there is a large standard error on this estimate, then that implies we are relatively less certain about the true value for this question (where the true vale would be given by a very large sample size of individuals in that occupation).

Now, let's go back to our case study of security guards. Let's create a dataset that restricts only to the security guard SOC code so that we can understand this occupation in more detail. Below, we should the Work Activities that were rated most important for Security Guards.

O*NET-SOC Code	Element Name	Data Value	N
33-9032.00	Identifying Objects, Actions, and Events	4.76	16
33-9032.00	Getting Information	4.58	16
33-9032.00	Communicating with Supervisors, Peers, or Subordinates	4.54	16
33-9032.00	Documenting/Recording Information	4.48	16
33-9032.00	Establishing and Maintaining Interpersonal Relationships	4.45	16

One small aside to note here. You will notice that two of the top 5 work activities involves information. For a security guard, this is likely a physical task. They have to go to a place to record the information. Now, you may have seen headlines that reports X percent of jobs can be done by AI. Getting information is a task that AI can do very well. But it is fundamentally different than the type of information that security guards retrieve and document. In other words, within a given task or work activity, there is likely a lot of variation in exactly what that means. However, methods that don't take into account the context of the occupation may falsely label security guards as likely to be replaced by AI.

What might also be informative is to see the work activities that security guards deem the least important in their jobs.

O*NET-SOC Code	Element Name	Data V
33-9032.00	Repairing and Maintaining Mechanical Equipment	2.07
33-9032.00	Repairing and Maintaining Electronic Equipment	2.15
33-9032.00	Drafting, Laying Out, and Specifying Technical Devices, Parts, and Equipment	2.42
33-9032.00	Handling and Moving Objects	2.61
33-9032.00	Controlling Machines and Processes	2.61

Now, this only scratches the surface of what O*NET provides, but it gives us a starting point. If we are interested in how future technology may impact security guards, we can think about the work activities they perform. How do we think these work activities will be replaced (or augmented) by technology in the future.